

10

Aquaculture



Introduction

New Zealand's aquaculture sector is a significant primary industry, currently estimated to have revenue in excess of \$400 million, of which about 75 per cent is exported to 79 countries.¹ Māori own an estimated 50 per cent of the aquaculture industry. The annual aquaculture production and revenue is shown in Figure 10.1.

The most commonly farmed species in New Zealand are the indigenous green-lipped mussel, which is not farmed anywhere else in the world, the introduced Chinook salmon (also called king salmon) and the Pacific oyster. Mussels are predominantly farmed in the Marlborough Sounds, as well as at Golden Bay, the Coromandel Peninsula, Great Barrier Island, Canterbury and Stewart Island. King salmon are farmed in the Marlborough Sounds, Canterbury and Stewart Island. Pacific oysters are mainly farmed in the Far North, as well as at Coromandel, Auckland and Marlborough. New Zealand also has growing industries in pāua and seaweed. Other species, including kingfish, eels, geoduck clams and hāpuku have been farmed or trialled as possible species of the future.² Land-based aquaculture is also a rising sector. The location of aquaculture production in New Zealand is shown in Figure 10.2.

	Mussels	Salmon	Oysters
Harvested product (greenweight tonnage)	101,311	14,037	1804
Export revenue NZ\$ (millions)	218.1	63.4	16.6
Domestic revenue NZ\$ (millions) (estimated)	35.0	65.0	8.0

Figure 10.1 New Zealand aquaculture production and revenue for 2011 (*Aquaculture New Zealand*)³

The high quality of New Zealand's coastal waters, the abundance of plankton, and the prevalence of sheltered harbours and inlets means that the country has considerable aquaculture potential. The consumer demand for seafood (including shellfish) is rising internationally at a time when global wild stocks are under pressure, creating growing markets for farmed seafood products. In 2006, members of the industry working with the former Ministry for Economic Development released a strategy aimed at making aquaculture a billion-dollar industry in New Zealand by 2025.⁴

The Government's 'Aquaculture Strategy and Five-year Action Plan to Support Aquaculture', developed in 2012, sets out a 'whole-of-government pathway' to support growth in the aquaculture industry. This includes developing new farming space, improving the use of existing space, and increasing the value gained from production.⁵



Oysters such as those shown here on racks in the Whangaroa Harbour contributed over \$16 million in export revenue to New Zealand during 2011 (*Raewyn Peart*)

Aquaculture methods

Mussels

The main harvestable mussel species is the native New Zealand green-lipped mussel (*Perna canaliculus*), the shell of which can be up to 260 millimetres long, 110 millimetres wide and 90 millimetres deep.⁶ These mussels have been harvested for eating since humans first settled in the country.⁷ However, wild populations of the mussels were significantly reduced after heavy dredging during the 1960s. Commercial production then focused on farming the mussels.

A mussel farm consists of a series of rope 'backbones' suspended in, or on top of, the water column by a series of buoys. A continuous rope (the 'longline'), on which the mussels are grown, is suspended from the backbone line.⁸ The line is anchored to the seafloor at either end. A 3 hectare farm would typically have nine 110 metre lines. Each line is supported by 50 to 70 large plastic floats. Each float may support one tonne of mussels. One 110 metre line will generally support 3,500 to 4,000 metres of crop rope.

Most crop lines are seeded with mussel spat (juvenile mussels), with the majority sourced from Ninety Mile Beach in the Far North. Here, at irregular times throughout the year, large quantities of spat attached to seaweed are washed up on the beaches. This spat is collected and transported by air or truck to farmers in other parts of the country. The collection of this spat is managed under the QMS based on a management plan prepared in association with local iwi. Local spat catching also occurs on a smaller scale at some mussel farm sites in Golden and Tasman Bays, in Jackson Bay and in some bays in Marlborough.

Traditionally, virtually all mussel stock farmed in New Zealand was collected as spat from the wild. However, in recent years there has been an increasing focus on the opportunities offered by the use of hatcheries, where mussels can be selectively bred for specific qualities such as higher meat yield or faster growth. Hatchery spat are grown from a 'family' where the offspring are from two known parents. Each family may consist of several thousand siblings, grown together until they are large enough to transfer to the farm, with about 50 to 60 families per cohort needed to establish an effective breeding programme.⁹

A team of scientists from Shellfish Production and Technology New Zealand Limited and the Cawthron Institute have been rearing 80 families as part of a Primary Growth Partnership programme between the company, Sanford Limited and the Ministry for Primary Industries.¹⁰ Grow-out trials commenced in April 2013, with about 60,000 mussels from 50 families being sent to several sites to assess how they perform. Work has now begun on building a pilot-scale mussel hatchery, with the first commercial spat scheduled to be produced from 2015.¹¹

Once the spat are at the mussel farm, approximately 1,000 to 5,000 juveniles are wrapped onto each metre of rope using biodegradable cotton stocking mesh. This mesh gradually disintegrates, by which time the spat have attached themselves to the rope. The mussels are left to grow for three to six months, and then these nursery lines are lifted. The young spat are stripped from the ropes, reseeded onto a final production rope at approximately 150 to 200 per metre, and then returned to the sea to grow. Mussels take between 12 and 18 months to grow to a harvestable shell size of 90 to 100 millimetres.¹² The mussels are harvested by pulling the ropes up onto barges and removing the mussels.



Mussel farming, such as shown here in the Waiheke Channel, Hauraki Gulf, is the most common form of aquaculture in New Zealand (Raewyn Peart)

impact of the Ostreid herpesvirus (OsHV-1). Aquaculture New Zealand and the New Zealand Oyster Industry Association, through the Ministry for Primary Industries' Sustainable Farming Fund, are undertaking a selective breeding programme which has been designed to assist the oyster industry overcome the virus. The programme is using selective breeding, and survival trials, to enable hatchery supply of reliably resistant juvenile oysters.¹⁸ The oysters will be on-grown to bigger sizes, before they are transferred to locations where the water temperatures and environmental conditions create a risk of small oysters succumbing to the virus.

The endemic flat oyster (*Tiostrea chiensis*) is referred to by a number of names, including Chilean oyster, dredge oyster and Bluff oyster. These oysters are found all over the coast, but particularly in extensive beds in the Foveaux Strait, Golden Bay and Tasman Bay, where they have been dredged since the late 1800s.¹⁹ They generally reach the harvestable size of 58 millimetres within four to six years. The Cawthron Institute is undertaking breeding programmes for the species.

Salmon

New Zealand is responsible for approximately 70 per cent of the global king salmon (*Oncorhynchus tshawytscha*) production. Sea pen farming of finfish has become widespread internationally, driven by a growing demand for fish and global pressure on wild stocks.

New Zealand has no native salmon and only king salmon, which is native to the cool waters of the northern Pacific rim, have been successfully farmed here. After their introduction, king salmon became established on the east coast of the South Island, and to a lesser extent also on the South Island's west coast. Commercial farming of salmon started in the 1970s, both in inland ponds and in coastal waters. Today, most commercially-sold salmon are grown in pens in the sea.²⁰

No imports of live salmon have been allowed into New Zealand for the last 50 years – instead eggs and milt are collected from captive broodstock. Fertilised eggs are incubated in a freshwater hatchery, where newly hatched fry are reared for another six to 12 months. The fish are then transferred to the sea pens, where they remain until they reach harvestable size, in about two to three years.²¹

Whereas shellfish feed on naturally occurring plankton, farmed fish such as salmon are top-end carnivores. They are fed an artificial diet of food pellets. Historically, much of the protein and oil in the feed was sourced from wild fisheries. However, in recent years, land-based proteins have increasingly been used to reduce the wild fish component. For example, Skretting Australia now produces fish pellets where only 28 per cent of the protein is from wild fisheries and 60 per cent is from land-animal proteins, with the balance derived from vegetables. Of the wild fishery component, just under half is sourced from fish processing wastage, with the rest consisting of wild-caught Peruvian anchovy.²² Fish pellets are not currently manufactured in New Zealand because of the small size of the finfish aquaculture industry here.



Salmon are reared in sea pens such as shown here in Akaroa Harbour, Banks Peninsula (Raewyn Peart)

Eels

Freshwater eels are found in many river systems and lakes throughout New Zealand. Worldwide eel aquaculture is estimated to be worth over US\$1 billion, accounting for 65 per cent of total eel production. Asia and Europe are the largest eel markets, but a decline in glass eel (returning juvenile stage) stocks in both regions has opened up opportunities for exploitation of species in other areas.²⁸

Eel farming involves catching juvenile eels when they enter freshwater from offshore spawning areas and on-growing them to a marketable size. Glass eels can be grown to maturity in two years in captivity. Eel aquaculture requires knowledge of upstream glass eel migration patterns, together with effective collection methods and appropriate water quality and temperature conditions for culture.²⁹

As there is also a commercial and customary eel fishery in New Zealand, there is some opposition to the farming of eels if it is to be at the expense of the wild fishery, the management of which is currently being reviewed by the Ministry for Primary Industries.

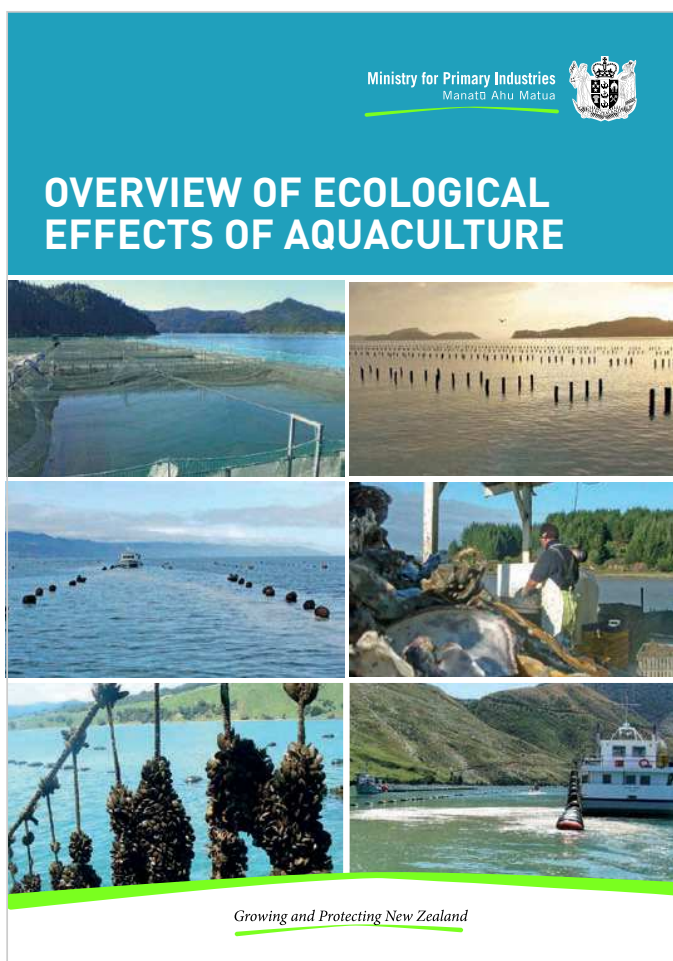
Potential environmental effects – general

There are a number of potential environmental effects that can result from aquaculture operations, both positive and negative, which are discussed below. Management options for mitigating the negative environmental impacts are described at the end of the chapter. The aquaculture industry has developed Environmental Codes of Practice specific to each species, which are designed to minimise potential effects on the environment. These are discussed further in the section on best practice.³⁰

In 2012 a risk assessment workshop run by the Ministry for Primary Industries produced a ranking of relative ecological risks from aquaculture activities. This involved an expert panel approach using an independent facilitator and internationally-recognised methods, with input from scientists, regional council and industry. The produced ranking of the potential risks included:

1. Biosecurity threats
2. Pelagic (water column) effects
3. Marine mammal interactions
4. Benthic (seabed) effects
5. Seabird interactions
6. Additive effects
7. Escapee effects
8. Wildfish interactions
9. Hydrodynamic alteration of flows

An 'Aquaculture Ecological Guidance Package'³¹ has been developed by the Ministry for Primary Industries in association with the Cawthron Institute, NIWA, Department of Conservation, regional councils, the aquaculture industry and others. This package provides 'information and advice on the ecological effects of marine-based aquaculture to assist in planning and managing aquaculture development'. It contains science-based information on the ecological effects of marine-based aquaculture to assist local authorities, the aquaculture industry and other stakeholders with their planning for and management of aquaculture activities.



This report, published by the Ministry for Primary Industries, provides useful information about the potential ecological effects of aquaculture

Marine mammals and seabirds

The effects of marine farming on marine mammals and seabirds are not fully understood. There is concern that marine farms could adversely affect such species if they reduce or block access to food sources, limit the space available for hunting, or interfere with normal behaviour such as migratory routes. In addition, marine mammals can become entangled in marine farming gear. In addition, underwater noise generated by activities can attract or exclude marine mammals, with whales and some species of dolphins being more sensitive to this kind of disturbance.³⁴ Currently, the limited extent of marine farming in New Zealand means that only a few such effects have been observed here, but this may become an issue as the industry grows.

In other countries, where marine farming is much more widespread, more serious problems have been experienced.³⁵ High stocking densities for finfish, in particular, has been identified as a potential risk to marine mammals.

In New Zealand, there have been no reported cases of dolphins or seals becoming entangled in mussel farm lines. Between 1996 and 2003, one Bryde's whale was reported as a casualty of entanglement in mussel spat catching lines.³⁶ Entanglement for whales may become more of a problem if large offshore marine farms are established in areas which overlap with the migratory paths of humpback and southern right whales.³⁷

New Zealand fur seals are attracted to salmon farms. Most farms use predator exclusion nets to reduce the risk of entrapment. In approximately 25 years of sea pen salmon farming in New Zealand, there have been five reported cases of marine mammals becoming trapped in farm nets, two cases involving seals and the other three involving dusky and Hector's dolphins.³⁸ Improvements in net design, and refined net changing and other practices, can reduce the risk of entanglement.

In the case of mussel farms, small cetaceans (whales and dolphins) can enter farm areas and swim between lines, and have often been observed to do so.³⁹ However problems can occur when marine farms are located in areas which are important habitats for specific marine mammal species. For example, research has indicated that mussel farms located in Admiralty Bay in the Marlborough Sounds likely interfere with the co-operative hunting of dusky dolphins, with the dolphins observed to avoid the mussel farms. This is probably because the floats and lines physically obstruct the ability of the dolphins to jointly herd fish. In addition, the structures may interfere with the dolphins' echolocation capabilities and thus their ability to 'see' and navigate within their surroundings. These concerns resulted in the Environment Court turning down applications to expand marine farming in the bay.⁴⁰

The main negative impacts for seabirds from marine farms are entanglement, habitat exclusion, displacement from feeding areas, increased potential for pathogens and changes to the food web. The way that marine farms attract small fish means that the structures may enhance feeding opportunities and enable more efficient foraging.⁴¹

Compared to mussel farms, habitat exclusion resulting from fish farms is more absolute and care is needed to ensure that they are not located in areas which are important habitats for marine mammals and important roosting areas for seabirds.



Mussel farms may disrupt the co-operative feeding behaviour of dusky dolphins, shown here off Kaikōura (Raewyn Peart)

Potential environmental effects – shellfish

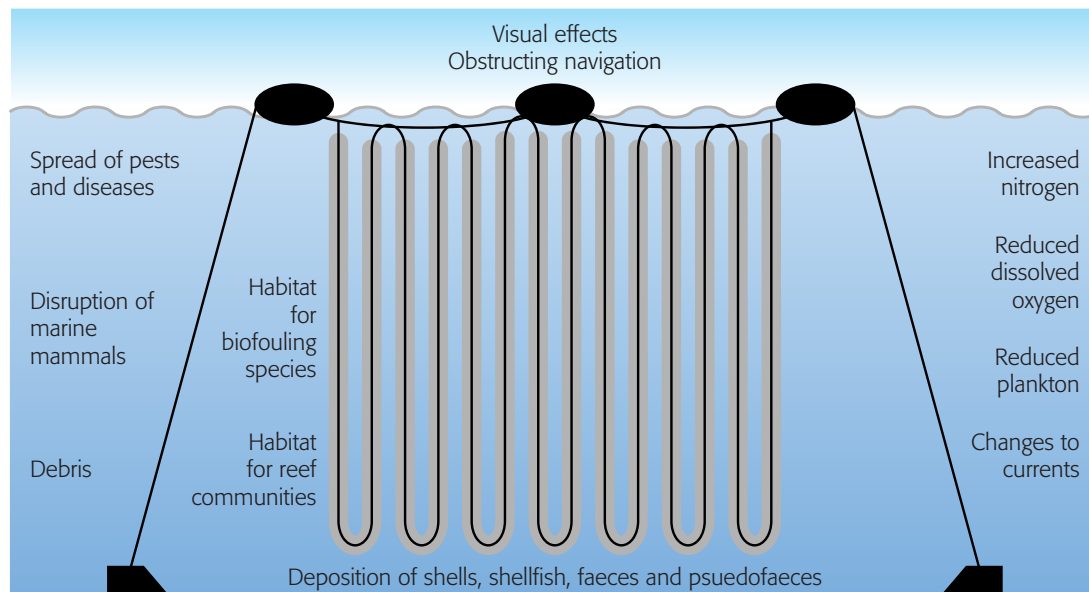


Figure 10.4 Summary of potential environmental effects of mussel farming

Seabed habitats

The principal ecological effect of mussel and oyster farming is localised changes to the seabed caused by shells, shellfish, faeces and pseudofaeces (particles which are not digested bound in mucous) dropping from the mussel lines and oyster racks. These items can accumulate on the seabed and alter its physical, chemical and biological nature. The waste products are nutrient-rich and thus increase microbial activity, which can lead to oxygen depletion within the sediment, affecting species that live there and reducing biodiversity.

Changes to the seabed are particularly pronounced where the seabed is sand or pebble, rather than mud. In these areas, benthic species are less able to cope with smothering by fine material from the farm. The extent of these effects will largely depend on the type of substrate under the farm, the flushing capacity of the site, and the depth between farm structure and substrate. Shellfish farms are less likely to be situated over or near to rocky reefs where such effects could be even more pronounced.

The creation of a nutrient-rich environment may give rise to future opportunities to farm other species (polyculture) underneath such farms. Species such as sea cucumbers have been the subject of trials. A community-based aquaculture research project between the Hongoeka Development Trust and NIWA focused on using the waste products from one species as a food input for another, thereby reducing waste and maximising energy flows throughout an integrated system. In this particular project, the polyculture was investigated to see whether the waste produced by a pāua farm could be recycled to produce other high-value foodstuffs relevant to Māori.⁴⁶

Water column

Farmed shellfish extract phytoplankton and organic particulates from the water column, and can thereby cause their depletion, particularly in sheltered bays. This depletion tends to be localised and its significance will depend on the density of the farms and the level of flushing in the area. Phytoplankton are the building blocks of the marine food web, and the impact of their depletion on the broader marine ecosystem is poorly understood.⁴⁷ Farmed shellfish release ammonia and organic particles into the water column which promote some additional phytoplankton and macro-algal growth. In addition, the shellfish eat fish eggs so could potentially have an impact on fish stocks if very large areas of shellfish farms are established.

The structures required for mussel farming create obstructions in the water column that can alter the direction and speed of currents in the area, which can in turn alter the rate at which plankton and nutrients are replenished and waste dispersed. The extent of these effects will depend on the particular characteristics of the area, the scale and density of the farm, and the orientation

Potential environmental effects – finfish

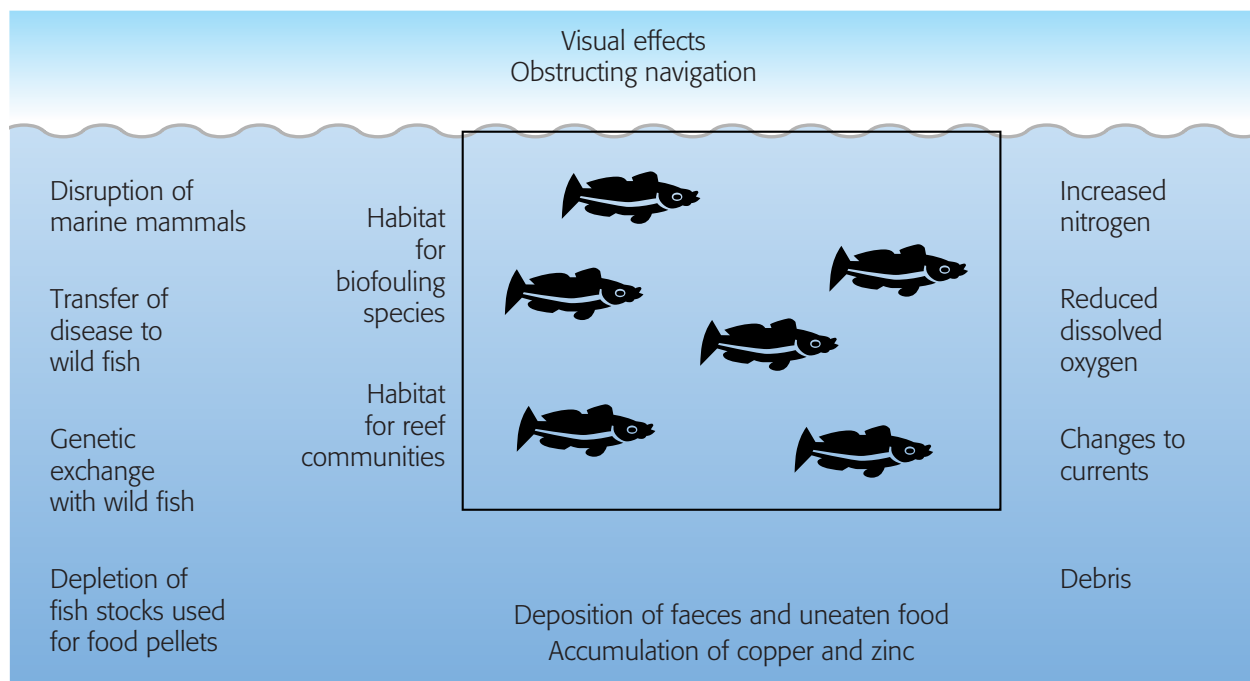


Figure 10.5 Summary of potential environmental effects of finfish farming

Seabed habitats

Deposition of faeces and uneaten feed causes changes to the seabed beneath finfish farms. These particles are organic-rich and can alter the physical, chemical and biological nature of the seabed, increasing microbial activity. This in turn can decrease oxygen levels in the sediment and lead to changes to the biodiversity of the area.

These effects tend to be greater in finfish farming than in shellfish farming. However, they can be minimised by siting farms in deep, well-flushing locations and by good farm management such as using higher-quality feed to gain an improvement in feed conversion rates (meaning that less feed is needed to grow a fish), and using automated feeders with shut-off signals linked to underwater cameras that detect waste feed.⁵⁴ Farmed areas can also be rotated so that areas of the seabed are regularly fallowed to allow recovery. Such approaches are also important for reducing the effects that uneaten food and fish waste have on the water column.

Water column

Finfish farming results in the release of nutrients into the water column from uneaten feed and fish wastes. These can stimulate phytoplankton growth and at excess levels may result in algal blooms and oxygen depletion. Nutrient enrichment can also cause changes in the mix of phytoplankton species with potential impacts on food webs. Such impacts can be reduced through siting farms in deep and well-flushed waters.⁵⁵

Local biodiversity

Finfish farms can cause changes to local biodiversity, in particular a decrease in those benthic species that are intolerant of the enriched conditions and a corresponding increase in species that are adapted to them. Wild fish and other species can be attracted to finfish farms because they feed on the dropped fish feed and use the structures for shelter.

Toxic chemicals

Therapeutants (vaccines, antifouling paints and feed) are all potential sources of chemical pollutants to the marine environment. Farmed finfish are very susceptible to diseases and large numbers of chemical treatments can be needed to combat these.

Method	Details
Addressing the effects of occupation of the coastal marine area	• A regional coastal plan may include provisions to address the effects of occupation of the coastal marine area and to manage competition for the occupation of space (including for aquaculture activities) • Plans may include rules which: - specify that no application can be made before a certain date - state that the consent authority may process and hear applications relating to the same space or spaces in close proximity together - place limits on the character, intensity, scale or size of activities
Alternative allocation mechanisms	• The default allocation mechanism under the RMA is 'first in, first served' • This means that a complete resource consent application lodged with the consent authority must be determined before any resource consent applications lodged later in time for the same or overlapping space • Alternative allocation mechanisms may be provided for through a regional coastal plan or through <i>Gazette</i> notice
Regional coastal plan	• Alternative allocation mechanisms (public tender of authorisations or any other method of allocating authorisations) can be set out in a regional coastal plan • Before setting out an alternative allocation mechanism, the council must be satisfied that such a rule is necessary or desirable in the circumstances of the region and that the proposed method is the most appropriate for allocation of space in the circumstances of the region, having regard to its efficiency and effectiveness, compared to other methods of allocating space
<i>Gazette</i> notice	• As a result of the 2011 reforms, regional councils may also ask the Minister of Conservation to approve an alternative allocation method for activities in the coastal marine area through <i>Gazette</i> notice • The regional council will need to provide evidence demonstrating high or competing demands for coastal permits • While a <i>Gazette</i> notice is in place, no person can apply for a coastal permit authorising occupation of space unless the person is the holder of an authorisation that relates to that space and activity
Suspension of applications	• Regional councils may ask the Minister of Aquaculture for a stay on new applications for consents to occupy space for specified aquaculture activities • The regional council will need to demonstrate that there is actual or anticipated high or competing demand • The suspension must be for no more than 12 months
Refusal of similar applications	• A consent authority can refuse to receive an application for a coastal permit to occupy space in the coastal marine area if, within one year before the application is made, the consent authority has refused to grant an application for a permit for a similar or same activity in respect of the same space or space in close proximity

Figure 10.6 Tools available to councils to allocate marine space

Undue Adverse Effects test

The effects of proposed marine farm areas on fishing are assessed through the Undue Adverse Effects test which is undertaken by the Ministry for Primary Industries.⁶² A proposed marine farm cannot proceed if it would have undue adverse effects on recreational or customary fishing, or commercial fishing for non-QMS stocks. In addition, unless an aquaculture agreement or compensation declaration is registered, a proposed marine farm cannot proceed if it would have undue adverse effects on commercial fishing for QMS stocks.

There are a number of matters that the Ministry for Primary Industries considers in applying the test, being:⁶³

- The location of the proposed marine farm in relation to fishing areas

Case law on marine farm conversion in an outstanding natural landscape

The Environment Court recently considered whether a resource consent application to convert a mussel farm (located in Pelorus Sound) to a salmon farm should be granted.⁶⁸ The site was identified as an 'Area of Outstanding Landscape Value' in the Marlborough Sounds Resource Management Plan and therefore section 6(b) RMA and Policy 15 NZCPS were applicable. Section 6(e) regarding Māori interests was also applicable. The application was turned down on the basis that section 6(b) and (e) (as particularised by the NZCPS) were 'sticking points' (at [215]) and '[p]lacing all relevant considerations in the balance ... we consider the purpose of the RMA, and the instruments made under it, is better achieved if consent is refused (at [217])'.

Protecting indigenous vegetation and fauna

Under Policy 11, a marine farm applicant will need to show that there will be no adverse effects on indigenous vegetation and fauna which are threatened, at risk or naturally rare. In relation to other indigenous species and ecosystems, an applicant must show that significant adverse effects have been avoided and other adverse effects have been avoided, remedied or mitigated. These matters are discussed further in Chapter 2: Marine management.

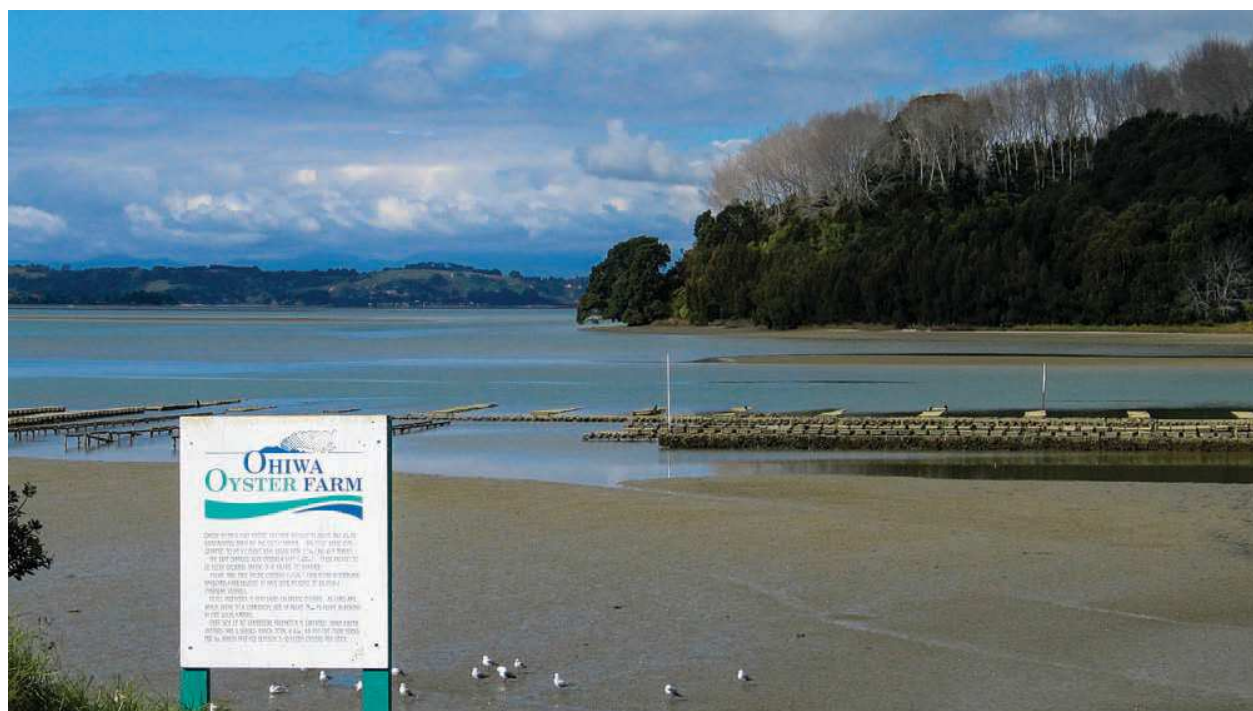
Harmful aquatic organisms

Policy 12 addresses the management of harmful aquatic organisms and requires councils to provide in regional policy statements and plans, as far as practicable, for the control of activities that could cause harmful aquatic organisms to be released or otherwise spread including 'the establishment and relocation of equipment and stock required for or associated with aquaculture'. This is discussed further in Chapter 5: Marine biosecurity.

Protecting water quality

Policy 23 of the NZCPS addresses the discharge of contaminants into the marine environment, which includes those by aquaculture. It directs particular regard to be had to a number of matters including the sensitivity of the receiving environment, the nature of the contaminants to be discharged and the capacity of the receiving environment to assimilate the contaminants.

In addition, Policies 8(c) and 21 of the NZCPS require councils to consider the effects of other resource use on the water quality required for aquaculture, and to work to improve coastal water quality where aquaculture growing areas are significantly adversely affected.⁶⁹



The NZCPS requires councils to consider the effects of other activities on water quality required for aquaculture, such as for the oyster farm in Ōhiwa Harbour shown here (Raewyn Peart)

Effects	Best practice response
Debris from poorly maintained farms can be a hazard to other water users and to marine fauna	All non-biodegradable waste material should be removed from the coastal marine area, and structures should be adequately maintained to ensure their integrity. Lines should be kept under tension (this reduces likelihood of nuisance for other users and reduces likelihood of marine mammal entanglement).
Degradation of landscape values and natural character of the area	Marine farms should be sited away from areas with outstanding natural landscape or character values Marine farms should be sited away from areas where the visual effects of the farm could be significant on landscape or natural character values
Marine farms can pose a hazard to watercraft and navigation	Marine farms should be adequately lit at night to ensure safety of navigation Marine farms should not be located in areas which obstruct navigation or safe anchorage, such as shipping routes and popular recreational vessel routes and anchorages
Shellfish	
Effects on water quality through excretion of faeces and alteration of naturally occurring currents	Marine farms should be designed to minimise effects on currents, should be sited in areas with adequate flushing, and should not be stocked at densities that result in significant build-up of faecal material in the water column
Depletion of naturally occurring plankton leaving less available for wild species	The effects of shellfish farming on the amount of plankton available should be carefully managed, including the cumulative effects of having more than one shellfish farm in the same area
Effects on light-sensitive marine flora as a result of shading by the farm structure	Marine farms should be located away from areas where they can have negative impacts on seagrass and other light-sensitive species such as seaweeds
Changes to the seabed through deposition of shell, shellfish and faeces	Marine farms should be sited in areas where benthic habitats are less sensitive to impacts such as areas with a muddy substrate rather than rocky or sandy areas
Effects on wild species as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or of ecological importance
Changes to patterns in the current and natural flow of water as a result of farms, particularly oyster racks, causing sand build-up under the farm	Proposals for marine farms should be accompanied by an assessment of the hydrodynamic patterns at the subject site, with current speeds and directions, flushing times and water-dispersion patterns and modelling of how this will be affected by the proposed farm Marine farms should not be sited in poorly flushing areas
Abandonment of marine farm sites and structures when farming ceases	If marine farming activity ceases, all structures associated with its use should be removed and the site rehabilitated. The consent authority should require this as a condition of consent or require a bond or payment to cover the cost of removal by the local authority.
Finfish	
Increased likelihood of disease from high-density stocking rates for fish farming	Good husbandry, stress reduction and appropriate nutrition should be used to help prevent disease outbreaks Fish should not be farmed at high intensity to minimise the likelihood of disease There should be single age-class fish farms Farm spacing should be at least 1 kilometre within the same management plan and at least 2.5 kilometres if not within the same management plan Equipment should be regularly checked
Depletion of fish stocks when farmed fish are fed wild-captured fish or feed pellets made from wild-captured fish	The proportion of wild-captured fish used to provide protein and oils in fish feed should be minimised Feed which maximises the conversion rate between the amount of wild fish consumed and the amount of farm fish produced should be used Only wild fish from sustainable capture fisheries should be used

Guidance for aquaculture monitoring in the Waikato region

The Waikato Regional Council, with funding from the Ministry for Primary Industries, is developing a regional environmental monitoring framework to integrate consent-related environmental monitoring with state of the environment monitoring. The aim of this initiative is to rationalise and improve all environmental monitoring conducted in the Waikato coastal marine area. Aquaculture has been chosen as a first case study, in anticipation of the development of fish farming and increase in shellfish farming in the region.⁷³

The purpose of the framework is to achieve a more consistent and transparent approach to environmental monitoring that better integrates consent-related environmental and state of the environment monitoring, and accounts for cumulative effects. A regional monitoring framework has the potential to facilitate these types of improvements, and to provide a broader context for understanding the effects of existing and future anthropogenic activities against the background of natural variability.

The framework is based on the understanding that the environmental effects of consented activities are best managed if environmental monitoring is aligned with broader state of the environment monitoring. In addition, it is accepted that environmental quality standards should be established prior to the commencement of the consented activity, so that consent monitoring can be designed to verify compliance with these standards.⁷⁴

The first phase of this project focused on reviewing current monitoring approaches, reviewing relevant consents, assessing locally relevant ecological effects and their significance, and developing the structure for the monitoring framework. The second phase focused on developing the detailed framework, identifying priority issues that need to be addressed either through industry best practice and reporting or through monitoring, and identifying environmental indicators and indicative standards. All information is currently being collated in a series of three technical reports. These will be distributed to the aquaculture industry for feedback prior to being finalised.

Ultimately this project will produce a guidance document for the aquaculture industry and other stakeholders which provides clear information on environmental monitoring requirements for aquaculture consents in the Waikato region. As a result, the outcome of this project should be practicable and directly applicable to the resource consenting process.

Within an integrated monitoring framework, state of the environment monitoring and consent-related environmental monitoring should be more effective in improving understanding of the overall state of the environment (including cumulative effects). This project supports the concept that having well-designed environmental quality standards and monitoring approaches at the time of consent will enable meaningful adaptive management strategies to be designed for marine farms.⁷⁵



This mussel farm in the Coromandel Harbour will be part of the Waikato Regional Council's aquaculture monitoring programme (Raewyn Peart)

Using DNA technology to monitor environmental change around marine farms

Monitoring the environment in the vicinity of marine farms is a requirement of resource consents, but the challenge has always been having the technology to do this in an effective way that supports adaptive management. Traditional methods, such as chemical analysis of ecological changes, can be time-consuming, highly skilled and expensive, and the results can often take months to be delivered. This means that often the marine farm is continuing to operate in a void of information.

New technology developed by the Cawthron Institute called 'Next-Generation Sequencing' will enable the aquaculture industry to monitor the environment around marine farms in near real time. This means that there will be almost instantaneous knowledge about any changes in the marine ecosystems. From this information marine farm managers will be able to quickly adapt their management practices.

This project has been in the 'proof of concept' stage for the last two years. The study has been funded by the Ministry of Business, Innovation and Employment, with in-kind support from New Zealand King Salmon Company Limited through trialling around some salmon farms in the Marlborough Sounds and at Stewart Island. These trials have been completed, alongside the traditional monitoring methods to enable comparison and to assess the effectiveness of the tool. Scientists have now proven that with the new technology, using a liquid workstation 'robot' they can potentially analyse hundreds of species at the same time using a group of microscopic single-celled organisms known as foraminifera, which are highly responsive to environmental change.

With such fast results being possible, the aquaculture industry will be in a much better position to make rapid changes to management systems, ultimately minimising environmental impacts. 'Marine industries are very excited about this – they need a tool that allows them to provide an almost real-time result, reducing the cost and demonstrating environmental sustainability', said Cawthron Institute scientist Dr Xavier Pochon.

Phase two of this research will see monitoring results being compared annually over a three-year period. It will also explore the potential to look at other types of aquaculture such as mussel farming, as well as other marine industries like the oil and gas sector.



New monitoring technology is being trialled around some salmon farms in the Marlborough Sounds to provide near real-time information (Raewyn Peart)

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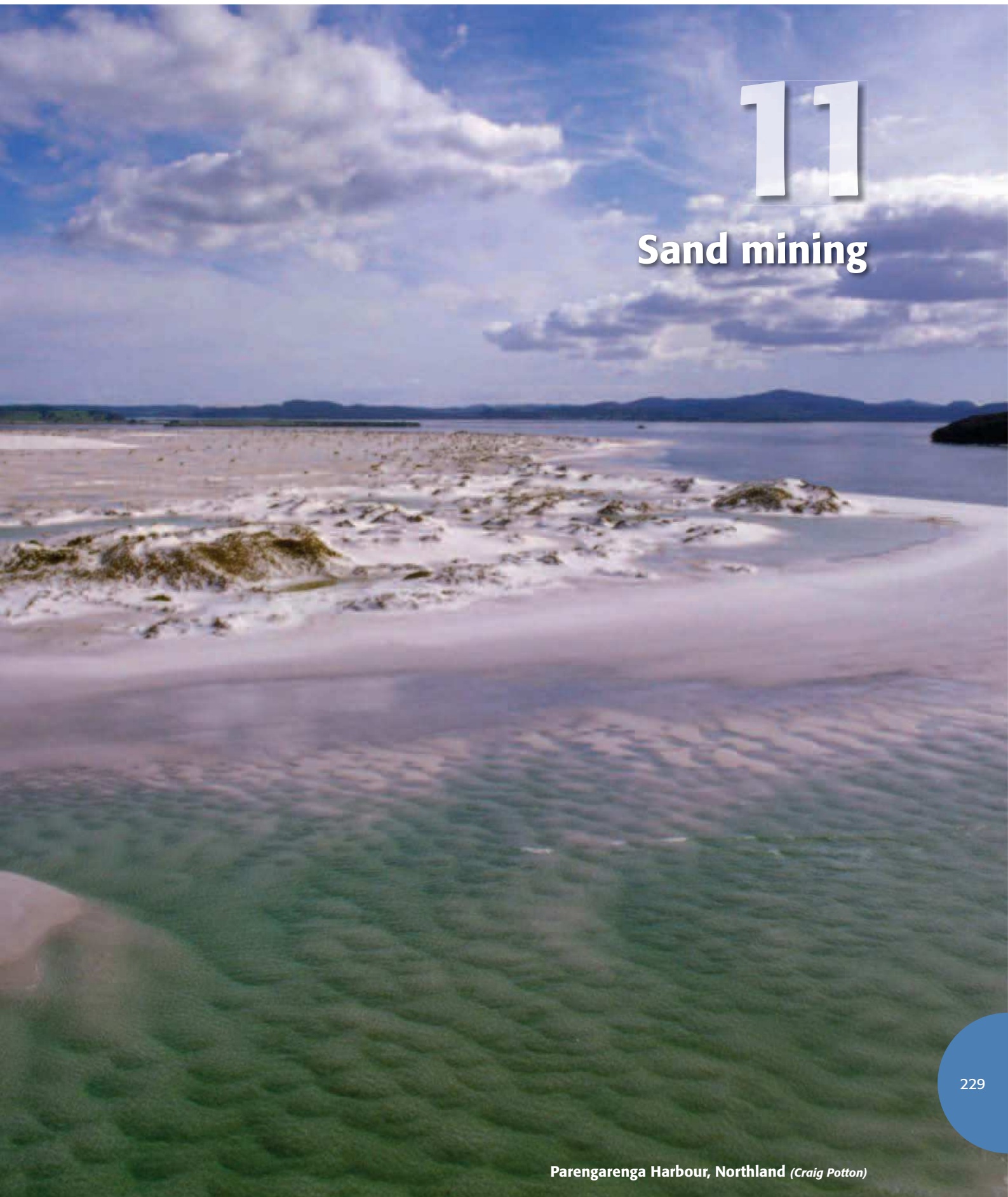
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11

Sand mining



Introduction

Mining of sand for commercial purposes has been occurring in New Zealand at a number of locations for many years. Extraction generally targets grain sizes of between 0.063 and 2 millimetres. Sand is commonly used as a component of aggregate in the production of concrete and as an ingredient for other cement-based products. The titanomagnetite contained in the black iron sand off the west coast is used for the manufacture of iron and steel. Sand is also used to restore popular beaches, such as Auckland's St Heliers and Mission Bay. More recently there has been a focus on the opportunities to mine iron sand from offshore areas.

This chapter discusses mining of sand from both inshore and offshore areas. Other forms of marine mining are discussed in Chapter 12: Deep-sea mineral exploration and mining, and Chapter 13: Oil and gas exploration and mining.

The two areas which are distinct in terms of the regulatory framework which governs sand mining are:

- Inshore sand mining – within the coastal marine area out to the 12 nautical mile limit (including onshore beach and dune deposits)
- Offshore sand mining – within the EEZ from the 12 to the 200 nautical mile limit

Inshore sand mining

New Zealand has extensive sand deposits spread over the continental shelf, particularly close to major rivers. In many areas, the sand is derived from land-based erosion, with the sediment being carried by rivers out to the sea. But some sand comes from other sources, such as shell-based sands made up of the crushed remnants of molluscs and bryozoan sands which are found in Northland. In the early days of European settlement, many local beaches were mined for sand until concerns about erosion stopped the practice. For example, Limeburners Bay in the upper Waitematā was where shells mined from beaches around the Hauraki Gulf were processed to make lime and cement. In addition, sand from Whiritoa was mined for over 50 years, with a total of more than 180,000 cubic metres of sand removed.¹



Mined sand has been used to replenish beaches such as at Point Chevalier, Auckland, shown here, which have been mined in the past (Raewyn Peart)

Resources Limited prospecting permit sets a limit of 600 cores in total over an area of 6,319 square kilometres.⁸ The intensity of sampling will typically increase when the exploration phase starts.

Method	Summary
Boat-towed magnometer systems	• Useful for mapping placer deposits or diffuse deposits where the magnetic minerals are spread over a wide area • Recent research has demonstrated that titanomagnetite can be mapped well using a seabed sledge developed by the University of Bremen that measures magnetic fields over a sediment thickness of a metre or so. This work is still in progress⁹
Vibracore sampling	• Undertaken in water depths of up to 200 metres and to a target depth of 10 metres below the seafloor • A single 2 to 10 metre long core tube (deployed and operated by a cable) is driven into the subsoil • A vibration allows the tube to penetrate the surface • When the core is removed, the hole created collapses in on itself, leaving a slight depression in the seabed • Sampling sites are generally discrete and around 200 metres from each other • A vibraborer does not work well if the material is too coarse, well-consolidated or cohesive
Reverse circulation sampling	• Used for unconsolidated sediment samples • Tube is generally around 70 to 80 millimetres in diameter • A series of water jets drive a hole into the substrate • Often uses a drilling 'mud' which is mixed with water and pumped down into the hole • Samples withdrawn using an air-lift system • Hole created collapses in on itself, leaving a slight indentation in the seabed around 250 millimetres deep and 400 millimetres in diameter • Used for sampling along the Taranaki coast where six discrete samples per hole are taken at a depth of approximately 6 metres
Sonic drilling	• High-frequency vibrations sent down a drill 'string' (or sampling rod) to the drill bit • Vibrations resonate and magnify the amplitude of the drill bit, which then fluidises the seafloor • Has the ability to recover samples to 100 metres deep • Typically uses a 100 millimetre drill bit within a 150 millimetre external diameter sleeve • No materials discharged at sea • Sampling generally occurs at discrete distances between 100 to 200 metres • Drilling 'muds' are sometimes used
Side-scan sonar and multi-beam sonar	• Maps topography and sedimentary environment of seafloor • Provides information about the surface sediments in terms of texture and reflectivity rather than the minerals that are present • Side-scan sonar uses a towed sonar device to emit a conical or fan-shaped pulse of sound through the water across a wide angle • Device can be towed or mounted to the hull of the vessel • Water-depth data is collected using multi-beam sonar • The frequency of the sound is generally between 100 and 500 kilohertz, but closer to 100 kilohertz
Low-frequency echo sounder	• Used to map sand in harbours and on the continental shelf • Systems such as a CHIRP Subbottom Profiler can be used • These generally operate at between 3.5 and 15 kilohertz, which gives a maximum range of between 20 and 50 metres depending on the type of sediment • This corresponds to a peak sound output of 149 to 158 decibels (compared to 176 decibels for a standard 200 kilohertz echosounder)

Figure 11.2 Main methods for sand prospecting and exploration (*Adapted from Thompson K, 2012, and De Lange W, 2014*)

to the Glenbrook Mill through an 18 kilometre long underground pipe. Freshwater used during this process is drawn from a lagoon adjacent to the Waikato River. The by-products from these processes are used for site rehabilitation after they have been dewatered.¹³

At the Taharoa mining site, sand is extracted from a pond by a floating dredge and then, because the area has no natural harbour, pumped as a slurry to a bulk cargo ship moored 3 kilometres from the shore plant to be processed and then exported.¹⁴ Between 200 and 300 tonnes of magnetic concentrate is produced per hour.¹⁵ The removal of sand from the Mangawhai-Pākirī embayment is generally completed using a suction dredge. This is typically a rotary cutter suction dredge where a rotating head disturbs the sediment, allowing it to then be entrained into the inflow of the suction pipe. Sand is then screened using mesh sizes of 3 millimetres, which removes gravel, shell and debris.

Potential environmental effects

There are a number of potential environmental impacts resulting from both inshore and offshore sand mining which are summarised in Figure 11.4 below and discussed in further detail following. Whether such effects will occur in practice, and their significance, is dependent on a number of factors including the equipment and methods used, the dredging frequency, the volume being extracted, and the physical characteristics of the area.¹⁶ Over the past two decades a number of Environment Court cases on seabed mining,¹⁷ and the recent Trans-Tasman Resource Marine Consent Decision (June 2014), have provided some important guidance on the significance of environmental impacts. Ways of reducing potential environmental effects are discussed below in the section on best practice.

Area	Potential impact
Destruction of benthic communities	• Mortality of organisms living on the seafloor, and within the seabed sediment, in the area mined • Smothering of benthic organisms inside and outside the mining area as a result of sediment plumes • Sediment in water column reducing light penetration, potentially affecting food webs
Disruption of sedimentary and coastal processes	• Physical changes to the habitat within the mined area • Changes to the seabed bathymetry affecting wave patterns, surf breaks, coastal stability and the sediment budget – dependent on vicinity of the mine site to shore, existing processes and the level of mining proposed
Disturbance of marine mammals, fish and birds	• Disruption to marine mammals that are dependent on sound for locating prey, navigating and communicating, from the acoustic impact of seismic surveys and use of mining equipment • Entanglement, injury or drowning in equipment • Birdstrike from the introduction of man-made light above the surface • Subsurface activity and lighting potentially interfering with the feeding and reproductive behaviour of marine mammals and seabirds¹⁸ • Increased risk of ship strike
Changes to natural character, landscape and amenity	• Reduction in the ‘naturalness’ of landscapes and the coastal environment due to the introduction of man-made structures, vessels and equipment • Reduction in recreational amenity
Other	• Increased biosecurity risks through overseas equipment acting as a vector for marine pest species • Reduction in fish stocks and spatial displacement of commercial fisheries • Disruption of marine traffic routes through exclusion zones around the mining area • Degradation of areas and resources of importance to iwi and hapū • Destruction or degradation of historic heritage (such as shipwrecks and other marine archaeology)

Figure 11.4 Potential environmental impacts from sand mining (Adapted from Secretariat of the Pacific Community, 2013, various Environment Court cases¹⁹ and Trans-Tasman Resource Marine Consent Decision, June 2014)

Sedimentary and coastal processes

Dredging may affect the physical environment by altering the bathymetry, which in turn can change the velocities of currents and sedimentary processes.²⁶ In what are known as 'closed' systems, there may be little or no new sand entering the sediment cycle, depending on the size of the system. If sand is removed from these systems, there can be less available to be deposited on the beach, and erosion of the coastline may result as seabed reserves are depleted.



If sand is removed from closed systems, coastal erosion can result, as occurred at Ōmaha where expensive groynes were required to remedy the problem (*Raewyn Peart*)

Case law on sustainability of sand mining

In 2009 the High Court upheld a decision by the Environment Court to reverse the former Auckland Regional Council's decision to decline consent to McCallum Bros Limited. The consent was to renew dredging consents in the Mangawhai-Pākirī embayment at a rate of 76,000m³ per annum, an increase of 6,000 cubic metres per year above the previous 20 years. The Environment Court had granted consent for 14 years on the basis that the total input of sediment into the system meant that the proposed extraction was not unsustainable. It also concluded that shoreline retreat and erosion could not be attributed to sand extraction over the last 80 years.²⁷ This indicates that sand mining activities should be directed to areas where the amount to be extracted is less than new inputs into the system.

In 'open' systems new sand is constantly entering the cycle. This can include sand that is being transported along the coast from other areas by marine currents, sand being carried down rivers into the sea, and sand being produced by erosion of nearby coastal cliffs and headlands. In some areas, such as along the west coast of New Zealand, sand may also be transported into coastal areas from further out on the continental shelf. If the amount of sand removed from open systems by mining is less than the quantity of new sand entering the system from outside sources, coastal erosion is much less likely to occur. Ultimately, to avoid erosion it is desirable to maintain a positive sediment budget.

Mining activity	Example of potential impact	Potential level of impact
Suction sampling	• Avoidance • Habitat degradation	Proximity dependent
Sonar mapping	• Habitat degradation • Behavioural changes • Avoidance of the area	Low ³²
Shallow seismic surveying (low-intensity subbottom profiling)	• Behavioural changes • Displacement from habitat • Changes in vocalisation	Low
Vessel activity	• Avoidance • General habitat degradation • Behavioural changes	Potentially high
Active mineral extraction	• Habitat degradation • Avoidance • Behavioural responses • Potential for indirect effect due to the negative effects on prey	Potentially high

Figure 11.5 Examples of the potential impacts of sand mining off the North Island west coast on marine mammals
(Adapted from Table 5, Thompson K, 2012)

Noise is a concern with marine mining activities, particularly in relation to its potential effects on marine mammals. Seismic surveys are known to impact on marine mammals. In addition, any drilling activities, suction and pumping, and routine maintenance will produce underwater noise. Helicopter overflight also has a potential to disturb marine mammals. Noise from vessel traffic, particularly that associated with active mining, is highly likely to significantly raise the local ambient noise levels.³³ The consequences on marine mammals and other fauna will vary from none to behavioural (may leave the area) to acute injury (ear drum damage) to serious (death) depending on noise level encountered, species and habitat. There may also be cumulative effects from repeated exposure.³⁴

Survey and support vessels create a risk of ship strike. The risk will depend on the speeds of the vessel involved, with greater speeds creating greater risk. During surveying, the vessels tend to move slowly (typically operating at around 5 knots) and special flags are flown to warn other vessels of the slow movement.³⁵

Dolphin conditions on sand mining consent

In the final decision of the Kaipara Harbour sand mining application in 2006 a new condition was inserted relating to increased vessel traffic in the area: 'Vessels used by the Consent Holder will, to the extent practicable, travel at a speed that results in no, or minimal, wake when within 300 metres of dolphins'.³⁶ In addition an Advice Note was inserted: 'Sightings of Maui's dolphins will be reported by the Consent Holder to the WWF Sightings Network. Where practicable detail on the sighting including GPS coordinates, number of dolphins, and date and time of sighting, will be provided'.³⁷ This new condition recognises the impact sand mining activities can have on marine mammals, particularly rare species.

Large-scale increases in water turbidity have been shown to impact some marine fish species inhabiting these areas, by lowering feeding rates.³⁸

Lights at night on vessels can attract and disturb seabirds and may cause them to collide with the above-sea-surface structure. Subsurface activity and lights may affect marine mammals by interfering with their navigation.³⁹

Natural landscapes, character and amenity values

Seabed sand mining can potentially impact on natural landscapes, character and amenity, particularly when it is close to shore or in the foreshore area. This includes unwelcome visual impacts from barges and tugs operating in the marine area, particularly in those

Decision-making framework

Allocation of mining rights

Any party wishing to undertake prospecting, exploration or mining activities must obtain a permit issued under the Crown Minerals Act. Applications are processed by New Zealand Petroleum and Minerals which is a branch of the Ministry of Business, Innovation and Employment. Under the Act, the Crown creates minerals programmes.⁴⁴ The programme that applies to the exploitation of sand resources is the 'Minerals Programme for Minerals (excluding Petroleum) 2013'. This sets out the policies and procedures to be followed for the allocation of mineral resources (including consultation with iwi and hapū). The requirements to be met by permit holders are defined in the regulations. It does not address environmental impacts, which are addressed under other legislation.

The current minerals programme provides that mining for sand in the coastal marine area does not require a permit under the Crown Minerals Act, but mining for minerals that are a component of sand, such as titanomagnetite, does require a permit. Hence iron sand mining is governed by a different regime to the mining of sand for aggregate.

Management under the RMA

The environmental effects of sand mining within the coastal marine area are managed under the RMA. The regional coastal plan will determine if a resource consent is required, and if so, what kind of consent. In practice, a proponent will normally need to obtain a resource consent to undertake sand mining. When an application is lodged, the consent authority will determine whether or not it should be publicly notified. A consent application must be publicly notified if the consent authority determines that its adverse effects on the environment are likely to be more than minor.⁴⁵

When considering an application for consent, the consent authority must also have regard to the relevant provisions of the applicable statutory instruments made under the Act, for example the NZCPS (shown in Figure 11.6), regional policy statements and the regional coastal plan.



The impacts of sand mining on seabirds, such as those shown here at Te Arai, Auckland, need to be considered as part of sand mining consent applications (Raewyn Peart)

Elements of best practice

As described above, sand mining can potentially have a number of negative impacts on the environment; as a result, it is important that it is only undertaken in appropriate areas. Sand mining activities should avoid, where possible:

- Closed sand systems – depending on how big the closed system is and how much sand is being extracted over what time period
- Sediment with a high silt content
- Benthic habitats of high ecological value
- Significant shellfish beds
- Fish spawning areas and areas of high fish density (can be time dependent)
- Areas frequented by threatened or endangered marine mammals including whales, dolphins and sea lions
- Important seabird areas
- Areas of high cultural or archaeological significance
- Areas of recreational significance including areas near high-quality surf breaks



Sand mining should not be undertaken in areas frequented by threatened or endangered marine mammals such as the New Zealand sea lions shown here at Sandfly Bay, Otago Peninsula (Raewyn Peart)

When sand mining is authorised, strict conditions are generally included as part of the consent, including a requirement for a robust monitoring regime to identify any negative environmental effects. Some of the Environment Court cases on sand mining found that the precautionary approach was appropriate, considering that the extractions were to occur within dynamic systems subject to constant change. However, these cases generally took the view that monitoring and other conditions could encompass this precaution.⁵⁷

With respect to offshore sand mining applications, there is limited knowledge of the EEZ, as only a small proportion of this expansive area has been scientifically investigated. Therefore, applicants will need to undertake significant scientific investigations before lodging a marine consent application. The applicant must have detailed information about the receiving environment, the proposed operation, and existing interests. But even given such baseline information, there are still likely to be considerable uncertainties about impacts. This means that the caution requirement contained in the EEZ Act will be a key consideration for decision-makers.

Decision on sand mining application within the EEZ

The Trans-Tasman Resources Decision Making Committee refused the marine consent application and stated, 'Overall, we think this application was premature. More time to have better understood the proposed operation and the receiving environment and engage more constructively with existing interests and other parties may have overcome many of the concerns we have set out in this decision ... Ultimately, the information we have to make our decision, while voluminous, was too uncertain and inadequate, and we did not have sufficient confidence in the adaptive management approach proposed to address that uncertainty to enable the activity to be undertaken. For all of these reasons, the application as presented to us does not meet the sustainable management purpose of the EEZ Act'.

Decision-makers will need to assess the extent to which an adaptive management approach can diminish the environmental risk and the uncertainty, prior to deciding if the activity can proceed. Decision-makers should consider whether:

- There will be good baseline information about the receiving environment
- The conditions provide for effective monitoring of adverse effects using appropriate indicators

South Taranaki Bight iron sand extraction

Trans-Tasman Resources Limited holds a mining permit to undertake iron ore extraction in the South Taranaki Bight. In October 2013 the company applied for marine consent under the EEZ Act to undertake iron ore extraction in a 66 square kilometre area in the South Taranaki Bight. In June 2014 the EPA Decision-Making Committee refused the marine consent application.

The project would have involved the excavation of up to 50 million tonnes (27 million cubic metres) per year of seabed material containing iron sand for processing on an offshore floating vessel. Approximately 10 per cent of the material would have been processed offshore into iron ore for export, with the remaining material returned to the seabed.⁵⁸

The company prepared an environmental impact assessment, as well as more detailed technical reports, to support its application for marine consent. The Decision-Making Committee considered the effects on the environment as detailed in Figure 11.8.

Area	Detail
Sediment plume	The sediment plume was an important consideration as many effects of the proposal would result from it. A sediment plume model was provided and this was relied on to analyse other environmental effects. The Committee accepted that the plume model was fit for purpose and had an accuracy of -50 per cent to +100 per cent.
Primary productivity	The sediment plume would reduce light penetration in the water column and therefore reduce primary production. Modelling predicted that there would be a three- to four-fold reduction in light penetration at the mining site and a two-fold change in light penetration 10 kilometres east of the mining site. It predicted that this would result in a 10 percent reduction in total primary productivity in the South Taranaki Bight. The Committee concluded that there was considerable uncertainty as to the effects of changes in primary productivity on the wider ecosystem.
Benthic environment	The application area is a high-energy environment characterised by soft-sediment habitats subject to regular disturbance and colonised by fast-growing, opportunistic species. The area to be mined would be significantly disturbed, with near total mortality of benthic fauna, and with recovery taking approximately 10 years. The benthic environment outside the mining area would be subjected to sedimentation and reduced light levels as a result of the sediment plume.
Marine mammals	A number of dolphins, whales and seals could be present in the proposed project area, including a number with threat classifications (such as Maui's dolphin, Southern/Antarctic blue whale and orca). Despite finding that the application area is unlikely to be primary habitat for Maui's dolphin, the Committee determined that the threat status of the Maui's dolphin meant that a very cautious approach was required. The Committee reached a similar conclusion in relation to blue whales. The activity had the potential to cause the following adverse effects on marine mammals: loss of food resources, displacement, noise effects, risk of collision and entanglement with vessels. The Committee considered that a comprehensive and longer-term baseline study of the marine mammals present in the South Taranaki Bight would have assisted it to understand the importance of the area to various species and what they use the area for. The absence of such information meant the Committee was uncertain as to the significance of the area to cetaceans. However, the consent conditions proposed by the company were comprehensive and may have addressed those concerns if consent was granted.
Fish	The species richness of the reef, demersal and pelagic fish assemblages in the South Taranaki Bight is moderate on a national scale. The activity had the potential to cause the following adverse effects on fish: effects of suspended sediment and sedimentation, effect of heavy metals released into the water column, loss of benthic food sources, displacement of fish and loss of habitat, loss of spawning nurseries, noise, entrainment of fish in mining apparatus, and light spill from mining operations. The Committee accepted that the effects on fish were unlikely to be significant, provided no significant food web effects arose due to primary production changes.
Seabirds	No field surveys were undertaken to inform the report on seabirds. As a result, the Committee could not reach a conclusion as to the importance of the South Taranaki Bight for seabirds. The activity had the potential to cause the following adverse effects on seabirds: direct effects of vessel lighting and noise, effects of sediment on visual foragers and effects on the food web, including habitat loss and the loss of foraging areas. The Committee concluded that it lacked an understanding of how important the South Taranaki Bight is for seabirds and therefore the significance of the potential effects.

The Committee determined that there was considerable uncertainty in the information provided to it as to both the nature of the environment and the way the mining operation might affect it. It also found information on how Māori existing interests would be addressed to be inadequate.⁵⁹ As a result, under the legislation, the Committee was required to favour caution and environmental protection and to consider whether an adaptive management approach would allow the activity to be undertaken (section 61 of the EEZ Act).

Trans-Tasman Resources did not propose an adaptive management approach that would enable the activity 'to commence on a small scale or for a short period so that its effects on the environment and existing interests can be monitored' for commercial reasons. Instead, the applicant proposed an approach which involved:⁶⁰

- Qualitative environmental performance objectives (specified in the consent conditions)
- Pre-operational baseline monitoring
- A process for developing quantitative 'trigger indicators or values' which would inform compliance with environmental performance objectives
- Operational monitoring
- Management of operations to achieve the identified environmental performance objectives

The Decision-Making Committee assessed the approach against the following criteria:

- The extent of the environmental risk
- The importance of the activity
- The degree of uncertainty
- The extent to which an adaptive management approach will sufficiently diminish the risk and uncertainty

The Trans-Tasman Resources' adaptive management approach prescribed a process for the establishment of quantitative triggers based on qualitative 'Environmental Performance Objectives'. This meant that the Environmental Performance Objectives were the 'cornerstone pillars' of the adaptive management approach.⁶¹ The Decision-Making Committee was concerned that the Environmental Performance Objectives were 'very general and non-specific'.⁶² They contained terms such as 'no more than moderate', 'moderate impacts' or 'no more than moderate and temporary adverse effects'. The Committee was not confident that these could be reasonably translated into trigger values by a third party. The Decision-Making Committee was also concerned 'that, without the option of a scale or staged implementation, the applicant has not provided any compelling adaptive management alternatives should the environmental objectives not be met'.⁶³

The Decision-Making Committee concluded, 'Ultimately, the information we had to make our decision, while voluminous, was too uncertain and inadequate, and we did not have sufficient confidence in the adaptive management approach proposed to address that uncertainty to enable the activity to be undertaken. For all of these reasons, the application as presented to us does not meet the sustainable management purpose of the EEZ Act'



The Trans-Tasman Resources proposal was found to be unlikely to have an effect on waves, surf, coastal stability or the sediment budget off the Taranaki coast (*Neale Wills*)

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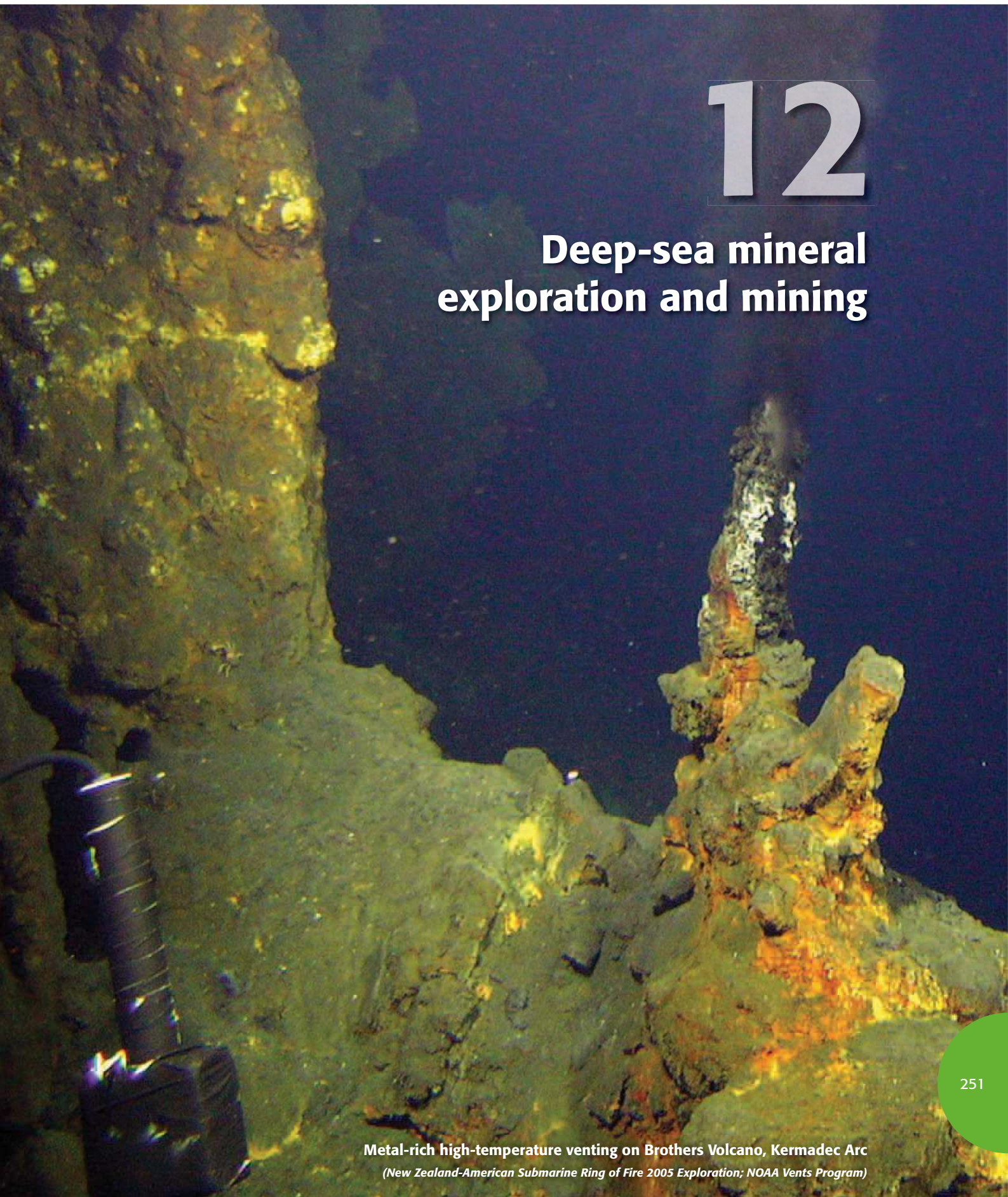
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12

Deep-sea mineral exploration and mining



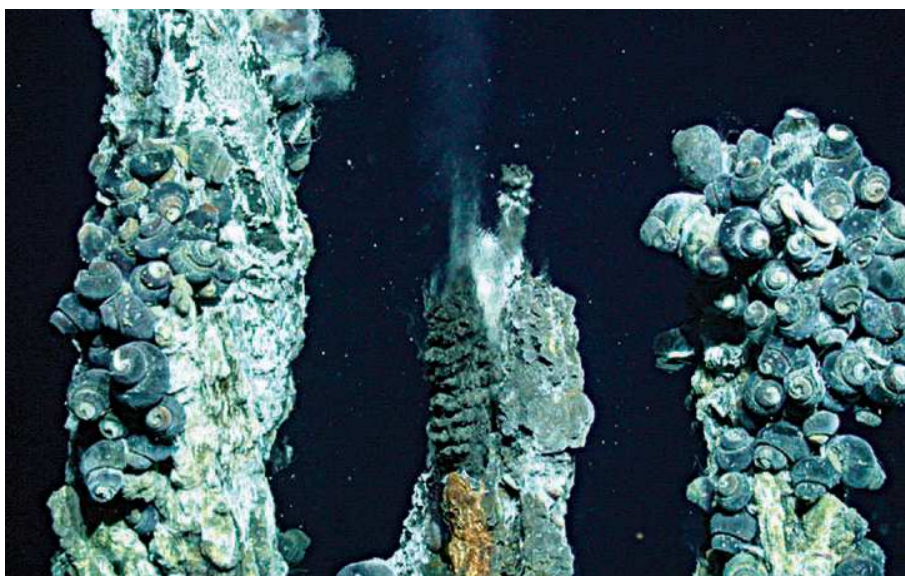
Metal-rich high-temperature venting on Brothers Volcano, Kermadec Arc
(New Zealand-American Submarine Ring of Fire 2005 Exploration; NOAA Vents Program)

Introduction

Natural gas and oil have been extracted from the sediments beneath the seafloor for decades, but the ores and minerals that lie under the ocean have been considered too difficult to extract except in shallow waters. However, as resource prices rise and deep-sea technology develops, the attention paid to these minerals has dramatically increased. The excavation of phosphate nodules, seafloor massive sulphides and manganese nodules is expected to be of increasing commercial interest within the next few years.¹

New Zealand's EEZ and extended continental shelf cover around 6 million square kilometres. Northwest, south and east of New Zealand are large areas of relatively shallow sea underlain by plateaus and tectonic ridges that border the deep ocean basins of the Pacific Ocean and Tasman Sea.² To the northeast there are volcanic arcs, including the Kermadec Arc. These areas offer significant potential as sources of minerals.

Sand mining and extraction, both inshore and offshore, are covered in Chapter 11. Oil and gas exploration and mining are covered in Chapter 13.



Aggregations of the snail *Ifremeria nautilai* on active hydrothermal chimneys (NSF Ridge 2000 Program)

Mineral resources

Figure 12.1 below highlights the potential large-scale sites of mineral resources found within New Zealand's marine area.

Mineral mining of New Zealand's deep-water seabed is an emerging industry, with no mineral mining yet being undertaken on a commercial basis within the EEZ or extended continental shelf. The Trans-Tasman Resources Limited marine consent application was the first EEZ seabed mining application to be processed (see Chapter 11: Sand mining). In June 2014, Chatham Rock Phosphate Limited lodged a marine consent application for phosphate mining on the Chatham Rise. In February 2015 this application was declined. This project is discussed in further detail in the case study section below. Figure 12.2 provides detail about the main deep-sea mineral resources found within New Zealand's marine area.



Fauna on an undisturbed hydrothermal vent (National Science Foundation Ridge 2000 Program and Charles Fisher, Penn State University)

Mineral	Detail
Manganese nodules and cobalt crusts	• Manganese nodules are rocky lumps found on the seabed in areas where sedimentation rates are very slow • They are created when manganese dissolved in seawater reacts with oxygen to create insoluble manganese oxide • They often contain metals such as cobalt, nickel and copper • Crusts that are rich in several elements, including cobalt, are found on the summits and flanks of seamounts, whereas manganese nodules are spread out over the abyssal plains and cover around 10 to 30 per cent of the seafloor • Some of the most extensive deposits of cobalt crusts in the world are found in the Pacific Ocean • Manganese nodules are abundant in an area over 250,000 square kilometres in extent, at water depths between 4,000 and 5,000 metres south-east of the Campbell Plateau and in the vicinity of Bollon's Seamount, 600 kilometres south of the Chatham Islands⁶ • The composition of nodules found within New Zealand's EEZ has been found to be relatively low in valuable metals; the value of the constituents varies with time • Since 2001 a number of exploratory permits have been issued by the International Seabed Authority for government institutions to survey manganese fields in areas of the deep sea outside national jurisdiction⁷
Methane hydrates	• Are a crystalline form of methane trapped in water, occurring in continental shelf sediments in many parts of the world⁸ • New Zealand has some large offshore gas hydrate provinces located along the east coast of the North Island and the south-west coast of the South Island⁹ • The methane hydrates in New Zealand are found in relatively shallow water¹⁰ • Research is currently underway to explore the most economical and safest way to mine methane hydrates in New Zealand¹¹

Figure 12.2 New Zealand's deep-sea mineral resources

Mining methods

Stages of work

Generally, there are some key stages of work involved in mining for marine minerals. These include:^{12,13}

- **Prospecting** – the purpose of this stage is to identify areas that are likely to contain mineral deposits. The work includes geological, geochemical and geophysical surveys (e.g. seismic surveying), and aerial surveys, including the taking of samples by low-impact mechanical methods.¹⁴ For example, multi-beam swath bathymetry can be used to map the seafloor, and remotely-operated vehicles or submersibles can be used to collect images of the seafloor, take core samples of the seabed at low concentrations, and collect targeted samples.



Manganese nodules are rocky lumps found on the seabed where sedimentation rates are low and they often contain metals such as cobalt, nickel and copper (*Secretariat of the Pacific Community, 2013b*)

In 2014 Chatham Rock Phosphate Limited applied for marine consent to mine phosphate nodules from the crest of Chatham Rise, about halfway between Banks Peninsula and Chatham Islands. The application was declined in February 2015 despite the company holding a mining permit for 820 kilometres on the crest of the Chatham Rise, and two prospecting licences for surrounding areas. The process proposed for the mining operation is discussed further in the case study at the end of this chapter.

Seafloor massive sulphides

Seafloor massive sulphide deposits are found at water depths of up to 4,000 metres, and the technology to extract minerals from the deposits is as yet unproven. Steps involved in the exploration and extraction of these deposits include:¹⁷

- Identifying potential targets
- Testing potential targets and delineating prospects
- Evaluating the resource
- Extracting minerals from the seafloor using remotely-operated mining equipment
- Transporting the ore, via a lifting system, from the seafloor to a production support vessel at the water surface
- Dewatering the ore on board a production support vessel or platform
- Transferring the ore from the production facility to a transport barge or bulk carrier
- Transporting the ore to land for treatment and/or processing
- Transporting concentrate and saleable products to market

Currently the only model for mining is that proposed by Nautilus Minerals Limited at Solwara 1, a high-grade copper-gold mining project which is located 30 kilometres off the coast, and about 1,600 metres deep, in the territorial waters of Papua New Guinea.¹⁸ This company proposes to use a 'Seafloor Mining Tool', envisioned to be a large-tracked remotely operated vehicle, incorporating tools for cutting up the vent chimneys and hydrothermal deposits, and suctioning the material to a Mining Support Vessel at the surface.¹⁹ The ore will be dewatered on the vessel, and transferred to a barge for transport to shore for processing.

Manganese nodules and cobalt crusts

Manganese nodules sit on a soft-sediment surface, and can be recovered by dredging through trawling or by suction. There is currently no known active extraction of manganese nodules anywhere in the world.²⁰ Since 2001, several permits have been issued to governmental institutions and private companies by the International Seabed Authority to survey manganese fields. These permits are not for mining, but for a detailed initial investigation of the potential mining areas. In 2013, several companies also submitted applications for the exploration of manganese nodule fields in the high seas in co-operation with developing countries (Kingdom of Tonga and Republic of Nauru).²¹ There is now interest in manganese nodule fields inside the EEZ of countries such as the Cook Islands.

Exploration and mining of manganese nodules and cobalt crusts are expected to involve the following basic processes:²²

- Recovery of minerals from the seafloor using remotely operated seafloor production equipment



Exploration of seafloor massive sulphides requires remotely operated mining equipment (MARUM, University of Bremen (Secretariat of the Pacific Community, 2013a))

- The logistical cost of exploration is high because of the offshore location of potential resources
- New Zealand's geology is complex, meaning the geological parameters such as source rock, migration pathways and reservoir rock are highly variable, both within a basin and between basins
- New Zealand has few exploration wells to date, so that the volumes of undiscovered methane hydrate reserves are difficult to predict
- The methane hydrate is under immense pressure, thus there are significant concerns about being able to bring the resource safely to the surface

Potential environmental effects

Deep-sea mining can cause a range of both direct and indirect impacts on the marine environment. Some of these potential impacts are insignificant and/or reversible. But others are potentially significant and could cause irreversible impacts on different components of the marine ecosystem. Figure 12.4 summarises the full range of potential impacts which need to be considered, but which will not necessarily result from any individual mining operation. Ways to mitigate these potential effects are discussed in the best practice section below.

	Potential environmental impact
Benthic and water column impacts	• Mortality of organisms on the seafloor, and within the seabed, in the direct mining area • Loss of sensitive and/or unique habitats if these exist in the area to be directly mined • Removal of habitat which underpins marine communities such as through the direct removal of a vent • Permanent change in the composition of marine habitats, such as through the removal of hard substrate, which species such as cold-water corals require for attachment • Smothering of benthic organisms outside the mining area by a sediment plume generated by tailings deposition • Reduced ability of visual predators to feed due to increased turbidity of seawater • Clogging of gills of fish and disruption of filter feeders from high levels of sediment in the water column • Reduction in primary productivity through reduced light levels
Marine mammals, seabirds	• Acoustic impacts from seismic surveys and other production equipment interfering with the way marine mammals locate prey, navigate and communicate • Lights causing seabirds to collide with vessels • Increased risk of ship strike
Biosecurity	• Increased biosecurity risk from introduction of overseas equipment acting as vectors for marine pest species
Other	• Other ecological impacts resulting from increased light, noise and vibration in the deep ocean environment. These are poorly understood and dependent on the spatial scale of the mining and the transient nature of methods used. • Accidents, such as oil spills and loss of equipment, having adverse impacts on the marine environment • Exclusion zones around mining areas disrupting shipping and fishing activity

Figure 12.4 Potential environmental impacts from deep-sea mining (*Secretariat of the Pacific Community, 2013a, b and c*)

Seabed and water column

Seabed mining inevitably results in significant physical changes to the area that is being mined. The removal of rock or sediment from the seabed will result in the mortality of all species living on or in the sediment layer, and those living immediately above.

Other impacts

There may be other ecological impacts, which are not yet fully understood, resulting from increased light, noise and vibration in the deep ocean environment.³¹ There is also the risk of accidents leading to spillages and loss of equipment within the marine environment.

Mining activities can displace other marine users, such as shipping and commercial fishing, as there are likely to be exclusion zones around the area of mining activity. This is only likely to cause significant problems if large areas are excluded, or if there are multiple mining operations undertaken in one region at the same time. If fishers are excluded from a significant portion of a quota management area, this could increase fishing pressure on the balance of the area. In addition, fish behaviour may be affected, with fish moving away from the area of mining influence.³²

Potential environmental impacts specific to types of minerals

Phosphates

Phosphate mining involves the removal of hard phosphate nodules from an otherwise (mostly) sedimentary environment. This means that suitable hard habitat, needed by sessile organisms such as corals, is greatly reduced. In addition, as a number of these organisms create habitat themselves (biogenic habitat), it has direct repercussions for associated species potentially resulting in a significant shift in community structure.

Once the phosphate nodules have been removed, the finer non-phosphatic sediments and seawater used to produce the slurry for the phosphorite nodule extraction will be discharged on or near the seafloor, resulting in a sediment plume. The range of potential impacts from phosphate mining is described in more detail in the case study below.

Seafloor massive sulphides

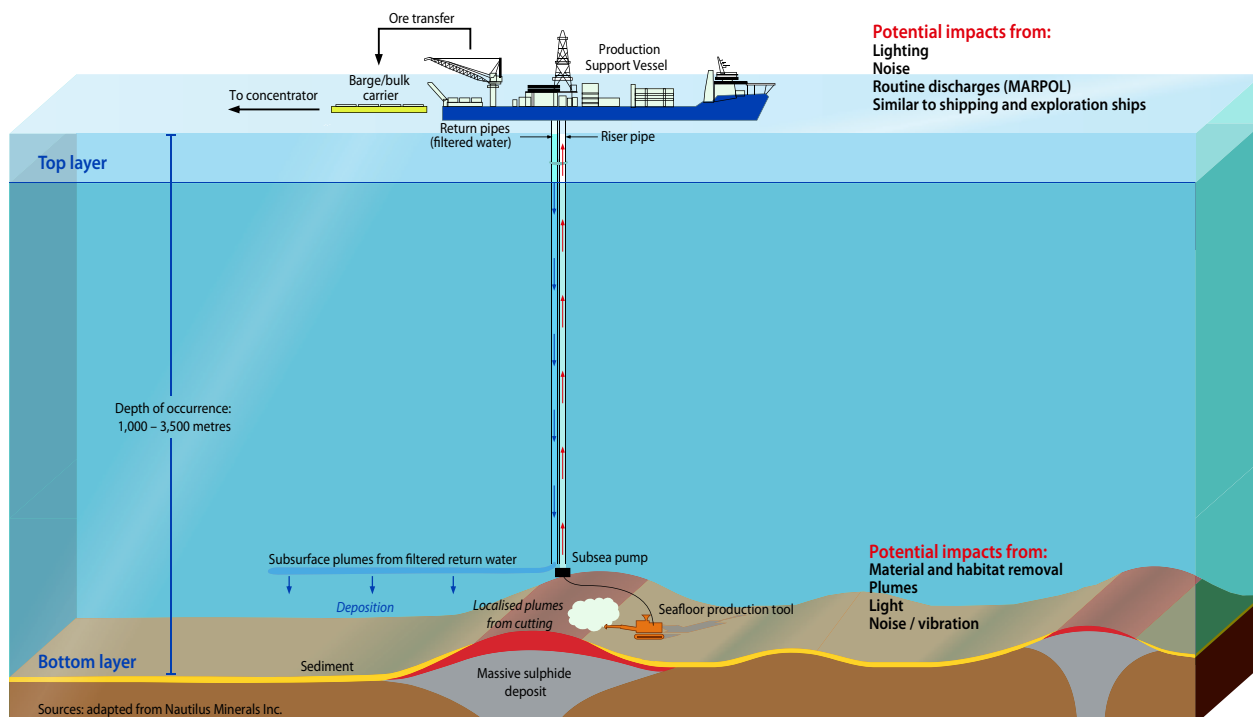


Figure 12.5 Example of seafloor massive sulphide mining system and related sources of potential environmental impacts
(Secretariat of the Pacific Community, 2013a)

Cobalt crusts

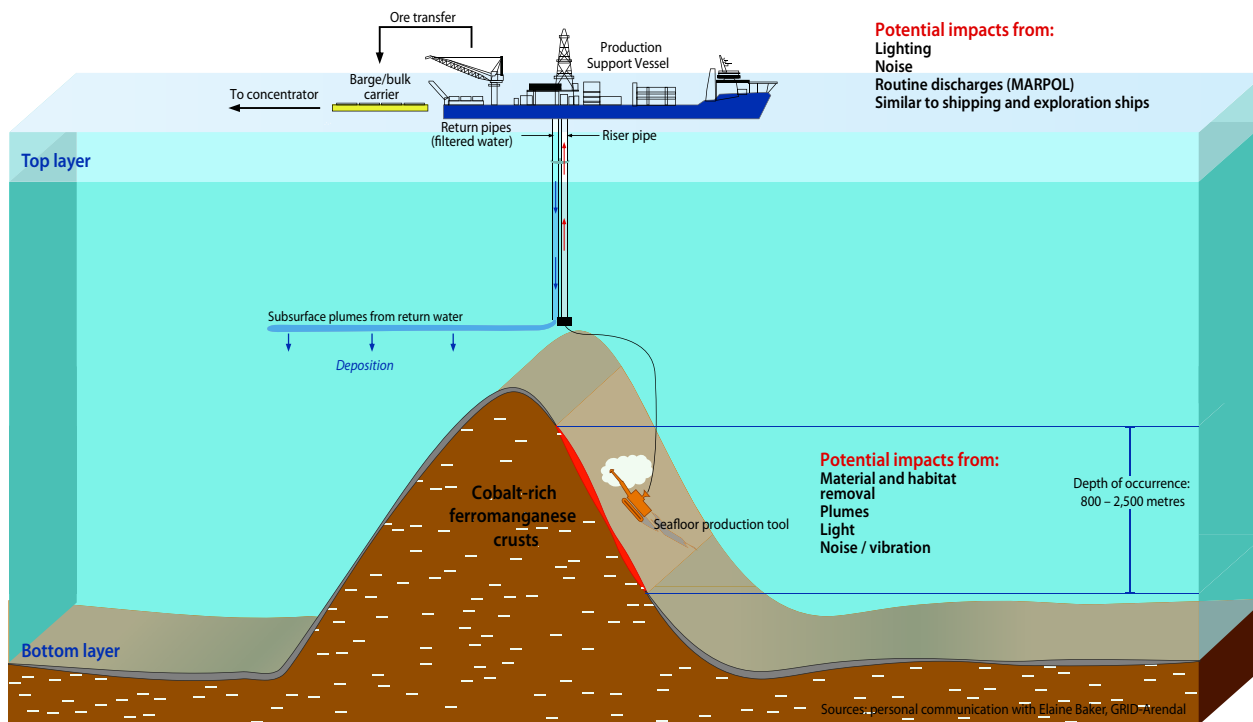


Figure 12.7 Example of seafloor ferromanganese crust mining system and related sources of potential environmental impacts (Secretariat of the Pacific Community, 2013c)

Seamounts studied to date show that there is an abundance of life associated with some of them, including corals and sponges and huge aggregations of fish. Seamounts have been described as ‘underwater oases’, and they can be important habitats for migrating species. Mining in the vicinity of these structures will destroy any corals and sponges which grow on the seamounts. This removal of habitat-forming species is significant, and recent studies indicate that recovery times would be decades to centuries.⁴⁰

The potential environmental impacts from exploration and mining for cobalt crusts are highlighted in Figure 12.7 and summarised as:⁴¹

- Destruction of sessile marine life due to the removal of material from the summit of seamounts, and the scraping of the crusts of underlying rocks
- Plumes from the mobilised sediment impacting life in the surrounding area
- Rock surfaces left behind will be recolonised over timeframes that are largely unknown (but likely to be slow)

Methane hydrates

The methane gas is trapped at such pressure that when released at the surface it expands 160 times, presenting both a massive technological challenge for extraction and a hazard.⁴² There is the potential for collapsing of areas of the seabed following the extraction of the methane. Such areas of collapse would devastate any marine life in the mining area.

There is still much work to be done on the potential environmental impact from methane hydrate extraction activities. Scientists from GNS Science, NIWA, and the Universities of Otago and Auckland have begun a Government-funded project for the Hikurangi Margin. NIWA will provide detailed information on the stratigraphic and geological architecture of the reservoirs and surroundings, and establish an ecological and biological baseline of potential exploration sites. These studies will be critical for management and environmental protection of any mining sites. The aim is to assess optimal sites by 2015, with test drilling scheduled by 2018.⁴³

Elements of best practice

The best practice considerations identified in Figure 12.8 can be applied to deep-sea minerals exploration and mining.

	Detail
Marine mammals	• Avoid areas frequented by threatened or endangered marine species, such as Hector's and Maui's dolphins, Bryde's whale and Southern right whale, as well as those identified on the 'Species of Concern' list within the Department of Conservation's 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals for Seismic Survey Operations'⁴⁵ • Use marine mammal observers • Use passive acoustic monitoring • Investigate options to cease operation if marine mammals are present • Limit unnecessary underwater sound generation and lighting • Reduce vessel speeds in areas known to be frequented by marine mammals
Seabirds	• Limit unnecessary lighting, and ensure that lights that are required are directed vertically onto work surfaces and/or utilise green lighting • Monitor bird strike and review mitigation measures in response to bird mortality above an identified level • Implement measures to reduce the likelihood of oil spills that may affect seabirds
Biodiversity	• Undertake sufficient research, to identify what species are likely to be present in the area to be mined, including their distribution • Avoid areas of high ecological value, including areas of high endemism, such as active vent sites • Only mine a small portion of any specific habitat • Consider the cumulative impacts of mining activity, and other activities such as deep-sea trawling, on benthic habitats • Consider trialling artificial hard surfaces on the seafloor to act as settlement and attachment surfaces for corals, sponges and similar organisms • Regularly check, clean and apply antifouling paints (preferably copper-free) to infrastructure that is brought to New Zealand from overseas (see 'Craft Risk Management Standard for Biofouling on Vessels Arriving to New Zealand' discussed in Chapter 5: Marine biosecurity) • Assess options for biodiversity offsetting, as appropriate, to protect seabed habitats
Water column	• Adopt mining methods that minimise the amount of sediment that is released into the water column⁴⁶ • Avoid areas where the sediment plume would disrupt areas with high ecological value • Deposit tailings close to the seafloor to minimise extent of plume
Monitoring	• Develop and undertake a robust monitoring regime – including biological and geomorphological monitoring • Undertake rigorous baseline monitoring, prior to mining commencing, so that ongoing monitoring data can be compared to the baseline • Undertake sufficient monitoring so any significant adverse effects on the marine environment can be identified and addressed in a timely manner • Publicly release monitoring information in a timely fashion • Consider involving tangata whenua and stakeholders in overseeing monitoring programmes
Other	• Avoid significant fish spawning and nursery areas, and areas of high fish density • Adopt an adaptive management approach where appropriate • Support the efficient use and recycling of minerals to minimise the need for mining activity in the marine area

Figure 12.8 Best practice considerations for deep-sea minerals exploration and mining activities

There are a number of mineral-specific best practices that were identified by the Secretariat of the Pacific Community's 2013 reports on deep-sea mineral mining. These are summarised in Figure 12.9 as they provide some additional guidance for specific mineral mining activities.

Chatham Rise phosphate mining

In June 2014 Chatham Rock Phosphate Limited lodged an application under the EEZ Act for a marine consent to mine phosphorite nodules from the crest of Chatham Rise, about halfway between Banks Peninsula and the Chatham Islands. In February 2015, the Decision-Making Committee appointed to consider the application released its decision, refusing consent.

The application

The crest of the Chatham Rise is characterised by low sedimentation rates (because of the distance from land and prevailing currents) and high productivity which makes it favourable for both nodule formation and fish production.

Chatham Rock Phosphate holds a mining permit extending over 820 square kilometres of seabed on the crest of the Chatham Rise. It also holds two prospecting licences for surrounding areas. The original application covered the entire area where the company held prospecting or mining permits, in total 10,192 square kilometres of the Chatham Rise. Chatham Rock Phosphate subsequently revised its application to reduce the proposed mining area to 5,207 square kilometres by excluding the eastern prospecting area.

The phosphate nodules were proposed to be mined using a trailing suction hopper dredger or drag head connected to the mining vessel by a flexible pipe. The nodules were to be sucked up the pipe using a series of pumps, and separated from the other seabed material on board the mining vessel. Waste tailings were to be discharged, using a sinker pipe with a diffuser, 10 metres above the seafloor.

Phosphorite nodules contain phosphate, one of the main components of commercial fertiliser. In its proposal, Chatham Rock Phosphate stated that the project could reduce phosphate imports by \$85 million per year and increase GDP by \$280 million per year. The company also stated that the phosphate has low cadmium content and could reduce nutrient runoff if applied as a direct application fertiliser (as it is less soluble than superphosphate).

The potential effects of the proposal on the marine environment, as identified in the Chatham Rock Phosphate application, are shown in Figure 12.10.

	Potential effect
Benthic environment	The mining operations will completely remove the upper layer of the seabed, including resident benthic fauna. Fauna directly above the seafloor – hyperbenthos – will also be removed/killed. There will be a long-term change in habitat from mixed nodule/soft sediment habitat to soft sediment habitat. Benthic fauna which live in soft sediments are likely to recover, taking at least 10 years to attain community complexity. Benthic fauna which depend on hard substrate are unlikely to recover due to the removal of the hard material. This fauna includes cold-water coral species and associated organisms. After the phosphoric material is extracted, the non-phosphoric material (approximately 85 per cent of the material extracted) will be returned to the seabed using a pipe. These tailings will be discharged approximately 10 metres above the seabed in the area that is being mined. The sediment will form a plume of suspended sediment. Depending on currents, the sediment plume will be transported from the site of discharge (with distance depending on the sediment particle size) and sediment will subsequently settle on the seabed. The suspended sediment plume will have sedimentation effects on the benthic environment adjacent to the mining site (deposition of 1 to 2 centimetres could extend 3 kilometres from the mining blocks). Chatham Rock Phosphate states that the suspended sediment concentration is predicted to be near background levels (around 1 milligram per litre) within about 15 kilometres of the mining site. Approximately half of the marine consent area overlaps with approximately 60 per cent of a Benthic Protection Area (the overlap is 5,236 square kilometres). These are areas which are closed to bottom trawling in order to protect seabed habitat. Chatham Rock Phosphate states that it has identified mining exclusion areas (1,822 square kilometres) for protection from seabed disturbance which better recognise the values associated with the central crest of Chatham Rise.

The decision

The Decision-Making Committee acknowledged that Chatham Rock Phosphate had gone to considerable lengths to provide information about the proposed mining operation. However, it was left with a number of uncertainties, many stemming from the fact that the proposal was the first of its kind in the world to be undertaken at such depths. Of particular concern was the reliance on modelling which had not been empirically and *in situ* validated. A staged adaptive management approach requiring extensive data collection and *in situ* validation was rejected as economically unviable by Chatham Rock Phosphate. Ultimately the Committee declined consent.

Benthic communities

The Committee found that the most significant adverse effect on the environment would be caused by the extractive part of the mining operation itself. This would result in the physical removal of the phosphate nodules and associated surface sediment along with all benthic biota living within and on the seabed. There was no dispute that this habitat and associated benthic communities would be destroyed over a large area. This included rare and vulnerable stony coral communities where were potentially unique to the crest of the Chatham Rise. The Committee found there was insufficient information verifying the existence of equivalent coral communities within other areas of the Chatham Rise outside the mining area. Referring to the Food and Agriculture Organization International Guidelines the Committee found these coral communities to constitute a rare and vulnerable ecosystem requiring protection.

The Committee concluded that recolonisation of the mined seabed by stony coral and other sessile organisms was unlikely because of their reliance on a hard substrate habitat. Re-establishment of benthic communities following mining was uncertain and at best would take decades. Further, the re-established communities would be based on soft sediment as opposed to a hard substrate and would likely be made up of different species. In addition, the benthic communities surrounding the worked mining block would be impacted by deposition of the returned sediment discharge. Although the extent of the sediment plume had been modelled, it had not been validated *in situ*. This meant that there was considerable uncertainty as to the extent of the area over which discharged sediment would disperse.

Benthic protection area

Part of the area in which Chatham Rock Phosphate proposed to mine the seafloor fell within an established benthic protection area where ground trawling and dredging by fishing vessels is prohibited. The Committee concluded that allowing mining of the seafloor in an area in which a directly comparable activity was prohibited was, at the very least, contradictory. The resultant effects of allowing mining to take place in the benthic protection area (or directly contiguous to it) were considered to be clearly contrary to its purpose, which was to protect the benthic environment, and would compromise its integrity.

The Committee did not accept that proposed mining exclusion areas could be considered compensation for the damage from mining because areas within the benthic protection area were already protected from bottom trawling and shellfish dredging. If the proposed exclusion areas were outside the benthic protection area there was no ability to ensure that they would be protected from activities other than the currently proposed mining activities.

Adaptive management

Chatham Rock Phosphate had proposed an adaptive management approach, restricting mining to a 820 kilometre area of the Chatham Rise for the first five years of mining operations, with the requirement that a set of conditions be met before mining could be expanded. Under the proposed approach, mining would be able to commence in the additional mining area only if:

- The applicant had investigated the bathymetry, physical characteristics of the seabed, and benthic ecology of the additional mining area
- The plume modelling results had been confirmed based on current data for the additional mining area
- The applicant had identified mining exclusion areas in the additional mining area
- The Environmental Reference Group (made up of representatives of key stakeholders and scientists) had reviewed information gathered
- The applicant had been granted a mining permit under the Crown Minerals Act 1991

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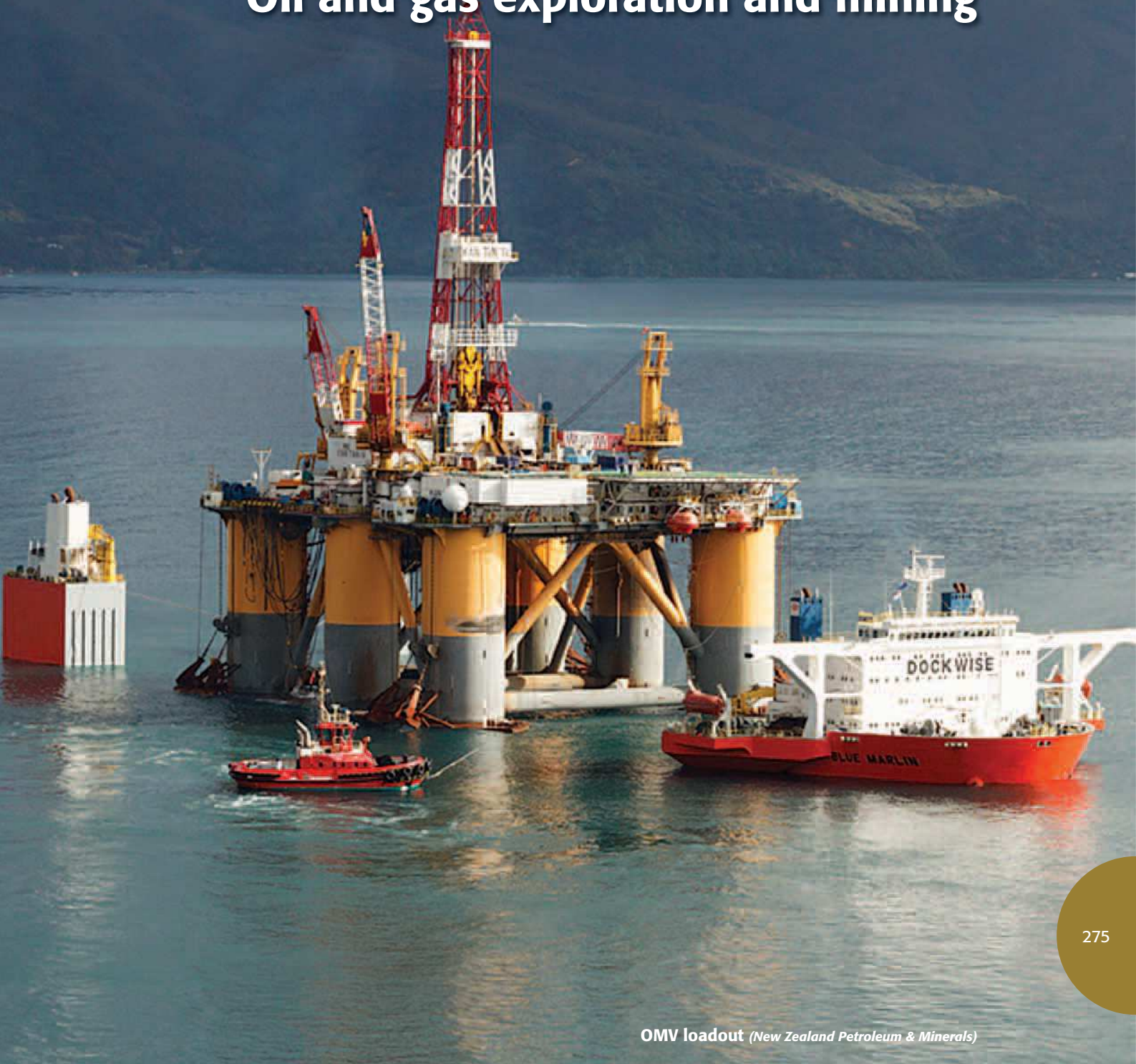
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13

Oil and gas exploration and mining



Introduction

Oil and gas are currently an essential part of this country's energy mix. They are used in transportation, production, manufacturing, construction, domestic and international travel and leisure. Oil is now New Zealand's fourth-largest merchandise export (after dairy, meat and wood), with a value in 2011 of nearly NZ\$2.2 billion.¹ This is despite the fact that production is limited to a relatively small number of fields and only one petroleum basin.² New Zealand has rights to exploit oil and gas over an area of seabed equivalent in size to the European Union, the North Sea and a quarter of the Mediterranean combined.³ There are multiple sedimentary basins with known or potential hydrocarbons onshore and underlying the extensive continental shelf of New Zealand, as well as several deepwater basins within the EEZ (shown in Figure 13.1 which has been prepared by the Ministry of Business, Innovation and Employment).⁴

While New Zealand has multiple basins with hydrocarbon potential, only one, the Taranaki Basin, is currently producing. Over 400 onshore and offshore exploration and production wells have been drilled to date in the Taranaki Basin and there are currently seven producing installations.⁵ No wells have yet been drilled beyond the continental shelf edge.

There are four other basins in the country which are known to hold petroleum – the Reinga-Northland Basin, East Coast Basin, Canterbury Basin and Great South Basin. New Zealand has at least 14 sedimentary basins of various ages with hydrocarbon potential.⁶

Most of New Zealand's marine area remains unexplored, but our knowledge of its geology suggests that over 20 per cent of it has the potential to hold petroleum resources.⁷ Currently, just over 30 offshore exploration permits have been issued by New Zealand Petroleum and Minerals.⁸ With the historical increase in petroleum prices, and the decline in existing fields, there was (until recent price declines) increasing interest in petroleum exploration within New Zealand's marine environment.⁹

The burning of oil and gas creates greenhouse gas emissions contributing to global climate change. For this reason, some argue that no further oil and gas exploration should be undertaken in New Zealand or elsewhere. This chapter focuses on the environmental impacts arising from oil and gas exploration and mining activity itself, rather than addressing the wider global impacts of burning the resulting product.



The Taranaki Basin is the only basin in New Zealand currently producing oil and gas, although four other basins are known to hold petroleum (Raewyn Peart)

Methods used for oil and gas exploration and mining

There are some key stages involved in oil and gas exploitation as outlined in Figure 13.2.



Figure 13.2 Main stages in the exploration and mining of oil and gas resources (www.cairnenergy.com)¹⁰

Prospecting and early exploration

Identifying a petroleum prospect to be drilled involves various steps, the most significant and costly of which is seismic surveying over a large area to obtain data about the topography of the seafloor and geology of the subsurface. Seismic surveying involves sending acoustic sound waves into the rock layers beneath the seafloor and then recording the time it takes for each wave to bounce back, as well as measuring the strength of each returning wave.¹¹ The various layers of rock reflect back different signals. Figure 13.3 illustrates the seismic survey at work. Underwater remotely operated vehicles or submersibles and camera systems can also be used to capture a wide range of data through both images and the collection of samples.

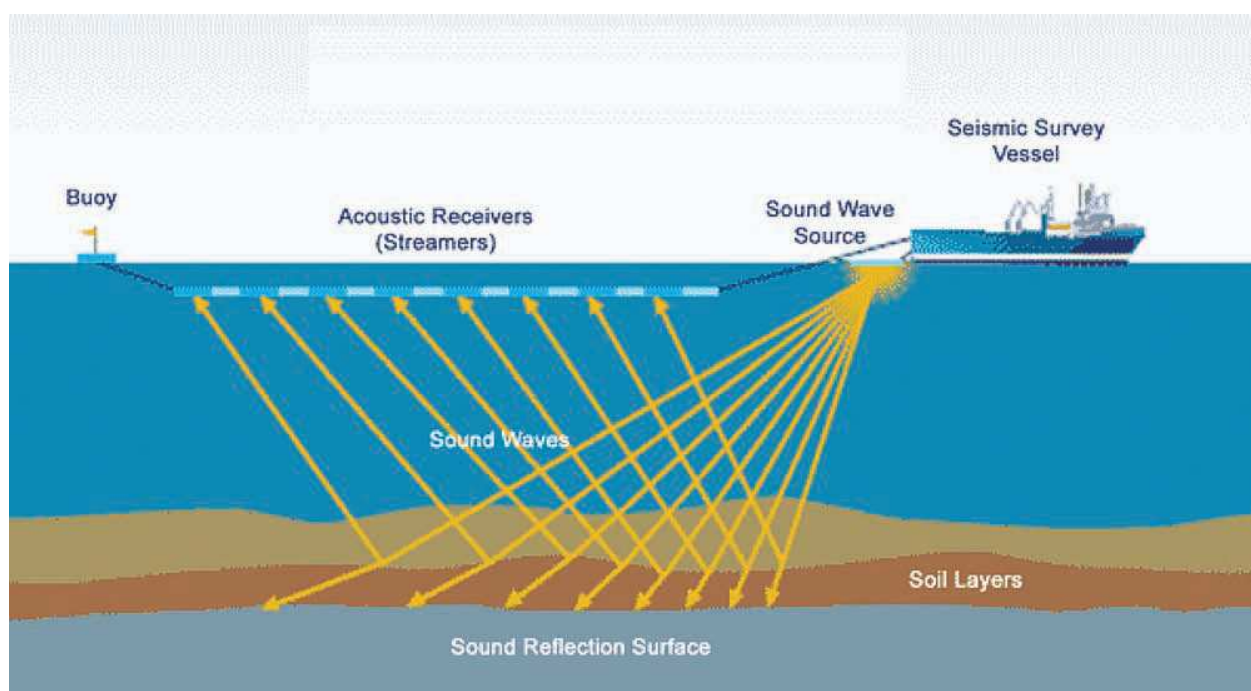


Figure 13.3 Seismic surveying (*Oil Spill Solutions*)¹²

If the exploratory phase is successful, then the operator will usually return later to drill appraisal wells to further understand the nature and extent of the field that has been discovered. In parallel to this, the operator will be working on development scenarios for the field, which will involve considering engineering, commercial, environmental and other factors.

Production

The development phase leading to production generally involves well drilling, construction of a platform structure or floating production vessels, and laying of underwater pipelines. Examples of production from existing offshore operations in New Zealand are:

- Māui A – two fixed platforms offshore with pipelines to shore and further processing onshore
- Tui field – the Umuroa floating production, storage and offloading facility
- Maari field – one fixed platform connected to the Raroa floating production, storage and offloading facility

Gas can be injected into an oil reservoir, which is known as re-injection. This process increases the pressure within the reservoir to help maintain pressure and production levels. In addition, it can assist with stimulating more oil recovery from a mature oil field. This has been done at the Kapuni field to extend its economic life.¹⁵



Māui A platform off the Taranaki coast (Shell Todd Oil Services Limited)

Figure 13.4 illustrates different kinds of production facilities.

Commissioning the Māui platform

To enable the development of the Māui gas field in 1973, one of the largest in the world at the time, a drilling platform had to be found which could withstand one of the most hostile environments of any offshore discovery. No existing designs were suitable for the extreme wind and waves. One of the largest steel manufacturers in Japan began construction in 1974. The tower was built at the Tsu shipyard before crossing more than 8,400 kilometres of ocean to New Zealand. Bad weather delayed the project for several months before the platform was officially handed over to the drilling teams in December 1977.¹⁶

There is a range of support vessels and helicopters which are located in the vicinity of the oil and gas field providing assistance to the drilling rig itself. In addition, production facilities and related infrastructure are required. These may be located on or offshore depending on the nature of the field (oil or gas) and the distance away from the shore. At the Tui and Maari facilities the oil is loaded directly onto oil tankers for export.

Decommissioning

Once all the oil and gas has been extracted, the wells and facilities will all be decommissioned. This



Figure 13.4 Various kinds of rigs and platforms above and below the water
(Oil Spill Solutions¹⁶)

Area	Potential impact from oil spill
Habitats	- Habitats become unsuitable for feeding, nesting and other services they provide for marine life - Intertidal areas, wetlands and mangroves become inhabitable - Nursery areas for fish and shellfish species become damaged
Animals	- Direct mortality of marine life - Oil coating limits the ability to swim or fly, and to maintain body temperature, feed properly and even reproduce - Oil harms the eyes, mouth, nasal tissue, immune system, red blood cells and organs of marine animals - Greater death of young animals which are usually the most vulnerable to the effects of oil
Toxic compounds	Release of toxic and carcinogenic compounds from the oil into the marine environment including trace metals and polycyclic aromatic hydrocarbons
Social/cultural	- Significant loss to customary and recreational fishing - Stress on subsistence communities which depend on fish and shellfish for food - Reduction in amenity and recreational values, particularly when oil reaches the shoreline
Economic	- Loss of incomes to tourism-based businesses and communities - Financial loss to the commercial fishing industry, both because of impacts on the fisheries themselves, and on market perceptions - High costs of clean-up which can reach billions of dollars, as seen in the Gulf of Mexico spill

Figure 13.5 Potential environmental impacts from an oil spill (*Adapted from Oceana, 2009*)



Petroleum offloading off the Taranaki coast (*New Zealand Petroleum and Minerals*)

Other potential impacts

Other environmental impacts can result from equipment and ship movements, the use of artificial light, increased noise levels and air emissions. Figure 13.6 identifies some of these potential direct and indirect environmental impacts, which are then discussed in further detail below. Many of these impacts are confined to a small area surrounding the activity. Ways of addressing them are discussed in the best practice section below.

Seabed habitats

Marine life present on the seabed where drilling occurs will be directly impacted, in particular by the legs of a jack-up rig or fixed platform, the well bore and wellhead itself, and any mooring chains and anchors.

The type of drilling lubricant used (oil-based, synthetic or water-based) has a large bearing on the toxicity of drill cuttings and the speed of impact if cuttings from the well are discharged into the water column and then settle on the seabed, rather than being collected and disposed of on land.²⁹ If a submerged pipeline is built to bring the oil or gas onshore, then the seabed will be damaged during construction and any associated trenching.

The impacts of an individual platform and associated infrastructure are localised, but there may be more significant cumulative impacts from multiple production facilities and pipelines in a single region. The level of this impact will depend on whether the area contains vulnerable benthic species, and whether these are limited in range or are rare.

The effects during decommissioning will depend on whether the infrastructure is dismantled and removed, sunk to the seafloor and abandoned, or left intact for another use such as carbon sequestration. If left intact, the effects of the structure being present will continue as for the production phase. Removal of the structure will have some impact on the seafloor and benthic organisms, as well as on the creatures which have become attached to the structure. In some cases the environmental impacts may be less if it is left *in situ*. If the infrastructure is abandoned, the oils and toxic substances will need to be cleaned off the structure to ensure that they do not dissipate into the marine environment. The platform will slowly rust and disintegrate, and this may cause negative effects over a long term.³⁰

Wildlife

Seismic surveys may have an acoustic impact on marine fauna. The acoustic impact of seismic surveys on marine mammals is particularly well documented. The consequences will vary from none to behavioural (may leave the area) to acute injury (ear drum damage) to serious (death) depending on the noise level encountered, the species and the habitat. There may also be cumulative effects from repeated exposure.³¹

Seismic streamers (as shown in Figure 13.3) may attract fur seals and sea lions during deployment and retrieval, creating a risk of entanglement and injury or drowning.

Lights at night (such as those on a drilling platform) typically attract and disturb seabirds and may cause them to collide with the above sea-surface structure. Subsurface activity and lights may affect marine mammals. Fauna may also interact with the infrastructure. For example, seal haul outs occur on the leg bracing struts of some platforms.³²

Survey and support vessels create a risk of ship strike. In addition, the installation of the production platform, pipeline construction and trenching, underwater inspection using remotely controlled vehicles or submersibles, drilling activities, and routine maintenance and pumping will produce some underwater noise that could affect marine mammals, mainly causing them to avoid the area.³³



Seals may become entangled in oil and gas exploration and mining equipment and may haul out on the struts of some platforms (Raewyn Peart)

Decision-making framework

Allocation of mining rights

Applications for prospecting, exploration and mining licences are processed by New Zealand Petroleum and Minerals. The Minerals Programme for Petroleum sets out the policies and procedures which are followed for the allocation of mineral resources (including consultation with iwi and hapū). The requirements to be met by permit holders are defined in the regulations.

The Crown Minerals Act separates activities into Tier 1 and Tier 2. Tier 1 includes high-commercial-return, high-commercial-risk projects, and most, if not all, marine petroleum operations will fall within this category. The Act requires a high-level preliminary assessment of the applicant's health, safety and environmental policies and capabilities to be undertaken for all applications for Tier 1 activities. This is not required to duplicate any assessment process under other legislation, nor does it affect the requirements under any health and safety or environmental legislation. Permits cannot be granted over marine reserves as these are in Schedule 4 of the Crown Minerals Act.

Resource Management Act

The RMA regulates the environmental effects of oil and gas activities within the territorial sea. The relevant regional coastal plan determines whether a resource consent is required. An activity cannot occur if the relevant regional coastal plan classifies it as a prohibited activity, unless the plan is changed. Generally, petroleum prospecting is classified as a permitted activity and mining as a discretionary activity, except in high-value areas where non-complying or prohibited activity status may apply.

When an application is lodged, the consent authority will determine whether or not it should be publicly notified. A consent application must be publicly notified if the consent authority determines that its adverse effects on the environment are likely to be more than minor.³⁷ This is likely to be the case for most production drilling proposals, so they should in most cases be publicly notified, enabling wide public involvement in the decision-making process.

When considering an application for consent, the consent authority must also have regard to the relevant provisions of the applicable statutory instruments made under the Act, for example the NZCPS, regional policy statements and the regional coastal plan. The NZCPS contains a number of policies that are relevant to applications for petroleum activities, including:

- Adopt a precautionary approach towards proposed activities whose effects on the coastal environment are uncertain, unknown, or little understood, but potentially significantly adverse³⁸
- Adverse effects on certain indigenous biodiversity, such as taxa which are threatened or at risk or naturally rare, must be avoided³⁹
- Adverse effects must be avoided in areas with outstanding natural character⁴⁰
- Adverse effects must be avoided in outstanding natural landscapes⁴¹
- In managing discharges to water in the coastal environment, have particular regard to [the need to] avoid significant adverse effects on ecosystems and habitats after reasonable mixing⁴²

Often the scientific evidence supporting an application contains considerable uncertainties because the marine environment is a dynamic system subject to constant change, and obtaining scientific data can be expensive and difficult. Demonstrating a cause and effect linkage can be problematic. In such cases the NZCPS dictates that a precautionary approach should be adopted.

The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

The EEZ Act regulates the environmental effects of petroleum activities beyond the territorial sea. The EEZ Permitted Activities Regulations permit seismic surveying if the person undertaking the seismic surveying complies with the Department of Conservation's 2013 'Code of Conduct for Minimising Acoustic Disturbance to Marine Mammals from Seismic Survey Operations',⁴³ which is discussed further below.

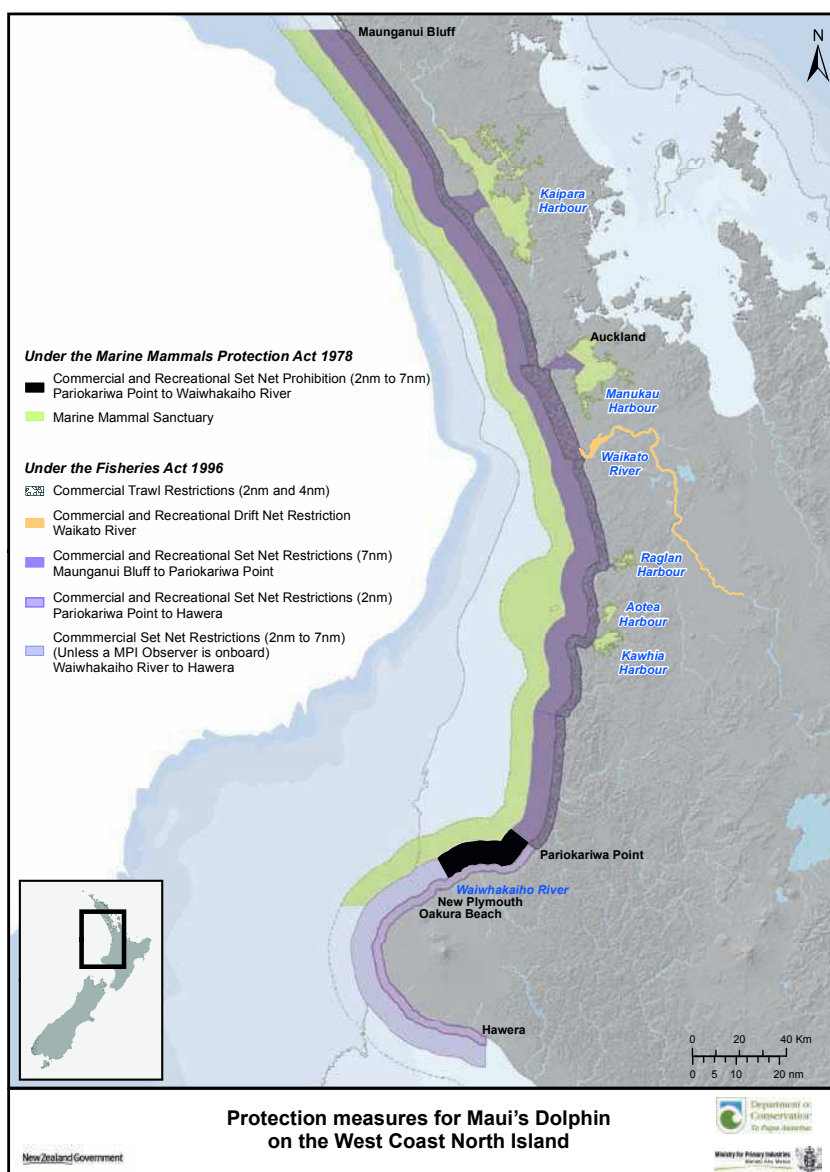
The regulations also permit prospecting and exploration for petroleum (except for any drilling for petroleum). Petroleum exploration drilling is non-notified discretionary meaning that consent is required but there is no opportunity for public submissions. Petroleum drilling for mining is a discretionary activity and applications must be publicly notified.

disturbance of marine mammals during seismic surveys. The Code has been endorsed as industry best practice by the Petroleum Exploration and Production Association of New Zealand.⁴⁸ This Code is discussed further in Chapter 3: EEZ legislation.

There are requirements in this Code for carrying out seismic surveying in a Marine Mammal Sanctuary (as defined in section 22 of the Marine Mammals Protection Act), where compliance with the Code is compulsory. The Code states that ‘particular attention needs to be given to minimising effects where surveys are planned in any of the Areas of Ecological Importance (which includes the marine mammal sanctuaries)’ as shown in Figure 13.8. In addition, no person may carry out a marine seismic survey within a Marine Mammal Sanctuary

unless he or she has, at the earliest opportunity but not less than three months before commencing the survey:

- *Notified the Director-General in writing of his or her intention to carry it out*
- *Submitted a written environmental impact assessment, and subsequently*
- *Agreed to comply with any additional conditions, such as increasing the mitigation zones or number of qualified observers required, imposed by the Director-General for operating in a Marine Mammal Sanctuary.*



Stringent conditions are placed on seismic surveying in marine mammal sanctuaries including the West Coast North Island sanctuary shown here (Department of Conservation⁴⁹)

Maritime Transport Act

Section 226 of the Maritime Transport Act states that ‘harmful substances shall not be discharged or escape, otherwise than in accordance with the marine protection rules, from any ship, offshore installation or pipeline into the sea within the exclusive economic zone or over the continental shelf of New Zealand’.

Part 200 of the Marine Protection Rules requires operators of offshore installations to develop a discharge management plan which must be approved by Maritime New Zealand. The purpose of a discharge management plan is to establish procedures and practices that will minimise environmental impacts from operational or accidental discharges of any harmful substances, including oil.

The discharge management plan must set out a number of elements as shown in Figure 13.9 and outlined below:

- Steps to control operational activities that may harm the marine environment
- Steps to minimise activities that present a risk of accidental discharge of harmful substances, including oil
- An emergency response plan that demonstrates that the operator can respond rapidly to minimise the extent and impact of a spill of harmful substances

Part of plan	Elements for inclusion
In relation to a potential oil spill	A risk assessment of the likelihood and consequences of a marine oil spill An inventory of all oils stored or produced at the installation Oil spill trajectory modelling results to indicate the likely fate of spilled oil A description of the surrounding environment highlighting sensitive areas Emergency response procedures to address oil spills in the marine environment
Emergency response procedures	Ensure the safety of personnel Stop the discharge and prevent further discharges Contain and prevent the spill from entering the sea if possible Notify relevant parties and authorities of the spill Evaluate the extent of the spill and appropriate response options Clean up spilled oil where possible Dispose of waste responsibly Review the plan and restock response equipment

Figure 13.9 Elements that need to be included in a discharge management plan

There is no public submission process for the approval of the discharge management plan, which is undertaken in-house. A discharge management plan may be approved for a period of up to three years. This means that a new approval is required at least every three years. Maritime New Zealand generally audits installations annually. Offshore installations must hold a valid International Oil Pollution Prevention Certificate which is issued by Maritime New Zealand.

Under the EEZ Act, the EPA is required to take into account ‘the nature and effect of other marine management regimes’ when considering a marine consent application.⁵¹ This includes those established under the Maritime Transport Act. So although Maritime New Zealand approves discharge management plans, the EPA can take them into account when determining if marine consent should be granted or refused.

Part 21 of the Maritime Transport Act provides for regulation of deliberate disposal into the sea of platforms or other man-made structures. Maritime New Zealand’s position on this matter mirrors Annex II of the London Protocol from which the regulation comes, which notes a hierarchy of preferred waste management options in increasing order of environmental impact:

- Re-use
- Off site recycling
- Destruction of hazardous constituents
- Treatment to reduce or remove the hazardous constituents

New Zealand. Maritime New Zealand is responsible for the national marine oil spill contingency plan. Tier 3 is the highest level of response and occurs when the marine oil spill exceeds the capacity of the Tier 1 and 2 responses. If additional response resources are required, international assistance will be requested.

Maritime New Zealand maintains a response capability of sufficient size to counter an oil spill of 3,500 tonnes, which is deemed to be a 'one in a hundred year' event. The Deepwater Horizon oil disaster in the Gulf of Mexico released around five million barrels of oil (each barrel contains just over one tonne of oil). The 2011 *Rena* disaster in Tauranga spilled approximately 350 tonnes of heavy fuel oil.

If the scale of an incident is beyond the nation's domestic capability, arrangements are in place to secure overseas assistance. This relationship is reciprocal — New Zealand will be expected to assist its neighbours if requested. Each region has resources to deal with the smaller spills that they are most likely to experience, calling on national or international assistance in the case of larger spills.

Oil spill preparedness is funded by an industry levy, the Oil Pollution Levy, which is required under the Maritime Transport Act. The Levy is collected to run New Zealand's maritime oil pollution preparedness and response system. The Oil Pollution Levy applies to all commercial vessels over 100 gross tonnes and 24 metres or more in length (except those operating in freshwater), offshore oil installations, exploration wells and oil pipelines.⁵⁵ The intention behind Maritime New Zealand's oil pollution preparedness and response is that the polluter pays. In circumstances where the polluter is not identified, the Levy is used to pay expenses from financial reserves set aside for that purpose.⁵⁶



Onwater oil recovery vessel using booms to contain oil on the stranded *Rena* off Tauranga (Maritime New Zealand)

Elements of best practice

The best practice guidelines prepared by the Ministry for the Environment and Maritime New Zealand in 2006 recommend that, while undertaking petroleum exploration, development and production activities in the EEZ and extended continental shelf, industry should adopt and apply the best practicable options to minimise or prevent adverse effects on the environment.⁵⁷ Best practice considerations for mitigation measures identified in Figure 13.10 can be applied to oil and gas activities in the marine area.

International recommendations for improving well incident prevention, intervention and response capability

Following a number of international well accidents, the International Association of Oil and Gas Producers established the Global Industry Response Group to learn from the accidents that had happened and identify recommendations for applying these learnings within the industry. The Group was divided into three core areas:⁵⁹

- Prevention – developing better capabilities and practice in well engineering design and well operations management
- Intervention – improving well capping response readiness in the event of an incident and undertaking further study into the need for, and feasibility of, global containment solutions
- Response – delivering effective and fit-for-purpose oil spill response preparedness and capability

In May 2011 the Group released its report. There were a number of key findings which are outlined below. Integral to these was the setting up of three groups:⁶⁰

- A Wells Expert Committee tasked with looking at all the data on well incident reports, and identifying opportunities for developing consistent standards, and promoting good practice
- The Subsea Well Response Project (which includes a group of nine major oil companies) with the purpose of designing a capping toolbox and additional hardware, and looking at global containment solutions
- A Joint Industry Project to develop new recommended practices, improve understanding of oil spill response tools and methodologies, and enhance co-ordination between key stakeholders internationally

The key findings from the investigation and recommendations are outlined in Figure 13.11.

Prevention: Well engineering design and well operations management recommendations

- Creation of a new, permanent, Wells Expert Committee
- Introduction of a three-tier review process
- The promotion of human competency management systems to ensure individual staff and management teams always have the skills they need
- The promotion of a culture that fosters the adherence to standards and procedures
- Recognition of existing internationally and nationally agreed standards as a baseline for industry improvements
- New and improved technical and operational practices for the overall governance of well construction
- Recommend to industry and regulators that a 'two (independent and physical) barrier' policy is in place during the life of the well
- During the drilling, completion, and abandonment phases of a well a blowout preventer is regarded as a barrier for the purposes of such a policy

Intervention: Capping and containment recommendations

- Creation of a consortium to investigate and deliver improved capping response
- Consortium to manage the concept selection and design of caps and associated equipment that could be deployed internationally
- Consortium to manage the selection and design of subsea dispersant hardware that could be deployed internationally
- Equipment to supplement existing intervention capability
- Consortium to study further the need for, and feasibility of, global containment solutions
- Consortium to investigate and develop procedures for specific operational issues related to the capping and containment of hydrocarbons released from a well blowout event
- Activity to be co-ordinated with the other initiatives

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14

Major marine developments

Introduction

New Zealand's geographic position means that the country must strongly rely on an effective port system to enable goods to be both brought into the country and exported. Close to 99 per cent of New Zealand's imports and exports are transported by sea, with over 31 million tonnes of exports and some 19 million tonnes of imports passing through the country's ports¹ Consequently, the infrastructure supporting the movement of goods is of great importance to New Zealand.

Coastal shipping services, operated by both local and international shipping companies, provide intra and interisland links and play a key role in the distribution of bulk cargos such as petroleum products and cement. There are around 30 global and regional shipping lines calling at New Zealand ports.² Six ports handle over 90 per cent of container volumes – Auckland, Tauranga, Napier, Wellington, Lyttelton and Otago.³ Large and essential marine infrastructure at these ports, such as wharves, jetties and terminals, supports commercial shipping.

Marinas, moorings, floating pontoons and boat ramps are necessary infrastructure to support recreational boating activity. New Zealand has one of the highest boat ownership rates in the world, and boats need to be stored, both on and off the water. Pressure for marinas has been greatest on the east coast of the northern part of the North Island where the larger proportion of New Zealanders live and holiday, and where the weather and coastline is particularly favourable for boating. In areas such as the Coromandel Peninsula, there have been high levels of conflict over marina proposals.

Canal developments create new coastal edges for waterfront houses and associated boat moorings. This kind of development is relatively new in New Zealand, with the first canal development being built at Pauanui on the Coromandel Peninsula during the early 1990s and others subsequently being developed at Whitianga, Gulf Harbour and Marsden Cove.

Wastewater treatment plants are large infrastructure developments which are generally located by the coast. This is because the wastewater system relies primarily on gravity to move sewage from urban areas to the treatment plant and then to release the treated water into a waterway. With these developments, there is a need for a range of supporting infrastructure, including pipes, outfalls and treatment ponds.

Other infrastructure which is commonly placed in the coastal area includes bridges, causeways, pipelines, airport runways, groynes and seawalls, all of which support human use of these environments. Despite the disparate uses for which these developments are intended, such projects often involve similar types of work. These include the placement and ongoing occupation and use of structures within the marine area, the dredging of the seabed with associated dredge material disposal, and the reclamation and declamation of land.

In the future, other types of marine infrastructure may start appearing in New Zealand's waters, including offshore wind farms and wind, wave and tidal energy devices.

These types of developments generally involve placing large structures in the marine environment. Another potential future use of the marine environment is for carbon sequestration. This involves CO₂ being separated from flue gases, and then being transported by pipes or vessels to a suitable site where the gas is stored in seabed rock.⁴ Depleted offshore oil and gas fields could possibly be used for this purpose.

Major marine developments associated with some specific activities are discussed in Chapter 10: Aquaculture and Chapter 13: Oil and gas exploration and mining.



Marsden Cove marina and canal development, Whangarei (Craig Potton)

Marinas are often built in shallow coastal waters adjacent to land. Initial dredging is usually required to create a basin deep enough for boats to moor and navigate within. Such dredging is also commonly required to create a navigable channel to connect the marina basin with open water. In general, marinas and ports should not be located in places that trap sediment, because this creates a need for regular maintenance dredging.

Canal developments in New Zealand have, to date, been constructed on low-lying land adjacent to estuaries, inlets and harbours. Some areas, historically, were coastal wetlands. The canals are dug out of land above mean high water springs, lined with seawalls, and then flooded when the area is opened up to the sea. This approach has avoided the need for canals to be dug within the existing marine environment. Seabed dredging is usually required, however, to create a navigation channel between the canal development and the open sea.

To accommodate the growing size of international ships, capital dredging has also been undertaken at commercial ports to increase the navigable depths within major shipping channels and near wharves. This mirrors an international trend in deepwater port developments. A number of the larger ports are in the process of major development to support the next generation of container ships, including the Port of Tauranga and Port Otago. The Port of Tauranga, the country's largest port by volume exported, is currently dredging to deepen the shipping channel from 12.9 metres to 16.0 metres depth at low water. This is to provide sufficient draught for larger ships which carry many more containers.⁶

Maintenance dredging is often required to maintain access to ports, wharves, marinas and canal developments. This is particularly the case in enclosed harbours and sheltered bays, where there is a large quantity of sediment running off the land into waterways, and little energy from waves or currents to disperse the sediment before it settles onto the seabed.

Reclamation and declamation

Reclamation and declamation of coastal land alters the shape of the coastline to provide for development by adding or removing material from the coastal marine area. Reclamation has been undertaken extensively in many parts of the country to provide for extensions of the urban environment. Indeed, in cities such as Auckland, Wellington and Dunedin reclaimed land forms a significant part of the central business district. Approximately 10 per cent of the area of Auckland's Waitematā Harbour has been reclaimed since 1840. Today, there is a greater understanding of the negative environment impacts of reclamation, and little is undertaken for urban expansion. Reclamation is still undertaken, however, for port and marina developments. Some of these reclamations have included dredged material, such as at the Fergusson Container Terminal.⁷

Declamation has been an integral component of canal developments in New Zealand, where land is removed, allowing the sea to expand into new areas. Such projects tend to occur in estuaries, wetlands and other parts of the coastal marine area where the water is shallow and the new structures will be protected from the worst of the coastal weather.



Works under-way at Auckland's Viaduct Basin (Raewyn Peart)

Landscapes, natural character and amenity

The placement of large structures into the marine environment and changes to the natural form of the coastal edge through the construction of reclamations and seawalls can have a major impact on natural coastal landscapes. Such man-made elements can create a significant visual impact (depending upon perception), replace natural rounded shapes and forms with straight lines and hard edges, and disrupt natural coastal patterns and processes. Many of the impacts are irreversible in practical terms.

Large single structures, such as wharves and boat ramps, may be intrusive when seen from both the land and from the sea. They are also likely to bring with them increased activity in the form of cars, boats and people. But it is the cumulative effect of many structures along the coastline that can have the most detrimental impacts on natural coastal landscapes.

Ports and marinas, in particular, often require reclamations and numerous structures to be placed within the coastal marine area. Some are occupied by a high density of boats and supported by a wide array of facilities. Most ports have historically been developed in city centres or near major land-based infrastructure that supports the effective distribution of imports and exports. City ports are often dominated by a combination of container cranes, stacked containers and vessels. They are commonly accompanied by commercial or residential development. Natural landscapes in such locations are often already highly modified and urbanised, and this can reduce the landscape impacts of further expansion.

Canal developments also have a major impact on natural coastal landscapes, through the creation of canals, high-density housing, numerous boats and floating pontoons and commercial facilities. The marine area and coastal edges within these developments are often entirely artificially created.

Because of the major impact that these kinds of developments can have on natural coastal landscapes, and the irreversibility of such impacts, it is important that the developments which do occur are carefully located, and that where possible they are confined to areas where natural coastal landscapes have already been compromised, such as disused port areas.



Canal developments, such as the one shown here at Whitianga, include artificially created coastal edges and significant infrastructure including houses, gangways and floating pontoons (Raewyn Peart)

Marine ecosystems

Reclamation causes the direct loss of coastal habitat in the marine area that is reclaimed. It can also disrupt the sensitive ecosystems of transitional zones between the land and the sea which, in their undisturbed state, contain a diverse range of organisms and are very ecologically productive. These areas can provide important nursery areas for marine life and habitat for seabirds. Natural systems in these transitional zones can also be important for trapping sediment runoff and controlling erosion. For these reasons, reclamations and major structures should be located away from ecologically important intertidal and marine habitats such as salt marshes, shell banks, seagrasses, shellfish beds, mangroves and major fish habitats. They should also be located away from bird roosting, feeding and nesting sites.

Causeways can result in the direct loss or relocation of marine habitat through burial. They can also reduce wave action and the amount of flushing on the inland side of the structure, resulting in the build-up of sediment, and sometimes eutrophication. This can create new coastal wetland habitat, but if the trapped sediments are contaminated by land-based discharges, the resulting habitat can be of poor quality. The amount of flushing will be affected by the size of the culverts built under the causeway.⁹ Bridges allow much freer flushing to occur and therefore should be used instead of culverts where possible.

Construction of offshore wind farms would disturb the seabed in the immediate vicinity of the structures. However, the structures can also provide new habitat for marine life by creating shelter and surfaces on which vegetation and benthic communities can establish themselves. Large fixed structures such as wharves and breakwaters may shade areas of water. The impacts of

Some animals are more sensitive to siltation than others. Shellfish, which have delicate feeding and breathing apparatus, may be affected by increases in sedimentation. Similarly, sediment can become trapped in the gills of young fish causing increased fatalities, and smothering of spawning or nursery areas for fish can result in the death of eggs and larvae. This could potentially reduce the level of recruitment for harvestable fisheries such as snapper. Lastly, smothering of intertidal areas may result in decreased availability of food for the birds and fish that feed there.

There have been concerns over the resuspension of sediment and impact on local ecosystems from disposal sites in the inner harbour areas. Historically, dredged material from the Pine Harbour Marina in the Waitematā Harbour has been disposed of locally. In 2009 the marina sought a resource consent to dispose of 3,000 cubic metres annually off Motukaraka Island near to the marina. Local concern about the impact of the disposed sediment on cockle, pipi and seagrass beds played a part in the consent being declined. The marina now has permission from Maritime New Zealand to dispose of the sediment off Great Barrier Island in deep water.¹²

In some areas where dredging is commonly undertaken, such as at the entrances to ports and marinas, the sediment on the seabed can often have an elevated level of contaminants. Dredging can remobilise contaminants accumulated in these areas and potentially redisperse them further away. The level of risk from this is based on the:¹⁴

- Degree of remobilisation
- Patterns of dispersal
- Sediment size and quality
- Bioavailability and toxicity of the contaminants
- Sensitivity of species in the downstream area

The potential for dredged sediment to cause toxic effects was assessed in 2012.¹⁵ This research reviewed monitoring and assessment data on sediment concentrations of five heavy metals (arsenic, chromium, copper, lead and zinc) in seven representative sites around the Auckland region (including two marinas, a port approach and a port area). The work highlighted that dredging material in this region ranged from low contamination to extremely contaminated. It concluded that the extent of contaminant dispersal is expected to be similar to sediment dispersal, being localised, but cautioned that care was needed with extremely contaminated sediments adjacent to sensitive areas.¹⁶



Maintenance dredging at the Westpark Marina removes up to 10,000 cubic metres of sediment a year (Raewyn Peart)

Clean marinas programme

The Clean Marinas programme is an industry-led programme that has been developed by the New Zealand Marina Operators Association. Its aim is to support marina operators, boatyards, contractors and recreational boaters to engage in environmentally sound practices. These include regular boat engine inspection and maintenance, proper waste disposal and reduction of discharges.²² There are a number of steps involved before a facility can gain 'Clean Marina' status, which include signing the Clean Marina Pledge, and having an assessment undertaken by an independent environmental expert. Clean Marina status has been awarded to the Marlborough Sounds Marinas, Opuā Marina, Westhaven Marina and Ashby's Boatyard. A number of other marinas have signed the Clean Marina Pledge.²³

When a marina or canal development site is located in a poorly flushing area, or the design of canals and structures prevent adequate flushing, the area can become a trap for sediment and pollutants, causing the water and sediment quality of the area to be affected.²⁴ An international report on canal estates highlighted that the majority of environmental problems arise from canal systems that are not designed to allow a certain amount of flushing, ultimately resulting in:²⁵

- Additional nutrient loading in the water
- An effect on biological diversity

The problem can be minimised through the use of wide canals which have good linkages, to allow a circular flow of water throughout the entire development. This helps to ensure that the development is well flushed by tidal action and that any contaminated runoff is collected and treated. Ongoing vigilant management is also required to ensure that spillages into the marine area are minimised through the adoption of good practice.



Westhaven Marina has been awarded Clean Marina status, which requires signing the Clean Marina Pledge and undergoing an independent environmental assessment (Raewyn Peart)

Relationship of tangata whenua with the marine environment

The impacts of development on the marine environment is a critical issue for coastal Māori. Gathering of kaimoana, both for sustenance and to provide for visitors, is an integral component of life. When those resources are compromised, such as when structures are placed over shellfish beds or in a place that obstructs access to them, the Māori way of life is also compromised. Activities which contaminate water and sediment can result in kaimoana being unsafe to eat and in a reduction in the abundance and diversity of marine life.

The impacts of dredging and reclamation are identified by the Te Awanui Tauranga Harbour Iwi Management Plan 2008 as significant cultural issues, in particular the impacts on traditional relationships in the area, disturbance to kaimoana and potential re-establishment of kaimoana.²⁶

Mahaanui Iwi Management Plan

The Mahaanui Iwi Management Plan 2013 identifies that 'dredging, reclamation, sedimentation and structures in the Whakaraupō/Lyttelton Harbour are affecting mahinga kai, including the potential effects of breakwaters on the ability of tidal flows to flush the harbour of sediment, and the resultant accumulation of sediment on kaimoana beds at Rāpaki'.²⁷ The iwi is concerned that dredging in Whakaraupō and the deposition of silt is increasing pressure on kaimoana resources of Te Ara Whānui o Makawhiua.

Coastal works such as reclamations and the construction of seawalls can result in the loss of intertidal areas, which reduces the areas where shellfish can be harvested. The Kāi Tahu ki Otago Natural Resource Management Plan 2005 identifies that such activities and structures restrict physical access to the foreshore and seabed for the collection of kaimoana.²⁸ Construction activities can also result in the destruction of culturally important sites such as wāhi tapu and traditional waka linkages and mooring places.

in any specific case will be affected by the ownership of the seabed on which the structure is located and the conditions of the coastal permit authorising the use and occupation of the structure. Under the Marine and Coastal Area Act (discussed in Chapter 2: Marine management), no person owns or is capable of owning the common marine and coastal area,³³ but this area expressly excludes freehold titles currently held over the seabed.³⁴

Reverse sensitivity

The very nature of major marine developments, particularly ports and marinas, means that there are operational activities which result in issues beyond the immediate boundaries – such as noise and light from port operations. Adequate lighting is a necessary part of providing for a safe working port. In addition, many major marine developments require land-based transport links that can impact on the surrounding community. Such issues are often referred to as ‘reverse sensitivity’, which has been defined by the Environment Court³⁵ as ‘the legal vulnerability of an established activity to complaint from a new land use. It arises when an established use is causing adverse environmental impact to nearby land, and a new, benign activity is proposed for the land. The ‘sensitivity’ is this: if the new use is permitted, the established use may be required to restrict its operations or mitigate its effects so as not to adversely affect the new activity’.

Application of reverse sensitivity to expansion of Port Otago

During the council hearing on Port Otago’s Project Next Generation resource application in 2011 to extend the wharf (which included dredging), a number of residents from the Careys Bay and Deborah Bay communities submitted in opposition, based on the potential visual effects of the proposal. Their particular concerns related to the berthing of larger vessels, stacking of containers, and the use of container cranes. Some submitters expressed ‘genuine and deeply felt concern for the way both the Multi-purpose Wharf and the fishing jetty would intrude into Careys Bay, which some likened to a Cornish fishing village’. Evidence by Port Otago stated that it was essential that operations were integrated over the whole port site, without restrictions on operations in certain or specific areas, and that any restrictions could ‘undermine the purpose’ of the extension. The decision of the Commissioners appointed by the Otago Regional Council and Ministry of Conservation stated that ‘in deciding to grant the consents sought we are also mindful that the port, along with channel dredging, has been in existence for close to 150 years and no serious evidence of adverse effects occurring in that time, as a result of port operations, were brought to our attention’.

Marine heritage

Marine developments can negatively impact on marine heritage if reclamations or structures are built in the vicinity of, or over, archaeological sites, wāhi tapu and other sites of historical significance. As the location of many marine heritage sites is poorly recorded, research will often need to be undertaken in respect of individual marine development proposals to determine whether a historical site may be impacted upon. Sensitively designed marine developments can help to preserve and profile marine heritage.

Jellicoe historical precinct, Auckland waterfront

The 3.7 hectare Jellicoe Precinct at the Auckland waterfront profiles the historical and gritty experiences of a working waterfront through the adaptive reuse of structures such as warehouses, slipways, silos and shipping containers; the retention of artefacts like rail tracks, bollards and salvaged seawall stone; and the use of structures, public art, planting, play equipment and events that depict the history, culture and nature of the coastal environment.



Jellicoe Precinct, Auckland waterfront (*Waterfront Auckland*)

New Zealand Coastal Policy Statement

The NZCPS contains a number of policies relevant to major marine developments. These policies attempt to strike an effective balance between providing for activities in the coastal environment and ensuring that important natural resources and community and cultural values are protected in doing so. Policy 6 directs councils to recognise the benefits of the use and development of the coastal marine area, whilst at the same time discouraging activities that do not have a functional need to locate there.

NZCPS, Policy 6: Activities in the coastal environment

1. In relation to the coastal environment:

- a. recognise that the provision of infrastructure, the supply and transport of energy including the generation and transmission of electricity, and the extraction of minerals are activities important to the social, economic and cultural well-being of people and communities;
- b. consider the rate at which built development and the associated public infrastructure should be enabled to provide for the reasonably foreseeable needs of population growth without compromising the other values of the coastal environment;
- e. consider where and how built development on land should be controlled so that it does not compromise activities of national or regional importance that have a functional need to locate and operate in the coastal marine area;

2. Additionally, in relation to the coastal marine area:

- a. recognise potential contributions to the social, economic and cultural wellbeing of people and communities from use and development of the coastal marine area, including the potential for renewable marine energy to contribute to meeting the energy needs of future generations;
- d. recognise that activities that do not have a functional need for location in the coastal marine area generally should not be located there

The Department of Conservation has prepared a guidance note which provides detailed assistance to decision-makers on implementing Policy 6, which is specifically focused on coastal activities and addressing their impacts. Through this policy, the Department encourages local authorities to identify what activities have a functional need to be in the coastal marine area, and to provide for those activities in appropriate places by employing an efficient use of space.³⁷ There is a direction in Policy 6 for decision-makers to consider 'reasonably foreseeable needs' when determining the rate at which built development and associated public infrastructure should be provided for.

Port development

Policy 9 outlines the need to ensure development in the coastal environment does not adversely affect the efficient and safe operation of ports. It also requires councils to consider where, how and when to provide for the efficient and safe operation of ports in policy statements and plans.³⁸ While focused on maintaining the natural state of the environment, the NZCPS seeks to ensure that urban waterfront development does not adversely affect the ability of the country's national and international ports to operate and connect efficiently and safely with other ports and land-based transport modes, as well as to expand in a strategic manner.

NZCPS, Policy 9: Ports

Recognise that a sustainable national transport system requires an efficient national network of safe ports, servicing national and international shipping, with efficient connections with other transport modes, including by:

- a. ensuring that development in the coastal environment does not adversely affect the efficient and safe operation of these ports, or their connections with other transport modes; and
- b. considering where, how and when to provide in regional policy statements and in plans for the efficient and safe operation of these ports, the development of their capacity for shipping, and their connections with other transport modes.

The Department of Conservation's 'NZCPS 2010 Guidance Note Policy 9: Ports' is a valuable resource for any local authority incorporating provisions for port facilities into its planning documents and for submitters on such provisions. The guidance note recognises that a proactive and strategic approach to the management of activities and resources in and around ports is required in order to provide for their ongoing efficient operation and development.³⁹ In addition, identifying and understanding the reverse sensitivity issues, and controlling the establishment of activities in the vicinity of ports, are necessary to minimise the pressure on local authorities to limit port activities. Identifying a buffer zone around ports, and then including policies and rules in that zone that provide for the operation of a port, is one way to manage reverse sensitivity issues.

Protecting indigenous vegetation and fauna

An applicant for a major development will need to show that there will be no adverse effects on indigenous vegetation and fauna which are threatened, at risk or naturally rare. In relation to other indigenous species and ecosystems, an applicant must show that significant adverse effects have been avoided and other adverse effects have been avoided, remedied or mitigated. These matters are discussed further in Chapter 2: Marine management.

Marine biosecurity is an important issue for ports and marinas, because of the type and number of vessels arriving from overseas and from other New Zealand ports, which can be a significant source of non-indigenous marine organisms. This is discussed further in Chapter 5: Marine biosecurity and Chapter 15: Vessels. The Department of Conservation's policy guidance recommends that in policies and plans, providing for ports will have to be integrated with managing the risk of any release and spread of harmful aquatic organisms.⁴⁵

Protecting outstanding natural character, features and landscapes

The NZCPS requires adverse effects on outstanding natural features and landscapes and on areas of the coastal marine area with outstanding natural character to be avoided. This effectively means that major marine developments will not be able to proceed in locations identified as areas of outstanding natural features and landscapes in the coastal environment. They should instead be directed to localities where the natural character and landscape values of the area have already been compromised. This is supported by the recent Supreme Court decision that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes.⁴⁶

Protecting water quality

Policy 23, in addition to addressing the general management of discharges, specifically addresses what is required for managing discharges from ports and other marine facilities. A consent authority will consider whether the structure (both its construction and operation of activities on it) will have adverse effects on water quality, and where relevant, whether tidal flushing rates are adequate to ensure that pre-existing water quality is maintained or enhanced in the vicinity of the development.

The NZCPS directs decision-makers to have particular regard to avoiding significant adverse effects on ecosystems and habitats and minimising adverse effects on the life-supporting capacity of water as a result of discharges of contaminants.⁴⁷ Research in the Auckland region on the release of contaminants during dredging indicates that including policies that 'require adverse effects to be minimised within a specified mixing zone, and avoided beyond the mixing zone, would provide more guidance to applicants and decision makers'.⁴⁸



Councils need to consider whether discharges into the marine environment, such as the outfall from the Bell Island Sewerage Treatment Plant in Nelson shown here, will have significant adverse effects on ecosystems and habitats after reasonable mixing (Raewyn Peart)

protect public health and safety, or to ensure a level of security consistent with port activities. In such cases, applicants may need to identify opportunities to enhance or restore public access in other areas of the coastal marine area, or provide connecting access around the development, if such access cannot be provided in the development itself.⁵⁰

Cumulative effects

When considering applications to undertake development of structures in the coastal marine area, the consent authority will need to consider the cumulative effects of allowing the development, in recognition of the fact that it may well be the impact of several small structures in the coastal marine area taken together that has the most effect on the natural character or landscape values of the area.

Elements of best practice

Those managing the development of the coastal marine area must walk a fine line between promoting and facilitating the close bond that New Zealanders have with the sea and coastal activities and preventing overdevelopment that ruins the values for which it is enjoyed. The coastal environment is a finite and fragile resource, so the development of structures in it should be carefully planned and limited to those which are strictly necessary.

A first step in achieving best practice coastal management for large marine developments is to carefully plan for their location in a strategic manner. This can include delineating port management areas and marina areas in regional coastal plans, with provisions clearly indicating that developments proposed outside the delineated areas will not obtain consent. The recent Supreme Court decision that plan changes must give effect to Policies 13(1)(a) and 15(a) of the NZCPS and avoid adverse effects on outstanding natural landscapes sets a clear precedent in support of this.⁵¹

There are a number of best practice elements that can be implemented to manage the adverse impacts of major marine developments and these are outlined in Figure 14.3.



Marinas and canal developments, such as the one shown here at Pauanui, need to be carefully designed to ensure there is a free flow of tidal water and good flushing (Raewyn Peart)

Potential adverse environmental effects	Best practice response
Degradation of water quality	- Carefully design marinas and canal developments to allow the free flow of tidal water and only site them in well-flushing areas - Where possible, avoid dredging in areas where there are contaminants concentrated in seabed sediment - Use the best available technology to reduce the amount of sediment released into the water column during construction - Adopt environmental management programmes to avoid discharges from marinas into the marine area from ongoing operations - Adopt voluntary certification programmes, such as the international Blue Flag accreditation⁵³
Disrupting the relationship of tangata whenua with the marine area	- Work closely with Māori to ensure that developments do not impact on wāhi tapu sites or other sites of significance to Māori - Ensure that the development does not obstruct access or adversely impact on kaimoana areas of importance to Māori - Ensure that development does not degrade water and sediment quality
Degradation of historic heritage	- Undertake sufficient research to identify any marine heritage sites in the vicinity of the proposed development area - Locate marine developments, where possible, away from any sites of heritage significance - Incorporate, where possible, heritage items into waterfront redevelopments
Public access and amenity impacts	- Make provision for public access around the coastal edge and to the marine area wherever possible - Provide public access to wharves and jetties unless exclusion can be justified for public health or safety reasons - Where public access needs to be excluded, provide for enhanced access elsewhere along the coastal marine area, close to the development site or around the development site to link up with coastal accessways - Apply no-complaints covenants to restrict new activities from complaining about adverse impacts from existing activities. For example, the Auckland District Plan Central Area Section 14.6.6.1 provides that building for accommodation in the Britomart Precinct will be a permitted activity where the site is subject to a no-complaints covenant in favour of the Ports of Auckland.⁵⁴

Figure 14.3 Best practice considerations for major marine developments



Marine structures should be designed to 'fit in' with the natural coastal environment by adopting natural shapes, such as this groyne at Kohimaramara Beach, Auckland, which has been designed to look like a rocky reef (Raewyn Peart)

Contaminated water has been entering the Westhaven Marina basin via a number of sources, including through the outfalls to the harbour and an outfall in St Mary's Bay. Waterfront Auckland is reducing and minimising any contamination of the water through marina-based activities, including:

- Mandatory sewerage removal system for any live-a-board vessel within Westhaven Marina so that vessels can remove black and grey water rather than discharge into the harbour
- Provision of sewerage reticulation in the recently completed Y Pier development for live-a-board vessels
- Stormwater filters to control the runoff from the motorways and car parks, removing hydrocarbons, sediments and gross pollutants
- Moving away from using tanalised piles towards using hardwood and then steel, which will be encased in sleeves
- Replacing all whaler boards to reduce the discharge of heavy metals on the water
- Replacing all galvanised through-bolts, which hold the berths and also rust, with pultron fibreglass rods and nylon nuts, resulting in no discharge of contaminants
- Discouraging boat owners from using coprous oxide antifouls and encouraging the use of Teflon as antifouls on appendages such as propellers
- In addition, the Government has banned tri-butyl tin antifouls which were commonplace until 1994 and the use of zinc as cathodic protection on vessels is reducing

Westhaven Marina and Z Pier are Blue Flag-certified, which is in part due to a number environmental solutions, including:

- Providing and promoting recycling facilities close to every pier, which include oil and oil filters
- Collecting and disposing of rubbish from the sea wall that drifts in from the harbour
- Implementing a 'no discharge' zone where sewage or pollutants are not allowed to enter the water
- Not allowing hull maintenance, in particular scrubbing or scraping, to occur on the marina
- Providing a berth specifically for the Watercare Harbour Clean-Up Trust



Westhaven Marina currently houses more than 2,000 boats, four yacht clubs and 50 commercial charter operators (Raewyn Peart)

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15

Vessels



Introduction

Every day New Zealand's marine environment is being utilised by a wide variety of vessels. These include fishing boats, foreign cargo vessels, oil tankers, cruise ships and pleasure craft. Vessels transport the majority of New Zealand's import and export goods. They underpin the commercial fishing industry. They also enable extensive recreational activity along our coasts.

Each year there is a vast array of vessels visiting New Zealand waters, including:¹

- 1,300 fishing boats operating in coastal waters
- 770 foreign cargo vessels
- 90 oil tankers transporting over 10 million tonnes of oil
- 500,000 pleasure craft
- One million passengers crossing the Cook Strait
- 97 cruise liners with almost 140,000 passengers which make nearly 500 visits to New Zealand ports²

Potential environmental effects

The high level of vessel activity in New Zealand can negatively impact the marine environment. One of the key risks is biosecurity, where invasive species are introduced to the country on hulls of vessels or through the discharge of ballast water. This issue is discussed in Chapter 5: Marine biosecurity. The three other main factors considered in the following section are:

- Pollution
- Harm to marine animals and birds
- Shoreline erosion



Vessels, such as this car ferry operating in the Hauraki Gulf, can impact on the marine environment in a number of ways
(Raewyn Peart)

A regional risk profile has been prepared by Maritime New Zealand which highlights the levels of risk associated with oil spills from three different sizes of vessels. The 2010 risk profile is shown in Figure 15.1. The risk profile includes an assessment of the likelihood of a spill happening and the effects of the spill. Mapping of environmental sensitivity is combined with the risk assessment to calculate the probability of the potential impact of oil on the coastline.⁵

Sewage discharge

The Resource Management (Marine Pollution) Regulations require that untreated sewage not be discharged within 200 metres of a marine reserve, within 500 metres of land, a marine farm or a mātaihai reserve, or in water less than 5 metres deep. Outside of these limits, some vessels (such as fishing vessels) can discharge raw sewage into the sea from onboard toilets. This is only likely to constitute an issue in sheltered areas close to the shore where there is poor flushing, where there are recreational users of the marine area, or within the vicinity of marine farms or marine protected areas. It is, however, of significant concern to tangata whenua for spiritual and cultural reasons. While the environmental impact from the discharge of sewage from one boat is likely to be minor, the cumulative effects of the whole maritime sector can be significant.⁶

Antifouling

Antifouling coatings are commonly used to protect submerged surfaces and prevent or slow down the growth of fouling organisms. Historically, tributyltin was the main component of antifouling paints but the use of this was finally banned in 2008 due to the severe impact on marine organisms (such as causing sterility in some gastropods).

Today, antifouling coatings can be grouped into two main categories based on whether they rely on the release of toxic (biocidal) compounds to prevent biofouling. Toxic coatings release chemicals such as copper, iron and zinc that aim to prevent settlement or survival of marine organisms. Toxic-free coatings instead rely on their physical nature, such as having smooth surfaces, so organisms cannot attach. All toxic coatings must be registered and permitted for use as an 'antifoulant' by the EPA before they can be applied.

The chemicals in antifouling paints may contain harmful substances that pose a contamination risk if leached into the surrounding seawater. In fact, recent modelling by NIWA suggests that in many New Zealand marinas, copper concentrations are likely above the guidelines for protection of marine aquatic life.⁸



Exclusion zone for boating around *Rēna* wreck, Tauranga (Daniel Sharp)



Boat being antifouled (Daniel Sharp)

Shoreline erosion

Vessels moving at fast speeds through coastal passages can create a large wake, which in turn can impact on the marine environment. This issue of shoreline erosion is particularly associated with fast ferries which can travel up to 40 knots, much faster than conventional vessels. This faster speed produces a longer-period wake that disturbs the seabed at greater depths than conventional shipping. This can result in changes to the marine habitat and the composition of the communities that live there. It can also alter the shape of the shoreline and result in accelerated coastal erosion. In confined areas such as the Marlborough Sounds, vessel wake has been associated with a reduction in marine species in the intertidal and shallow subtidal areas. Tangata whenua are concerned that the ferries could impact on kaimoana resources.¹²

Greenhouse gas emissions

Direct contributions to climate change through shipping activities and through the production of vessels and machinery is an increasing concern. The global shipping industry is believed to be responsible for around 1 billion tonnes of greenhouse gas emissions per year, which equates to approximately three per cent of the world's total emissions.¹³

There is now a major global effort being placed on reducing the impact of shipping on greenhouse gas levels. This will ultimately have an impact on New Zealand's shipping industry. For example, the European Commission is proposing that from 2018, large ships using European Union ports will need to report their verified emissions. Some of the major shipping lines entering New Zealand waters are investigating alternative power sources for their vessels. For example, using liquefied natural gas can potentially reduce CO₂ emissions by 20 per cent¹⁴ or powering container ships by kites can reduce fuel consumption by as much as 20 per cent.¹⁵



Powering container ships by kites can help reduce CO₂ emissions (SkySails¹⁶)

Resource Management (Marine Pollution) Regulations

These regulations address pollution from vessels and offshore installations within the territorial sea. They are designed to implement the provisions of the international conventions. The provisions include the following:

- The incineration of any waste in a marine incineration facility is a prohibited activity
- The discharge of plastics, dunnage, lining and packaging materials from a ship is prohibited
- The discharge of garbage from any offshore installation is prohibited
- The discharge of other ship garbage is permitted so long as the particle size is 25 millimetres or less and the discharge occurs at least 3 nautical miles from low water mark and more than 0.27 nautical miles from an offshore installation

A rule may only be included in a regional coastal plan for discharges included under the regulations if it increases the distances seaward or increases the depth specified for any area, or increases the distances from a marine farm, marine reserve or mātaihai reserve. A rule in a regional coastal plan cannot relax the requirements in the regulation.

Monitoring and enforcement of discharges from ships within the territorial sea (12 nautical miles) is undertaken by regional councils. Councils also consider resource consents for the dumping of waste at sea within this zone. Such discharges and dumping within the EEZ are controlled by the EPA. This is described further in Chapter 3: EEZ legislation and the RMA is discussed further in Chapter 2: Marine management.

Local Government Act

Regional councils also control the movement of vessels under the Local Government Act. The legislation empowers the council to appoint a harbourmaster who generally controls vessels within the harbour limits.²² The harbourmaster or a council enforcement officer can give directions, for the purpose of ensuring navigation safety, that regulate the entry, departure, navigation and mooring of a vessel.²³ In addition, the council can make bylaws to regulate shipping, including controlling speed and navigational routes.²⁴ These bylaws are, however, constrained to those aimed at achieving navigation safety rather than environmental outcomes directly.

The Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act

The EEZ Act regulates the discharges of harmful substances from offshore structures and production facilities on board mining ships and the dumping of waste within the EEZ and continental shelf. This is discussed further in Chapter 3: EEZ legislation.

Maritime Transport Act

Marine protection rules made under this Act provide for both marine safety and pollution prevention within the coastal marine area and EEZ. They deal with a wide range of matters pertaining to vessels including the design, construction and operation of vessels, discharges of oil and waste, collision prevention and carriage of dangerous goods.

The Act provides for the protection of the marine environment from harmful substances.²⁵ This includes identifying harmful substances that are not to be discharged into the sea or on the seabed of the EEZ or continental shelf, and requiring notification of the arrival of a ship carrying oil or a noxious liquid substance or the transfer of such substances. New Zealand operates a voluntary code for commercial vessels carrying oil or other harmful liquid substances in bulk. The code, implemented in 1993, makes recommendations to ship owners



Regional councils control the movement of vessels within harbour limits, such as the Police vessel shown here on the Waitematā Harbour (Daniel Sharp)

Elements of best practice

Figure 15.3 provides a summary of the best practice responses for managing the impacts of vessels on the marine environment.

Potential environmental effects	Best practice response
Biosecurity risks	• Exchange ballast water in mid-ocean or operate a type-approved ballast water management system • Install an antifouling system that is designed for the ship’s operating profile to maintain the ship in a clean state throughout its docking cycle • Regularly inspect and maintain the antifouling system for niche areas (such as sea chests, bow thrusters, propeller shafts and intake gratings) of a ship
Pollution	• The design and construction of ships and operational procedures should aim to prevent oil and other hazardous substances being discharged into the marine environment • The routes for vessels transporting oil and other hazardous substances should avoid marine areas with high ecological values including marine reserves • The discharge of oily waste into the marine area in sensitive environments including estuaries, harbours and sheltered coastal areas should be avoided • Up-to-date oil spill contingency plans should be in place • The discharge of untreated sewage into harbours and near marine farms and marine protected areas should be avoided • Recreational vessels with seawater flushing toilets should have holding tanks installed • Avoiding washing or scrubbing boats on the foreshore or intertidal area • Provide sealed boat haul-out areas where discharges can be contained • Use more environmentally-friendly compounds in antifouling paint • Use land-based disposal of onboard generated waste where possible • Avoid the disposal of contaminated waste into the marine environment
Harm to marine animals and birds	• Where possible, redirect shipping routes away from marine mammal habitat • Reduce ships speed in areas where marine mammals are likely to be present to less than 10 knots • Reduce the speed of vessels in areas susceptible to ecological impacts
Shoreline erosion	• Reduce the speed of vessels in areas susceptible to wake-induced coastal erosion
Greenhouse gases	• Investigate alternative options for powering of vessels, both while at sea and shoreside

Figure 15.3 Best practice considerations for vessels



The speed of ships passing through areas where Bryde’s whales are likely, such as the Hauraki Gulf shown here, should be kept to less than 10 knots *(Raewyn Peart)*

Rena

Early in the morning of Wednesday, 5 October 2011, the 236 metre container vessel *Rena* grounded itself on Astrolabe reef. The reef lies 12 nautical miles off the Bay of Plenty coastline, and New Zealand's fifth largest city, Tauranga. As the vessel struck the reef, it wedged its bow section deeply into the rock, leaving the stern precariously afloat and two cargo holds flooded. On board *Rena* at the time were 1,386 containers and 1,733 tonnes of heavy fuel oil.³⁸ Initially the oil leaking from the damaged vessel appeared to be minimal, but by daylight it was obvious the disaster was on a larger scale than first thought. This vessel grounding ultimately resulted in New Zealand's largest marine oil spill response.³⁹



Debris starts to emerge from the submerged section of *Rena* (Maritime New Zealand)

A national level (Tier 3) oil spill response was initiated by Maritime New Zealand. Such a response is in co-operation with the vessel owner, regional council and wider industry, as well as the local community, iwi and any other relevant agencies. According to Maritime New Zealand Tier 3 events usually call for substantial resources to be mobilised, nationally and internationally, and for these resources to be integrated into a well-organised and co-ordinated response.⁴⁰ When the *Rena* response was in full swing there were around 200 people working in the Incident Command Centre and many hundreds cleaning up the beaches and wildlife. In total, around 8,000 volunteers registered to assist the clean-up and contributed over 18,000 hours of work.⁴¹

To date, the salvage operation has recovered over three-quarters of the containers and much of the fuel and oils from the *Rena* wreck. It has also removed the accommodation block and ensured that the bow is below the water line. However, a debris field still remains around the wreckage area.⁴² The owners of the vessel have now sought resource consent to leave sections of the wreck and debris in place on Astrolabe Reef. A council hearing of the application is set down for June 2015.

Astrolabe Reef is a jewel in the Bay of Plenty's marine environment, identified as a site of outstanding natural character in the Bay of Plenty Regional Policy Statement. After the *Rena* disaster, the proposed regional policy statement 'Variation One Coastal Policy', released by the Bay of Plenty Regional Council in 2013, downgraded the classification of the reef from outstanding to high natural character.⁴³ This decision was appealed to Environment Court by residents of Motiti Island and was upheld by the Court so that the original outstanding classification was reinstated in 2014.

A \$2.4 million fund was set up by Government in 2011 to help achieve the goals of the '*Rena* Long-term Environmental Recovery Plan'. This plan describes the environmental issues associated with the disaster and outlines the way these will be addressed through six key projects, as shown in Figure 15.4.⁴⁴ For the latest updates on these projects go to <http://www.renarecovery.org.nz/rena-recovery-projects.aspx>.

In late 2013 an 'Independent Review of Maritime New Zealand's Response to the *MV Rena* Incident' was carried out by former Secretary of Foreign Affairs and Trade Simon Murdoch.

The review reported that Maritime New Zealand had underplanned and underestimated the scale of the disaster.⁴⁵ In response to the review, an additional \$2.05 million of government funding was provided to Maritime New Zealand for the coming three years to implement the recommendations. At the same time, the '*Rena* Environmental Recovery Programme 2011–2013' report was released. This report found that there were few long-lasting impacts on the marine habitats in the Bay of Plenty from the *Rena* disaster, except for heightened levels of contaminants contained to Astrolabe Reef and the north-east area of Motiti Island.⁴⁶

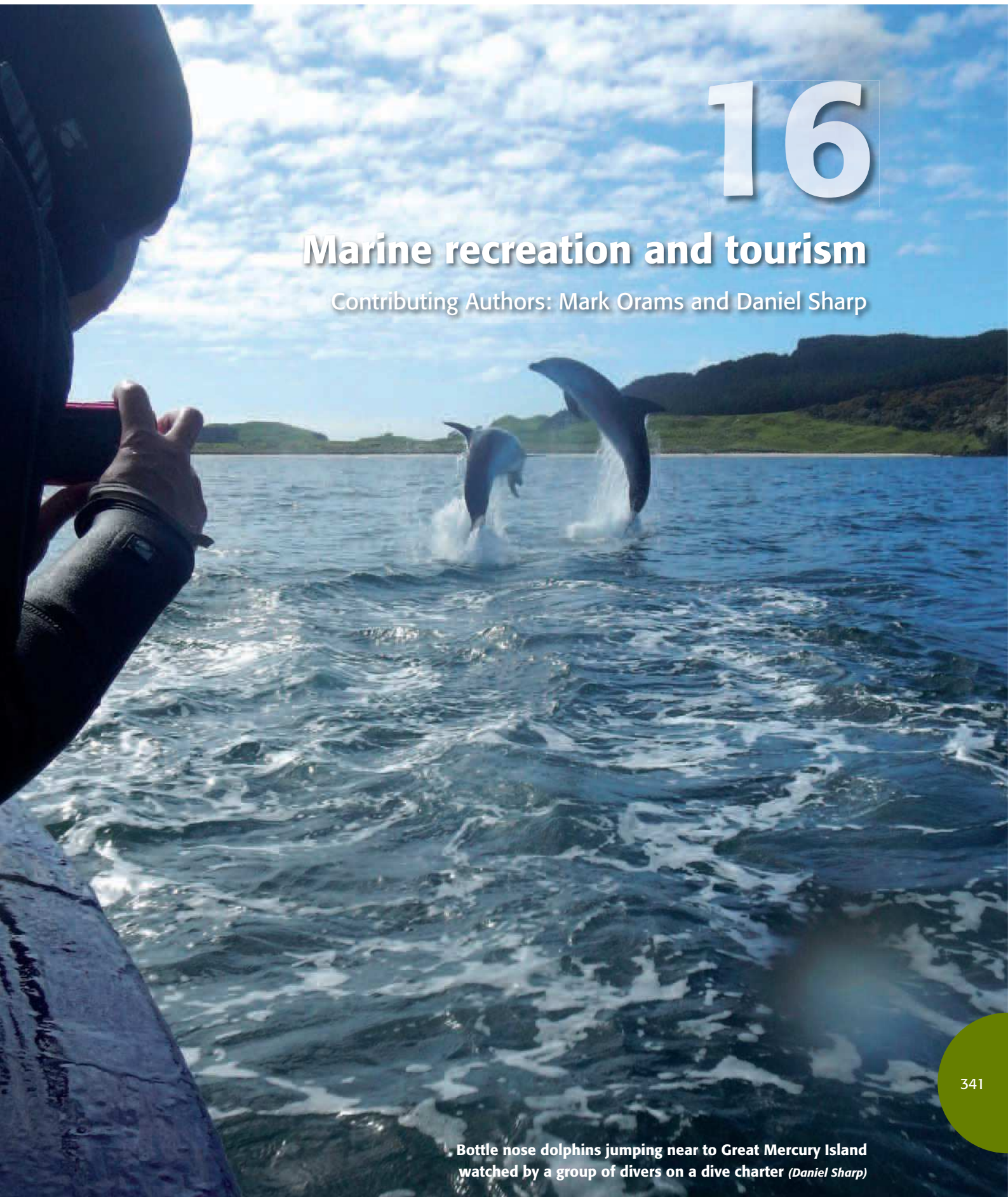
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Marine recreation and tourism

Contributing Authors: Mark Orams and Daniel Sharp



Bottle nose dolphins jumping near to Great Mercury Island watched by a group of divers on a dive charter (Daniel Sharp)

Introduction

In terms of its reputation and lifestyle, New Zealand is well known for its wide range of outdoor recreation activities (including tourism opportunities). These opportunities are supported by a relatively benign climate and a diverse range of outdoor settings which are in close proximity to each other. New Zealand also has a strong tradition of freedom of access to these settings and a culture which has valued outdoor recreation as an important part of the 'Kiwi way of life'. As an island nation, with a long and intricate coastline, the sea, beaches, estuaries, reefs and smaller islands are major recreation and tourism resources.

Tourism is one of the world's fastest-growing industries. In 2012, international tourism arrivals worldwide reached one billion, and are expected to further increase:¹

- Tourism is now one of New Zealand's largest export industries contributing nearly \$10 billion or 16 per cent of the nation's export earnings²
- International visitor arrivals to New Zealand doubled from 1993 to 2006, when there were 2.4 million arrivals, and reached 2.6 million in 2013³
- Domestic tourism (New Zealand residents travelling within New Zealand) is even more significant and combined with international tourism is responsible for \$24 billion of expenditure
- Tourism directly employs six per cent of the New Zealand workforce and indirectly employs a further 3.1 per cent
- In total, around one in 11 working New Zealanders is employed in the tourism industry⁴
- The growth of tourism is predicted to increase by four per cent every year for the next seven years⁵

Since the start of the 1970s, the use of marine ecosystems and adjacent coasts for recreation and tourism has grown enormously. There has been a continuous trend towards tourists seeking out more diverse and geographically remote coastal and marine tourism destinations.



Marine tourism has grown enormously over the past few decades, including that at Hot Water Beach, Coromandel Peninsula, shown here (Raewyn Peart)

In terms of patterns of use of marine areas for recreation and tourism, the model shows that both the intensity of use and the diversity of activities have an inversely proportional relationship with distance from both the shoreline (i.e. both intensity and diversity decrease further offshore) and places of human settlement. Thus urban beaches and near-shore locations are the most intensively used and offshore locations distant from cities and towns the least intensively used. This suggests that different management approaches will be required for the various types of marine environments as classified by proximity.

Research completed in 2012 showed that this pattern is generally true for New Zealand's marine environment with the notable exception of tourism 'hotspots', such as Milford Sound and the Poor Knights Islands, and locales that offer specific recreational experiences, such as the popular surfing breaks off the Raglan, Ahipara and South Taranaki coasts.⁸

Potential environmental effects

The growth and expanding influence of marine recreation and tourism is now well recognised in New Zealand. For example, NIWA noted in 2012 that ecotourism is a growing industry in the near-shore waters around New Zealand and is already starting to expand into the EEZ.⁹ Similarly, a study commissioned by the Department of Conservation also identified the increasing importance of the coast and near-shore coastal waters of New Zealand as resources for recreation and tourism.¹⁰

Historically, most of New Zealand's marine environment has been protected from intensive recreational and tourist use by its inaccessibility. This has changed over the past four decades and almost all parts of New Zealand's coastal environment are now accessible for recreation. While this has created opportunities for New Zealanders and visitors to the country, it has also resulted in increased environmental effects. While recreation and tourism may appear to be relatively benign and an inherently more sustainable use of natural resources than extractive activities such as fishing or deep-sea mining, it is irrefutable that coastal and marine recreation and tourism does have environmental impacts.¹¹ The continued growth in scale and distribution of this sector will, therefore, be accompanied by increasing environmental impacts.¹²

However, it is also important to recognise that there is huge potential for activities to be developed and managed in a way that have a positive impact on the very environment they rely upon.¹³ Marine and coastal recreation and tourism can also, potentially, foster a greater appreciation for the importance of marine environment and result in human actions that may benefit the ecosystem they depend on.

Figure 16.2 provides a summary of the main environmental impacts caused by different recreation and tourism activities in the marine environment. Each of these elements is discussed and explained in further detail below.



Marine tourism, such as that shown here in Milford Sound, Fiordland, can have a negative impact on the environment
(Raewyn Peart)

surfing and beach lifestyle, including Whangamatā, Mount Maunganui, Raglan, New Plymouth, Gisborne and Ahipara.

In 2013 the *New Zealand Herald* reported that more than 200,000 Kiwis and 30,000 tourists surf in New Zealand.¹⁶ It has become a way of life for many people living around the coast. As a consequence of the strong popularity of surfing, and threats to high-quality surf breaks from development, organisations such as the New Zealand Surf Break Protection Society have been advocating for legal protection of surf breaks. Some surf breaks now have statutory protection under the NZCPS. The idea of installing artificial surf reefs to promote and increase opportunities for surfing was popularised in New Zealand during the 1990s.



Board sports are very popular in New Zealand, including boogie boarding shown here (Daniel Sharp)

Windsurfing or boardsailing began in the early 1970s and, in a pattern typical of the invention of a new marine recreational activity, its popularity grew and the activity diversified. There is now general recreational windsurfing, wavesailing, slalom racing, speed sailing and Olympic competitions. Kitesurfing (also known as kiteboarding) has developed with similar speed since the early part of the 21st century and more recently stand up paddle boarding has grown rapidly. Competitive sporting events, recreational activities and commercial hire and tourism operations now exist for all of these marine recreational activities in New Zealand.

For many people it is hard to imagine that surfing might have negative impacts on the environment. However, there are a number of impacts, in particular accelerated erosion due to vehicle and foot traffic from surfers traversing over dunes and other sensitive coastal locations to access desired surf breaks. In addition, there are issues over increasing demand for infrastructure, disturbance of local wildlife, trampling on seafloor environments, leaving rubbish behind and competition with other existing recreational uses. Direct contributions to climate change through using vehicles and boats to access remote surf locations, and through the production of surfing equipment, are an increasing concern for people involved in this activity.

There are certainly psychological benefits from surfing with improvements to people's well-being, health and fitness and there are also proven economic gains from surfing. South Stradbroke Island in Australia is worth an estimated US\$20.7 million (NZ\$30 million) per year, via the 11,500 surfers who visit the island 64,000 times each year.¹⁷



Board sports, such as windsurfing in the Tauranga Harbour shown here, can impact on the marine environment (Lucy Brake)

Most people who enjoy diving and snorkelling, have an inherent passion for the marine environment, and in general the sport of diving and snorkelling has minimal adverse impacts. These activities certainly have a number of positive benefits, including improved well-being and health, economic benefits for a local community through special dive sites and educational opportunities. New Zealand's marine reserves, such as the Poor Knights Islands, are a prime spot for divers and snorkellers to learn about the marine environment and how to protect it.

The easy access that divers and snorkellers have to the marine environment can provide positive benefits, including protection of resources, raising awareness and education. An example is the Project AWARE Foundation, established by PADI as a non-profit organisation dedicated to preserving the aquatic environment and improving environmental outcomes. The work of such organisations and groups is discussed further in the section on best practice below.

There are, however, a number of challenges facing the industry. Snorkelling and diving practices can disturb wildlife, and damage marine plants and animals, through trampling or crushing by a person's body, fins and equipment. Snorkellers, in particular, enjoy fish feeding which can change the natural behaviour and diet of fish species. This can be particularly problematic in fragile environments. The anchoring of dive boats can inflict potential damage on the seabed ecology. Areas of seagrasses and reefs are particularly vulnerable.

Rubbish can be left behind in marine environments by snorkellers and divers. Moving dive equipment or support vessels from one site to another can encourage the spread of pest plants and animals, thereby posing a biosecurity risk. Other negative effects include diver-induced behavioural changes (e.g. fish feeding), damage or destruction of subtidal benthic species (e.g. repeated dislodgement) and impacts on water quality through stirring up sediments and benthos. These activities can have an increasing impact on remote environments through improved access to offshore islands.

People will often snorkel or dive to gather shellfish and harvest fish and crayfish, and in some cases this can result in over-harvesting of resources, particularly those that are close to shore or vulnerable from gathering pressure. The recreational fishing rules put in place by the Ministry for Primary Industries,²³ such as gear restrictions, maximum catch limits and minimum size limits, are designed to help reduce the pressure on fish stocks from recreational harvesting. But illegal harvesting is made easier by the use of snorkelling and diving equipment, especially of shellfish like paua and kina. Direct contributions to climate change through using boats and production of dive equipment (because of the types of materials used in the production and the emissions from transportation) is an increasing concern for people involved in diving and snorkelling.

Boating

The development of plywood and marine glues after World War II encouraged a generation of New Zealanders to build their own boats and head out onto the water in launches, sailing dinghies and cruising yachts. From the 1960s the invention and use of a range of materials such as fibreglass for hulls, alloy for masts and dacron for sails transformed boating in New Zealand by making craft more seaworthy, reliable and affordable. During the past two decades, the wide availability of much improved navigational equipment such as the GPS chartplotter, has facilitated safe cruising further afield. Boats now provide the platform for a wide variety of marine recreational activities including fishing, diving, waterskiing/wakeboarding, sightseeing, racing and wildlife watching. Boats also provide a means of transport to previously inaccessible islands, reefs, harbours and beaches.²⁴



Dinghy racing, as shown here off Takapuna Beach, Auckland, serves to introduce youth to marine sports (Raewyn Peart)

significantly contribute to the contamination of coastal waterways, particularly in crowded anchorages. Larger boats tend to have onboard holding tanks which are discharged far out to sea. But the vast majority of the smaller boats have no such facility. So whilst there may be rules in place to prevent it happening, raw human sewage is often discharged directly into some of this country's most beautiful bays and harbours.

Other impacts from boating on the marine environment include wildlife disturbance, including the disturbance of flocking, resting or feeding birds, disruption of roost sites (e.g. shag roost sites in overhanging trees and on cliffs), disturbance of larger marine animals (e.g. marine mammals and sharks) and accidental deaths of seabirds (e.g. little blue penguins). Increasing ocean noise levels and ship strike as a result of recreational boating can impact on marine mammals (this is discussed further in Chapter 15: Vessels). Regulations prescribe good boat behaviour around marine mammals.

The impact of anchoring on seafloor communities is becoming an increasing concern. The Department of Conservation produced a report in 2006²⁸ which stated that one of the most direct adverse effects on seagrass beds is the damage caused by recreational boating activities (e.g. cutting by propellers, propeller wash, anchor and mooring damage and boat groundings) which may result in significant, localised impacts on the physical integrity of seagrasses. The potentially long-term negative impacts of recreational boating activities on seagrass habitats has been recognised internationally, and the cumulative impacts can lead to the large-scale loss of seagrass beds from heavily trafficked areas.²⁹

Other issues include the conflict of vessel traffic with other users. Swimmers in shallow areas are often concerned about their safety and it is for this reason that there are clear rules about the speed that boats can travel in certain areas.

The growth in recreational boating drives demand for marinas, boat ramps and other large marine infrastructure which can have very significant impacts on the marine environment. This is discussed in Chapter 14: Major marine developments, which includes a case study on Westhaven Marina in Auckland.



Boating, such as shown here in the Hauraki Gulf, can have a range of environmental impacts (Raewyn Peart)

Recreational fishing

Kaimoana, or food from the sea, has always been an important component of Māori culture. It is significant as a source of sustenance, as a good to be traded, and as part of the hospitality provided when hosting guests. Other New Zealanders have also used the sea as a source of food, and have fished to both supplement diets and for pleasure from the time that Europeans first arrived on these shores.

Recreational fishing remains extremely important and this activity takes many forms, including:

- Fishing from rocks or from human made structures
- Surfcasting from beaches
- Using spears to catch fish such as flounder in shallow tidal areas
- Using set nets
- Hook and line fishing
- Trolling using lures or live baits from boats
- Spea-fishing (either freediving or using diving gear)
- Rock lobster (crayfish) hunting
- Shellfish gathering (in intertidal areas, using dredges, via digging and via diving and snorkelling)
- Taking marine life from rocks, rock pools and the intertidal zone

ability of the species to reproduce, resulting in lower productivity over the longer term.³² Recreational fishing can disturb marine wildlife, increase ocean noise and result in ship strike on marine mammals. Many of the environmental impacts relating to boating discussed above are also relevant to the recreational fishing sector, including biosecurity, anchor damage, rubbish, sewage and greenhouse gas emissions. In addition, conflicts with other users of the marine environment are becoming increasingly contentious.

Interacting with marine mammals and seabirds

Because of its isolation and huge latitudinal variation (extending from the subtropics to the subantarctic) the extended islands of New Zealand have been important breeding locations for seabirds (New Zealand has been termed the 'sea-bird capital of the world') and for migratory marine mammals, such as the southern right whale. New Zealand has a high level of endemism in seabirds (e.g. the yellow-eyed penguin and Fiordland crested penguin), coastal birds (e.g. northern dotterel and Auckland Islands cormorant) and in marine mammals (e.g. the Hector's and Maui's dolphins and the New Zealand sea lion). Early human settlement of New Zealand made extensive use of these animals and significant industries were based on whales, fur seals and sea lions and a variety of seabirds (e.g. shearwaters or 'muttonbirds'). In many cases, these animals were hunted until their populations became mere remnants of their pre-human numbers.



Marine wildlife-based tourism, such as viewing orca shown here, is now the dominant form of tourism in the country (Mark Orams)

More recent protection of these creatures has created, in some cases, the opportunity to develop a new wildlife industry based on the idea of visiting and observing them as a tourist attraction.³³ Such activities have become extremely popular in New Zealand over the past three decades and marine wildlife-based tourism is now the dominant form of marine tourism in the country.

In New Zealand, this is a wide-ranging sector, which includes dolphin watching and swimming (snorkelling with dolphins), whale watching, remote underwater sea life watching, seabird viewing and, increasingly, shark diving encounters. There are a number of permits issued for companies to provide marine mammal and seabird interactions from land, boats and air-based platforms around the country. There are dolphin-based tours nationwide, and whale-based tours in the Hauraki Gulf (Bryde's whales) and Kaikōura (sperm whales). It is worthwhile noting that New Zealand no longer permits the holding of dolphins in captivity, however some pinnipeds (seals and sea lions) remain in zoos (such as Auckland Zoo).

The NIWA risk assessment of activities in the New Zealand EEZ prepared for the Ministry for the Environment in 2012 identified that tourists were seeking close surface and underwater encounters with mega-fauna or densely shoaling smaller species. The assessment also noted the trend for charter boats to carry camera systems or remotely operated vehicles capable of encountering and viewing spectacular underwater scenery in the deep ocean. These trends are likely to increase over the next 20–30 years.³⁴

Swimming with dolphins, shark dive encounters and similar activities all have the capacity to disturb surface and shallow subsurface species. The NIWA risk assessment noted that the area of habitat likely to be affected at any one time is very small and the risk to ecosystem functioning is probably negligible. However, the risks to individual protected species and key species may be higher and affected individuals or populations may take months to recover.³⁵

Special events and sporting contests

Many coastal locations are deliberately establishing special events to encourage visitation and the commensurate economic and social benefits to their venues. Seafood festivals, fishing contests, surfing competitions, sand-sculpting contests, beach volleyball championships, sailing regattas, multi-sport races, music concerts and wildlife festivals are examples of a multitude of deliberately staged events which encourage tourism from participators and spectators alike.⁴⁰

The use of the coast for special events has intensified in recent years. From the Auckland Anniversary Regatta to the seafood festival in Kaikōura and fishing events in Bay of Islands, the country's marine resources are being used as a basis for a wide range of events. Many of these are competitive sporting events, such as the State Ocean Swim Series, or for festivals and hospitably events, such as the Whitianga Scallop Festival which is one of New Zealand's most popular seafood events. Smaller communities often host local music festivals using the beach as a 'party' base. The Ōhope Beach concert at Mahy Reserve is just one example of where locals come together to enjoy local musicians and a family picnic at the beach.

One particular growth area is the spread of marine adventure-type challenges that are extreme in nature. Rowing or paddling across significant bodies of water (such as Cook Strait and Foveaux Strait), traversing remote coastal areas on foot, surfing off-shore ocean reef waves, solo boating voyages and deep diving (including breath-hold diving) are examples of the growing range of extreme activities which attempt to provide adventure, challenge and deliberately place participants at a higher perceived risk of injury or death. Such activities have been growing in popularity and profile and this trend is likely to continue.⁴¹

Major events and contents held on the coast play an important economic and social role in the community. The negative impacts that come with them, such as rubbish, trampling and disturbance of wildlife, can be well controlled by the managers of the event.

General coastal and beach activities

There are many other activities that do not fall into the above categories. These include the increasing popularity of walking along the coast which has seen major infrastructure investment such as the development of coastal walkways (e.g. paved pathways, boardwalks and trails) in places such as Napier, Ōrewa, Ōmaha, and Auckland's Tāmaki Drive.

Sunbathing and relaxing at the beach is hugely popular and there are many destinations around the country, both busy and peaceful, where people head to enjoy this activity. Sandy beaches close to areas of population are the most intensively used coastal locations. A good example is the beach at Mount Maunganui, which was voted as New Zealand's 2014 top beach by Tripadvisor users, and which is a very popular place for sunbathing, people-watching, playing games and picnicking. Additional activities which take place on the intertidal zone include walking dogs, land yachting, beach volleyball, beach cricket, touch and other ball sports, horse riding, motorbike and offroad vehicle use, exploring rock pools, jogging/running, cycling and other forms of exercise, kite flying, sand castle and sand sculpture creation, photography, painting, tai-chi, yoga and many others.

Many of the impacts of more general coastal and beach activities are fairly benign on an individual basis, such as trampling and rubbish generation. But when these are looked at from a cumulative perspective they can have a serious impact on the coastal and marine environment. Trampling of dune vegetation can contribute to coastal erosion through the destruction of the native sand-binding plants. Cars and motorbikes being driven on beaches and sand dunes can damage important native vegetation and fauna (such as spiders, shorebirds and skinks).

Holiday homes

New Zealand has a long tradition of coastal holiday dwellings. Traditionally these were rudimentary structures with little infrastructure and few services (e.g. no electricity, telephone, sewage and waste water services, or 'town' water supply). The stereo-typical Kiwi bach (or crib) was a small basic structure built close to a beach, with an outside 'long-drop' toilet and tank water collected from the roof. Because New Zealand's summer coincides with the public holiday period of Christmas and New Year, and the long school holiday break, lengthy periods of living at the bach (often shared by extended family groups and friends) over this holiday period became synonymous with the Kiwi summer lifestyle.

Over the past three decades, the growing population, urbanisation and improved transportation (both roads and vehicles) has seen coastal properties within two to three hours' drive of a city become highly sought after. This demand has led to an escalation of coastal property prices, a commensurate investment in coastal dwellings, structures and infrastructure, and a major transformation of the New Zealand coast. Traditional baches within two hours' drive of a city are now rare. Larger, more luxurious holiday

ashore under Ministry supervision for recycling. However not all New Zealand ports can handle waste recycling, in which case waste is either stored on ships until they arrive in Auckland, or discharged at Australian ports.

Waste fuel oil and sludge from cruise ships is retained on board in ships' tanks and only discharged at approved New Zealand ports through approved waste oil contractors. The contractors receive the waste oil from the ship, separate the oil and water at their plant, and reuse the oil for local industry consumption. Most cruise ships have sophisticated grey and black water treatment plants, or holding tanks with filters, which remove all or nearly all of the toxic waste and sewage. The treated liquid is pumped out at sea more than 12 miles off land. The residue is compacted and either incinerated at sea or disposed of ashore.

Ships accidentally striking marine mammals can be a concern for the cruise ship industry. This is discussed above in the section on marine mammal interactions.

Decision-making framework

Resource Management Act

The control of recreation and tourism activities in the coastal marine area falls to regional councils under the RMA.⁴⁵ The relevant sections are discussed in Chapter 2: Marine management.

The NZCPS includes objectives and policies which regional coastal plans must 'give effect to'. Of particular relevance to recreation is Objective 4 which seeks to maintain and enhance the public open space qualities and recreation opportunities of the coastal environment. Many of the policies in the NZCPS relate to the provision of open and free access to the coast (particularly Policies 18 and 19). Specific recognition is given to surf breaks in Policy 16 which requires the protection of listed surf breaks of national significance by (a) ensuring that activities in the coastal environment do not adversely affect the surf breaks and (b) avoiding adverse effects of other activities on access to, and use and enjoyment of, the surf breaks.

NZCPS, Policy 16: Surf breaks of national significance

Protect the surf breaks of national significance for surfing listed in Schedule 1, by:

- a. ensuring that activities in the coastal environment do not adversely affect the surf breaks; and
- b. avoiding adverse effects of other activities on access to, and use and enjoyment of the surf breaks.

Regional coastal plans generally include few controls on recreation and tourism activities in the coastal marine area where they do not involve occupation of space or non-temporary disturbance.

The Resource Management (Marine Pollution) Regulations

These regulations set out rules relating to discharges from ships (including any boat or craft used in navigation with or without any means of propulsion) in the coastal marine area. Treated sewage can be discharged into the coastal marine area provided it is the prescribed distance from a marine farm or mātaihai reserve. Untreated sewage may only be discharged from a ship if it is more than 500 metres from mean high water springs and in water depths of more than 5 metres, as well as the prescribed distance from a marine farm, marine reserve or mātaihai reserve.⁴⁶



Regulations apply to discharges from recreational vessels, such as those shown here at Motuarohia Island, Bay of Islands, including the discharge of sewage, plastic, oily water, food waste and other garbage (Raewyn Peart)

Marine Mammals Protection Regulations

As a consequence of the growing interest in interacting with marine mammals in the wild in New Zealand, the Marine Mammals Protection Regulations 1992 were developed to regulate human contact with marine mammals by commercial operators, members of the public and others. Commercial marine mammal operations are prohibited unless a permit is issued by the Director-General of Conservation. Applications for permits are publicly notified and members of the public are able to make submissions.

The Regulations state that the Director-General of Conservation may not grant a permit unless satisfied there is substantial compliance with a number of criteria which include:⁴⁹

- The commercial operation should not have any significant adverse effect on the behavioural patterns of the marine mammals to which the application refers, having regard to, among other things, the number and effect of existing commercial operations
- That it should be in the interests of the conservation, management, or protection of the marine mammals that a permit be issued
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should have sufficient experience with marine mammals
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should have sufficient knowledge of the local area and of sea and weather conditions
- That the proposed operator, and such of the operator's staff who may come into contact with marine mammals, should not have convictions for offences involving the mistreatment of animals
- That the commercial operation should have sufficient educational value to participants or to the public

The Director-General has the final word over revoking or suspending any permits. In addition, the Director-General may place a moratorium on the issue of new permits when he or she has reasonable grounds to believe that it is necessary for the protection, conservation or management of any marine mammals.



Both recreational and commercial vessels coming into contact with marine mammals, such as the one shown here in the Bay of Islands, must comply with regulations including not causing the separation of an individual dolphin from its group (Raewyn Peart)

19 Special conditions applying to whales

In addition to complying with the provisions set out in regulation 18, every commercial operation and every person coming into contact with whales shall also comply with the following conditions:

- (a) no person in the water shall be less than 100 metres from a whale, unless authorised by the Director-General:
- (b) no vessel shall approach within 50 metres of a whale, unless authorised by the Director-General:
- (c) if a whale approaches a vessel, the master of the vessel shall, wherever practicable,—
 - (i) manoeuvre the vessel so as to keep out of the path of the whale; and
 - (ii) maintain a minimum distance of 50 metres from the whale:
- (d) no vessel or aircraft shall approach within 300 metres (1,000 feet) of any whale for the purpose of enabling passengers to watch the whale, if the number of vessels or aircraft, or both, already positioned to enable passengers to watch that whale is 3 or more:
- (e) where 2 or more vessels or aircraft approach an unaccompanied whale, the masters concerned shall co-ordinate their approach and manoeuvres, and the pilots concerned shall co-ordinate their approach and manoeuvres:
- (f) no person or vessel shall approach within 200 metres of any female baleen or sperm whale that is accompanied by a calf or calves:
- (g) a vessel shall approach a whale from a direction that is parallel to the whale and slightly to the rear of the whale:
- (h) no person shall make any loud or disturbing noise near whales:
- (i) where a sperm whale abruptly changes its orientation or starts to make short dives of between 1 and 5 minutes duration without showing its tail flukes, all persons, vessels, and aircraft shall forthwith abandon contact with the whale.

20 Special conditions applying to dolphins or seals

In addition to complying with the conditions set out in regulation 18, any commercial operation and any person coming into contact with dolphins or seals shall also comply with the following conditions:

- (a) no vessel shall proceed through a pod of dolphins:
- (b) persons may swim with dolphins and seals but not with juvenile dolphins or a pod of dolphins that includes juvenile dolphins:
- (c) commercial operators may use an airhorn to call swimmers back to the boat or to the shore:
- (d) except as provided in paragraph (c), no person shall make any loud or disturbing noise near dolphins or seals:
- (e) no vessel or aircraft shall approach within 300 metres (1,000 feet) of any pod of dolphins or herd of seals for the purpose of enabling passengers to watch the dolphins or seals, if the number of vessels or aircraft, or both, already positioned to enable passengers to watch that pod or herd is 3 or more:
- (f) where 2 or more vessels or aircraft approach an unaccompanied dolphin or seal, the masters concerned shall co-ordinate their approach and manoeuvres, and the pilots concerned shall co-ordinate their approach and manoeuvres:
- (g) a vessel shall approach a dolphin from a direction that is parallel to the dolphin and slightly to the rear of the dolphin.

Conservation Act

Any marine recreation or tourism business or organisation wanting to use public conservation land or marine reserves for their activities needs to get permission from the Department of Conservation in the form of a concession. This is a legal requirement under Part 3B of the Conservation Act. Some of the relevant activities that require a concession include guided activities (such as snorkelling, kayaking and canoeing), water transport services, commercial education and instruction and running a sports event.



Marine recreation and tourism businesses operating within public conservation land and marine reserves, such as kayaking in the Ōkārito Lagoon shown here, require a concession from the Director-General of Conservation (Mark Orams)

Activity	Best practice considerations
Swimming	• Identify areas for public swimming which are away from important habitats, especially shorebird feeding and nesting areas • Provide the minimal level of infrastructure (such as toilets, rubbish bins and car parking) to ensure impacts are reduced • Where access is over sensitive environments, such as dunes or wetlands, consider installing boardwalks or pathways to reduce impacts
Surfing, boardsailing, kiteboarding, stand up paddle boarding and surf skiing	• Provide the minimal level of land-based infrastructure (such as toilets, rubbish bins and car parking) to ensure impacts are reduced • Promote education as a way to modify user behaviour and attitudes towards the marine environment, such as that provided by local clubs and Surf Lifesaving New Zealand • Where access is over sensitive environments, such as dunes and wetlands, consider installing boardwalks or pathways to reduce impacts • Require certification for tourist operators to show environmental aspects have been addressed, for example Qualmark's Enviro-Gold is its top level in the Green rating for performers in Responsible Tourism⁵⁴
Diving and snorkelling	• Reduce the impacts on vulnerable species by rotating visits around a number of sites • Restrict the number of visitors to sensitive environments and habitats, for example at Golden Bay's Waikoropupu (Pupu Springs) diving is not permitted before midday and only four divers at a time are allowed to enter the water • Promote education as a way to modify diver and snorkelling behaviour, such as that provided by local clubs and diving organisations • Where access is over sensitive environments, such as dunes and wetlands, consider installing boardwalks or pathways to reduce impacts • Require certification for tourist operators to show environmental aspects have been given due consideration, for example Qualmark's Enviro-Gold is its top level in the Green rating for performers in Responsible Tourism⁵⁵ • In sensitive areas, control landing permits and only allow registered dive charter operators to land • Support diver awareness programmes, such as Project AWARE which is a non-profit organisation working in over 180 countries engaging the diving community to undertake voluntary work to care for the marine environment
Boating	• Install holding tanks in recreational vessels with seawater flushing toilets • Avoiding washing or scrubbing antifouling painted hulls on the foreshore or intertidal area • Provide sealed boat haul-out areas where discharges can be contained • Use more environmentally friendly compounds in antifouling paint • Minimise the production of onboard waste and use land-based disposal • Limit unnecessary use of any floodlights at night and ensure that those that are required are directed approximately vertically onto work surfaces • Reduce boat speed in areas where marine mammals are likely to be present and keep well clear of the animals • Reduce the speed of vessels in areas susceptible to ecological impacts from wake • In remote sensitive areas, control landing permits or enforce no public landing, such as at te Hauturu-o-Toi/Little Barrier Island in the Hauraki Gulf⁵⁶ • Identify public anchoring areas and install public mooring buoys in environments sensitive to anchor damage • Adopt the Clean Boating Programme recommendations for individual boaties, including guidance about simple household alternatives to harmful products to use while boating and ways to reduce the transfer of marine pests⁵⁷

Reducing recreational catch of seabirds

In association with a national recreational fishing expert and TV programme, Southern Seabird Solutions has developed a DVD identifying tips that recreational fishers can use to reduce the risk of catching seabirds and how to release them safely if they are caught, including:⁵⁸

- Making a seabird release kit and keeping it on board
- Keeping bait and offal in covered containers on board
- Sinking bait beneath the surface as fast as possible
- Covering baited hooks when not in use
- Weighing down a burly bag with sinkers
- Using soft baits as seabirds tend to spit them out without injury

Clean Boating Programme

The Clean Boating Programme has developed a series of recommendations for environmentally-sound recreational boating practices in New Zealand. These include:

- Whenever possible, refuelling at an approved refuelling station, where spill-kits are available, and not transferring fuel to a boat in containers
- Fitting a holding tank for onboard toilets
- Having a waste container on the boat and carrying out what is carried in
- Using shoreside facilities whenever possible when cleaning boats
- Ensuring the engine is kept clean and well-maintained so that it does not leak oil or fuel
- Disposing of unused bait well offshore



The Clean Boating Programme has developed a range of recommendations which apply to recreational boating such as shown here in Tauranga Harbour (Daniel Sharp)

Dive! Tutukaka

Many international and domestic divers and tourists are keen to visit the Poor Knights Islands. These 11 million year old islands, lying 22 kilometres offshore, provide visitors with some breathtaking scenery both above and below the water, and the opportunity to see a huge variety of marine life. A marine reserve was established around the Poor Knight Islands in 1998 and the area is currently being assessed as a World Heritage site (part of the Whakarua Moutere Islands application extending along the north-eastern coastline of the North Island which is registered with UNESCO by the Department of Conservation).⁶²

Dive! Tutukaka, founded in 1999, operates out of the Tutukākā Marina in the Bay of Islands and focuses its visitor experience around the Poor Knights Islands. It is New Zealand's largest dive charter company, taking over 12,000 people to the Poor Knights Islands every year on five boats.⁶³ The enterprise has won a number of national and international awards for environmental and business practices, including New Zealand Tourism's Supreme Tourism Award in 2006, Rankers Travel awards and PADI Green Star awards.⁶⁴

The voluntary routing of ships around the Poor Knights Islands has been an important part of protecting this very special marine environment. After two incidents when oily bilge water was released into the sea between the Poor Knights and the Tutukākā coast, threatening the area's world-renowned wildlife and marine ecology, the voluntary routing was set in place. Poor Knights Islands is a mandatory area to be avoided by all vessels greater than 45 metres' length overall, with only a few exemptions to this rule.⁶⁵ This is discussed further in Chapter 15: Vessels.

The evolution of the dive company has gone hand in hand with advocating for marine protected areas. This company was a key player in advocating for the exclusion of recreational fishing from within the reserve. This was highly controversial, and it was the value for tourism that was advocated by Dive! Tutukaka that supported the case for the exclusion. Due to the company's frequent and numerous trips to the area, it is actively involved in ensuring that the Poor Knights are respected by others, and also reports violations to the shipping ban restrictions surrounding the island group.



Dive! Tutukaka is New Zealand's largest dive charter company and it takes over 12,000 people to the Poor Knights Islands every year (Daniel Sharp)

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- 5 http://www.tianz.org.nz/content/library/FINAL_NZTS2015_HR.pdf
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17

Catchment-based activities



Introduction

Water flows in a constant cycle from the ocean to the sky and back again, often via a land mass, where it picks up other substances along the way back to the sea (as shown in Figure 17.1). Various substances, from both natural and human-influenced processes, are discharged from the land and transferred to the coastal marine environment as part of the natural water cycle.

The cycle can be thought of as starting when the sun's heat evaporates water from the oceans. The water vapour is carried along in air currents, which are forced up into cooler altitudes when they pass over a land mass. As the air cools, the water droplets condense and become visible as clouds. As the droplets become heavier, they eventually fall to the ground as rain (or snow).

Some rainwater is absorbed by plants and animals and is re-evaporated back into the atmosphere. Other rainwater flows back to the sea via streams, rivers, wetlands and estuaries (known generally as surface waters) taking with it dissolved material including nutrients and sometimes sediment. Yet other rainwater seeps down into underground rock and caves (known as groundwaters), where it may either flow underground to the sea, flow back up to join rivers or streams, or lie dormant underground. In addition, some rainwater falls onto urban areas where there are paved and other hard surfaces, and instead of percolating through the soil it is collected in stormwater drains and carried into the sea.

Human activity has greatly affected the amounts and types of substances that are transferred by water from the land into the sea, with corresponding effects on the health of marine ecosystems. Human-derived substances, and excess amounts of natural materials, impact on the marine environment by upsetting natural balances and introducing foreign or toxic elements into the ecosystem. Deforestation and intensification of land use has been a significant contributor to this problem, as has the destruction of wetlands, because vegetation fulfils an important role of filtering contaminants from water. Wetlands act as enormous sponges which absorb excess water during times of heavy rainfall, allow sediment to settle out, process and trap nutrients, and then slowly discharge cleaner water into rivers and estuaries.

In order of decreasing impact, land sources of increased sediment and contaminants in New Zealand's marine environment are generated mainly from agriculture, urban sewage, urban stormwater, industrial waste, agricultural waste, mining and forestry.¹



Water flows in a constant cycle from the ocean to the sky and back again, with much rainwater eventually ending up in the marine environment, such as at Doubtful Sound, Fiordland, shown here (Raewyn Peart)

Non-point source pollution

Although pollution from point sources has declined in the last 30 years, contaminants from non-point sources continue to be a major problem for the health of New Zealand's marine environment. Agricultural activities, forestry and urban earthworks can generate considerable quantities of sediment, and lead to an increase in the nutrient and microbial contaminants within sediment. Runoff from urban areas is another significant issue, including vehicle-derived zinc, copper and lead washed off roads and zinc from unpainted iron roofs.

Agriculture

Agricultural pasture occupies more than a third of the land in New Zealand, and agricultural activity affects nearly all catchments.² Since European settlement, agriculture has changed the landscape of New Zealand dramatically, through the removal of indigenous forest and wetlands, and replacement with exotic grasses. Due to the highly erodible nature of many New Zealand soils, this has resulted in significantly increased quantities of sediment entering the marine environment. In addition, agricultural chemicals such as DDT, and nutrients from fertiliser and stock urine and faeces, can make their way into the sea.



Urine from cows, such as those shown here on the Hauraki Plains, can result in elevated nitrogen levels entering the marine environment (Raewyn Peart)

Sedimentation

Forests intercept rainwater which eventually evaporates back into the atmosphere. This is through leaves and branches trapping the water before it reaches the ground, and tree roots sucking rainwater out of the soil, which then evaporates from foliage through transpiration. When forest is replaced by short-rooted pasture grass, the volume of water running off the land increases, and it travels at a faster rate. This enables it to scour out the soil and transport greater volumes of sediment.

In the absence of tree roots to hold the soil together, greater amounts of sediment made up of inorganic matter (rock and soil) and organic matter (mostly animal faeces) can be washed from the land into water bodies and make their way to the sea. When paddocks extend right up to the edges of rivers and streams, with no riparian vegetative barrier, livestock can damage the soil which leads to a mobilisation of fine sediment and ultimately sedimentation in the channel.³

The increased quantities of water runoff from pastoral land travelling at faster speeds results in higher peak flows. This can cause stream bank erosion which generates additional sediment, and often widens channels and raises the level of channel beds.

Pastoral catchments have a sediment load that can be two to five times greater than that of a forested catchment.⁴ Although most deforestation in New Zealand took place more than 80 years ago, the effects of these changes on the landscape continue to be felt, as the hills and riparian areas continue to be used for livestock production. In recent years, some councils have provided incentives for farmers to replant riparian areas.⁵ In the past, there have been assistance programmes for soil stabilisation, but these are now less common.

since lead was removed from petrol but zinc contamination continues to rise),¹¹ hydrocarbons from vehicles (these come from the road surface itself, exhaust emissions, brake linings and tyre wear), sediment and contaminated dust.¹²

Roof runoff is a major source of zinc contamination in catchments which have been urbanised for a long time, because of the large number of galvanised iron roofs in poor or unpainted condition. Zinc oxide used in tyres is another. In addition, urban earthworks have the potential to release large amounts of sediment into the catchment.¹³



Hard channelling of streams in urban areas, as shown here in Dunedin, can carry contaminated stormwater directly into the marine area (Raewyn Peart)

Point source pollution

Pollution of this type has generally declined over the last 20 to 30 years as treatment system technology has been upgraded and alternative methods of disposal have been developed. However, at specific sites in New Zealand it can still be a problem.

Agricultural and industrial enterprises

Meatworks, dairy factories, tanneries, fertiliser plants, stockyards, mills and many other types of agricultural and industrial enterprise all risk pollutants being discharged into the catchment. In the early years of European settlement, there were many such enterprises scattered across the country, all producing industrial or agricultural waste on a small scale. Some of these were connected to municipal treatment plants while others managed their own waste. In some cases, toxic waste was discharged directly into the marine environment.

Over the last 30 years, agricultural and industrial activity has changed, so that there are fewer, larger operations. For example, whereas in 1970 there were more than 200 dairy factories, there are now only about 30.¹⁴ These bigger businesses tend to deal with waste more effectively because of their relative financial strength and access to technology. Furthermore, the last 30 years have seen the implementation of stricter legal requirements and a shift in attitudes towards greater recognition of the importance of avoiding pollution, together with a greater interest in finding better technology to achieve this.

Agricultural enterprises generally now use two-pond treatment systems, often combined with land irrigation systems, to dispose of their effluent from dairy sheds and the like. Such systems pass the effluent through two separate ponds, each containing different naturally occurring types of bacteria, before discharging it into the catchment. This system removes 95 per cent of the organic matter from the effluent but may not greatly affect dissolved nutrients levels.

Land irrigation disposes of wastewater and nutrients to paddocks or forest rather than into the water. This is effective at protecting the catchment from some pollution, so long as the land is not overloaded. But if the capacity of the land to absorb the waste is exceeded, the nutrients will wash off into streams or leach into the groundwater, posing the same risk to marine ecosystems as they would have if they had been discharged directly into the catchment.¹⁵ Only some nitrogen pollution will be removed by plants, and only when they are growing. The remainder will pass into groundwater and eventually emerge in streams.

All treatment plants in New Zealand practise at least primary treatment, and a large number also use secondary and tertiary treatment. Only a small number use ultra-violet or equivalent systems for the reduction of pathogens in the final effluent. The treatment level is largely related to the sensitivity of the receiving waters. The effectiveness of wastewater treatment systems consequently varies. Some are ineffective because they are too small to cope with the volume of waste they receive. For example, some are put under pressure by new urban developments, by the increased amounts of effluent generated by an influx of visitors during holiday season, or by stormwater entering the systems and overloading them.

Septic tanks

Smaller settlements in New Zealand often do not have community wastewater treatment plants but rely on individual household septic tanks. Septic tanks are generally effective but can fail where the soil is unsuitable, or where the surrounding land area is not big enough for the soakage field. This is a particular problem in modern coastal developments, where large homes (rather than traditional baches) with modern conveniences requiring a large amount of water are built on small sections very close together. Overloading of the soakage fields, with resulting overland flow, is the most common problem. Septic tanks can also fail when they are inadequately maintained. Effluent that seeps untreated from septic tanks may get into streams or groundwater, and from there into the marine environment. Even with properly-operating systems, some nitrogen transfer to groundwater is inevitable.

Stormwater and sewage

Previous sections discussed the treatment of wastewater. This section considers environmental issues arising from the conveyance of wastewater to treatment plants.

Wastewater systems typically use gravity to move sewage discharged from houses and other premises along pipes towards the treatment works. When the sewage pipes reach the lowest point on land, gravity will no longer move the material, and pumping stations are required to pump the sewage up to the treatment plant. Many pumping stations are located near the coast, as this is often the lowest point to which sewage can be moved by gravity.

Pumping stations will usually have some holding capacity for sewage in the event that the pump fails, such as when there is an interruption in the electricity supply. When the pump stops for any reason, the sewage will keep flowing as a result of gravity and it will build up in the holding tanks at the station. When the holding capacity of the station is exceeded, raw sewage typically flows into the sea or a nearby river.



Wetlands and indigenous plantings assist with stormwater filtration at Wynyard Quarter, Auckland (Raewyn Peart)

Cumulative impacts

Cumulative effects are referred to in the definition of 'effect' in section 3 of the RMA as effects that can build up over time or occur in combination with other effects. The term refers to the impact of a number of contaminants or pollutants on a particular ecological system, which in some cases may be greater than the sum of their individual impact. It also incorporates the concept of flow-on impacts. Pollutants are dangerous due to their cumulative effect, as chemicals tend to have poor solubility, their effects are long term and they can travel long distances regionally.²¹ Toxic chemicals can enter marine food chains, and can accumulate to high levels in top predators. Changes in the marine environment from land-based contaminants can be the result of a single large event, or many small ongoing events. These are often unnoticed and occur over a long time.



The effects of multiple activities within a catchment can cumulative to cause significant change in the marine environment, as has been the case with the Mahurangi Harbour in Auckland (*Raewyn Peart*)

Sedimentation

An increase in the rate that sediment is transferred into the receiving environment can bring about profound changes in the marine ecosystem. An excess of sediment can reduce water depths, water coverage and change the type of sediment found on the seafloor (such as the size of the particles). In addition, when excess sediment enters the receiving environment it can become suspended in the water, reducing light levels and the ability of marine creatures to feed, and affecting the clarity of the water and its turbidity (cloudiness of a fluid generally invisible to the naked eye). Research being undertaken by NIWA suggests that severe impacts can result from fine sediment being suspended in the water column, because of the way the particles interfere with light penetration of the water (or light attenuation).²² This can lead to impacts such as the inability of visual predators to see prey.



Elevated levels of sediment, as shown here in the Wairoa River feeding into the Kaipara Harbour, can bring about profound changes to the marine environment (*Raewyn Peart*)

at much lower than lethal concentrations. Only a small proportion of heavy metal and pollutant contaminants are dissolved in the water column; the majority remain attached to particles of sediment, which settle in the inner parts of harbours and estuaries. In these areas, where wave and current movements are likely to be too weak to move the polluted sediments around, they can build up to toxic levels.²⁹

Scientific studies since the 1950s have shown that a high body burden of organochlorine pesticides (such as DDT) in animals can cause ecological damage, including abnormalities in the reproduction and development of fish-eating birds. Concerns have also been voiced about the impact that such chemicals are having on some marine mammals, including that they may be damaging their health, affecting their fertility, reproduction, immune system and growth.³⁰



Very high DDT levels have been detected in Hector's dolphins such as these shown here in Akaroa, Banks Peninsula (Raewyn Peart)

DDT and Hector's dolphins

DDT was used in the 1950s and 1960s to control agricultural pests. During this period, it was applied over wide areas of land, often being mixed into fertilisers and top-dressed onto pastures. The use of DDT in New Zealand was restricted in the 1970s and banned in 1989. Today, the residual DDT concentrations in the country's pasture soils are relatively low.³¹ However, there is increasing evidence that DDT from the soils is still washing into the marine environment and having a negative impact on the marine ecosystem. The near-shore habitat of New Zealand's threatened Hector's dolphins has exposed this species to a variety of pollutants and contaminants such as organochlorines and heavy metals. Hector's dolphin tissue has been found to contain high levels of organochlorines such as DDT, PCBs and dioxins.³² The DDT levels detected in Hector's dolphins are very high on a global scale and second only to the highly contaminated Canadian beluga whales.³³

Organochlorines are known to have detrimental effects on the reproductive success of some marine mammal species, in some cases resulting in sterility, and they may suppress the immune system in some species.³⁴ Persistent organic pollutants such as DDT, PCBs and dioxins are known to be particularly soluble in fatty tissues, and because the body is unable to excrete these substances, they accumulate over each animal's lifetime. In mammals, they are also passed onto the young in their mother's milk. When tissues of stranded and bycaught common dolphins in New Zealand have been analysed, traces of PCBs and a range of pesticides including DDT have been found.

Once these substances enter the food chain, they travel up the different trophic levels. The pollutants can have effects higher up the food chain as shellfish and other organisms are eaten by fish, birds and marine mammals. Marine organisms can be affected by the toxicity when they ingest the sediment or take up toxins that have dissolved into the water column. The effects that such toxins have on marine life varies widely but can be very serious. For example, even small amounts of lead in the water are very toxic to marine life.

Auckland sediment contaminant monitoring

The Auckland Council has been running the Regional Sediment Contaminant Monitoring Programme since 1998. The information this programme collects provides important data for Auckland's state of the environment reporting, management of stormwater quality, resource consenting, policy development and public education. Recent analysis of this data has shown that spatial patterns remain similar to those previously reported. The highest concentrations of contaminants were found in muddy upper estuarine areas where runoff from extensively urbanised and industrialised catchments is entering the coastal marine area, particularly in the Tāmaki Estuary and central Waitematā Harbour. Lowest contaminant concentrations are found in estuaries with rural or forested catchments and on open coastal beaches.³⁵

Organic material

Agricultural runoff, urban wastewater and industrial waste transfer organic material to the catchment. This can include faeces from humans and animals; proteins, vegetables and sugars from food preparation; and cleaning soaps. When effluent enters a healthy aquatic environment where there is plenty of oxygen dissolved in the water, aerobic (oxygen-using) bacteria, which occur naturally in the water, will eat the organic material, using up oxygen in the water to do so. Consequently their numbers grow beyond what would normally be expected in a healthy aquatic ecosystem.

This use of oxygen can have very serious effects on the health of the ecosystem, as fish and plants need oxygen levels to be maintained in order to survive. Even an occasional reduction of the oxygen level can have the effect of removing the marine species that rely on a continuous supply at a high level. If all the oxygen is used up, anaerobic bacteria (non-oxygen-using bacteria) will take over, decomposing the waste material and producing gases such as methane hydrogen sulphide and carbon dioxide, making the water toxic.⁴²

Thus, the potential harm caused by organic waste can be measured by its potential to remove oxygen from the water. There is a standard measure known as the 'biochemical oxygen demand' which is the amount of dissolved oxygen needed by aerobic bacteria to consume the waste (over a 5 day period where the water temperature is 20 degrees centigrade, measured in terms of the number of grams of organic material per cubic metre). Thus, if waste deposited in the catchment has a biochemical oxygen demand loading that is too high, too much oxygen will be removed from the water by the aerobic bacteria, jeopardising plant and animal life.

The surface sediment of the seabed is an important ecological zone and the home of some specialised forms of life. It is normally oxygenated. Excess sediments and low oxygen in the water above can cause it to become anoxic and unable to support its normal inhabitants.



Urban wastewater pipes, such as that shown here at Mangawhai, Northland, can transport a range of contaminants into the marine environment (Raewyn Peart)

Regional policy statements, regional plans and district plans are required to 'give effect to' the Freshwater NPS. Regional councils must implement it by 31 December 2025, or by 31 December 2030 if meeting the earlier date would be impracticable or would result in lower-quality planning.

There are a wide range of methods which may be used to ensure freshwater objectives and limits are achieved. This will require rules controlling land use changes, such as urban development and agricultural intensification. In areas where objectives and limits are not being achieved, this will require management of existing land use activities. For example, the preparation and implementation of farm environmental management plans and nutrient budgets can enable a range of environmental impacts from a farming operation to be identified and managed. Regional plans generally require stock to be excluded from freshwater bodies and the coastal marine area. Another mechanism which can be used to control non-point source discharges at a catchment level is a 'cap and trade' scheme which allocates tradable pollution allowances to individual land users.

Forestry generally causes a large amount of sediment runoff during the harvesting cycle which may have a significant impact on freshwater bodies and the coastal environment. Regional plans should include controls that ensure forestry operations are managed to minimize sediment runoff. This should include requiring riparian areas to be retained and restricting the proportion of a catchment being harvested at any one time. In highly erodable areas, clear-felling forestry may be inappropriate.

Estuaries, which lie at the interface between the freshwater and coastal environment, are not freshwater bodies under the definition in the Freshwater NPS. However, the policy statement directs regional councils to 'improve the integrated management of freshwater ... including the interactions between freshwater ... and the coastal environment' and make or change regional policy statements 'to provide for the integrated management of the effect of the use and development of ... land and fresh water on coastal water'. Therefore an important consideration when setting freshwater objectives and limits is the characteristics of the receiving coastal environment. In particular limits which protect the values of freshwater may not protect the values of estuaries, for example nitrogen limits. Wetlands are freshwater bodies and the Freshwater NPS specifically provides for those.⁴³



The Freshwater NPS requires councils to set limits for freshwater quality, and these will impact on the state of freshwater entering the marine environment, such as at Port Jackson, Coromandel Peninsula, shown here (Raewyn Peart)

Policy 23: Discharge of contaminants

1. In managing discharges to water in the coastal environment, have particular regard to:
 - a. the sensitivity of the receiving environment;
 - b. the nature of the contaminants to be discharged, the particular concentration of contaminants needed to achieve the required water quality in the receiving environment, and the risks if that concentration of contaminants is exceeded; and
 - c. the capacity of the receiving environment to assimilate the contaminants; and
 - d. avoid significant adverse effects on ecosystems and habitats after reasonable mixing;
 - e. use the smallest mixing zone necessary to achieve the required water quality in the receiving environment; and
 - f. minimise adverse effects on the life-supporting capacity of water within a mixing zone.
2. In managing discharge of human sewage, do not allow:
 - a. discharge of human sewage directly to water in the coastal environment without treatment; and
 - b. the discharge of treated human sewage to water in the coastal environment, unless:
 - i. there has been adequate consideration of alternative methods, sites and routes for undertaking the discharge; and
 - ii. informed by an understanding of tangata whenua values and the effects on them.
3. Objectives, policies and rules in plans which provide for the discharge of treated human sewage into waters of the coastal environment must have been subject to early and meaningful consultation with tangata whenua.
4. In managing discharges of stormwater take steps to avoid adverse effects of stormwater discharge to water in the coastal environment, on a catchment by catchment basis, by:
 - a. avoiding where practicable and otherwise remedying cross contamination of sewage and stormwater systems;
 - b. reducing contaminant and sediment loadings in stormwater at source, through contaminant treatment and by controls on land use activities;
 - c. promoting integrated management of catchments and stormwater networks; and
 - d. promoting design options that reduce flows to stormwater reticulation systems at source.
5. In managing discharges from ports and other marine facilities:
 - a. require operators of ports and other marine facilities to take all practicable steps to avoid contamination of coastal waters, substrate, ecosystems and habitats that is more than minor;
 - b. require that the disturbance or relocation of contaminated seabed material, other than by the movement of vessels, and the dumping or storage of dredged material does not result in significant adverse effects on water quality or the seabed, substrate, ecosystems or habitats;
 - c. require operators of ports, marinas and other relevant marine facilities to provide for the collection of sewage and waste from vessels, and for residues from vessel maintenance to be safely contained and disposed of; and
 - d. consider the need for facilities for the collection of sewage and other wastes for recreational and commercial boating.

Policy 21 requires 'priority' to be given to improving water quality in areas of the coastal environment where water quality has deteriorated so that it is having a significant adverse effect on ecosystems, natural habitats, or water-based recreational activities, or is restricting existing uses such as shellfish gathering. This policy also directs regional councils to identify such areas of coastal water and water bodies and include them in their plans. For example, the proposed Auckland Unitary Plan includes the spatial identification of degraded water areas. These were drawn from a report on marine degraded areas informed by benthic health, sediment chemistry and water quality monitoring programmes (see Figure 17.3).⁴⁴



The NZCPS sets out what local authorities need to do to effectively manage stormwater discharges, such as the one shown here at North Shore, Auckland, including adopting an integrated catchment management approach (Raewyn Peart)

Through Policy 21, regional councils must require that stock be excluded from the coastal marine area, adjoining intertidal areas and other water bodies and riparian margins in the coastal environment, within a prescribed time frame. The 'Australian and New Zealand Guidelines for Fresh and Marine Water Quality' promulgate a water quality management framework that could be useful for the purpose of implementing Policy 21 of the NZCPS. These water quality guidelines contain 'trigger values'. If concentrations of contaminants are below these values there is unlikely to be any concern. If concentrations are above, further investigation is required to develop site-specific criteria and if concentrations are above the site-specific criteria, action is required. These guidelines provide an important reference for water quality management in New Zealand, particularly in regards to toxic contaminants, and they offer methods for setting limits on pollutant concentrations in coastal and marine environments.⁴⁵

Policy 22 requires that subdivision, use or development will not result in a significant increase in sedimentation in the coastal marine area, or other coastal water. This policy requires councils to collect baseline information with which to assess the current levels of sedimentation in coastal waterways and the marine area. They are also required to identify the likely environmental impacts of current sedimentation levels, monitor changes in sedimentation levels and impacts, and ensure that there is no significant increase in sedimentation in coastal waterways and the marine area. Councils will need to deal with sediment generation on a catchment basis to address all activities that contribute to sediment loadings in runoff and stormwater systems.

Policy 23 requires particular regard to be given to avoiding significant adverse effects on ecosystems and habitats after reasonable mixing when managing discharges to water in the coastal environment. This particular policy arose from the recommendation of the Board of Inquiry that water quality policies in the former NZCPS be strengthened. This was to ensure that decision-makers who are considering discharges to water in the coastal environment have proper regard to how to prevent or minimise adverse effects, particularly on ecosystems, habitats and the life-supporting capacity of water.⁴⁶

Policy 23 provides clear direction that discharges of untreated human sewage into coastal water should not be occurring, including those from wastewater system overflows. In addition, the discharge of treated sewage into the coastal environment should only be permitted where there has been adequate consideration of alternatives and the effects on tangata whenua values has been understood and considered. There must also be early and meaningful consultation with tangata whenua when developing plan provisions which provide for such activity.

Policy 23(4) sets out what local authorities need to be doing to effectively manage stormwater discharges. This includes adopting an integrated catchment management approach which links the management of stormwater networks with that of other activities within the catchment, reducing contaminant and sediment loadings at their source through such actions as placing controls on land use activities, adopting good design, and requiring stormwater treatment. In addition this subpolicy requires that cross-contamination between stormwater and sewage systems is addressed. The Department of Conservation has prepared some brief guidance on implementing Policy 23. This makes it clear that when considering the discharge of treated sewerage into the coastal environment, councils will need to consider the location, frequency, duration, volume, level of treatment and the extent to which the discharge is the best practicable option (within a relevant time frame).⁴⁷

Regional council planning

Regional councils have responsibility for controlling the use of land for the purpose of maintaining coastal water quality and coastal marine ecosystems⁴⁸ and controlling discharges into water bodies.⁴⁹ This includes managing the negative effects of sedimentation and pollutants which are derived from land-based sources and which end up being discharged into the marine area. Regional councils prepare regional policy statements and regional plans to address the issue. Some regional councils prepare separate plans which specifically address issues such as sedimentation and discharges from dairy farms to help them effect their obligations under the RMA. In practice it has proved easier to deal with point source pollution under the RMA than non-point source pollution.

There are two main approaches that regional plans take to avoiding or minimising impacts of land use on the marine area:

- One method employed by a number of regional councils is to classify coastal receiving waters and to apply appropriate water quality or sediment quality standards to those waters. This is a more 'effects-based' approach, in that any consequent land use activity could be expected to at least consider the effects of the activity on the relevant receiving water quality or sediment quality standard. This allows innovation on behalf of the consent holder as to what site management practices or controls could be implemented. Water quality standards which are linked to rules will have a more direct effect on outcomes than standards linked to policies.



Figure 17.4 The wide range of issues addressed by the Whangamatā Harbour and Catchment Management Plans (Waikato Regional Council)⁵¹

Māori have long been advocates for planning, particularly in regards to coastal issues.⁵² There are a number of tools, including *taiapūre* and *mātaītai*, which are increasingly being used as part of managing a wider catchment. These tools are discussed in Chapter 4: Kaitiakitanga. More recently *iwi* have initiated catchment planning (see the third case study).

The success of catchment management planning lies in the engagement of the community and key agencies to support and implement the actions needed.⁵³ This is discussed further in the case study on Tairua Harbour and Catchment Management Plan below.

There are a number of best practice elements for catchment management planning that have been identified by research, including:^{54,55}

- Political leadership and multi-agency co-operation
- Cross-sector collaboration
- Inclusion of *iwi* and community aspirations and drivers
- Robust science
- Improved capacity building and action on the ground
- Provision of adequate resourcing for both planning and implementation phases
- Good governance and clear institutional frameworks
- Monitoring and evaluation of outcomes that leads to adaptive management
- The presence of a strong catchment manager or champion

A summary of the outcomes from the 10 year Integrated Catchment Management for the Motueka River programme identified a number of issues that were addressed during the programme⁵⁶ as shown in Figure 17.5.

Territorial authorities undertake monitoring and mapping to identify any issues associated with localised flooding and overland flow, as well as any pollution-related issues. The Local Government Act requires them to undertake assessment of the water and sanitary services in their district, including wastewater and stormwater, to assess amongst other things whether they pose a danger to public health.

Elements of best practice

Potential environmental effects	Elements of best practice
Agriculture	
Runoff from pasture causes increased sedimentation of the marine environment	• Exclude stock from waterways, wetlands and the coastal marine area • Create riparian buffer zones using native flora along waterways to protect the catchment from sediment, organic material and nutrients washed from pasture by the rain • Construct runoff management systems such as wetlands or pools along subsurface farm drains, where sediment and contaminants can settle • Restore old wetlands to create natural obstacles to sediment runoff • Construct farm tracks on sites where gradients are low • Reduce stocking densities or retire land on steep or highly erodible slopes • Develop and implement environmental farm plans • Provide financial incentives to encourage farmers to replant riparian zones with indigenous vegetation, and to retire land from productive use in particularly erodible areas
Nitrogen-nitrate and phosphate pollution derived from urine and fertilisers used to promote pasture growth encourage growth of algae and other marine vegetation, leading to eutrophication, oxygen depletion and ocean acidification	• Avoid using more fertiliser than can be taken up by the crop or pasture – soil nitrogen testing can indicate exactly how much nitrogen fertiliser a field requires, and therefore help avoid overfertilising • Ensure that catchments are not farmed overintensively, to avoid generating more pollution than the catchment can cope with • Use stand-off pads and barns for stock during wet seasons
Organic matter (animal faeces) enters the waterways, stimulating the activity of aerobic bacteria which deplete oxygen levels in the water	• Prevent stock from accessing waterways and the coastal edge • Have adequate farm treatment systems to cope with the volume and strength of effluent produced • Ensure land irrigation is not overused to avoid effluent sprayed onto fields running off
Forestry	
Felling of trees and construction of tracks results in increased runoff into the catchment	• Undertake tree planting, felling and track construction in plantation forests so that a riparian buffer is retained to protect the catchment from sedimentation • Avoid clear-felling on highly erodible soils • Avoid clear-felling large areas of land at one time, or in succession, in the same catchment • Install cut-off drains at regular intervals on tracks so that water runoff can be managed and directed away from waterways • Include in regional plans rules restricting the planting and harvesting of plantation trees from within a specified distance from rivers, streams and the coastal margin, and if necessary seasonally to avoid high rainfall times • Include in the regional plan rules controlling the harvesting of trees on steep or erodible slopes
Burning to clear land for forestry or after harvesting, and mechanical clearing of the plantation, cause intensive soil disturbance leading to increased runoff into the catchment	• Maintain riparian buffer zones along waterways to prevent large amounts of sediment from entering the catchment • Avoid highly invasive mechanical and burning techniques to clear land with a high erosion potential • Carefully manage the disposal of slash

Auckland's Sustainable Catchments Programme

The purpose of the Auckland Council's Sustainable Catchments Programme is to provide an integrated catchment planning and implementation framework to protect the long-term health of receiving environments through iwi and community interventions. With a byline of 'Weaving Science and Community into Action' the programme uses a range of tools to foster land use practice change including riparian restoration, education, creative engagement, farm plans, incentives, research and walkways.

The programme currently works in seven identified priority catchments across Auckland. These are: Mahurangi, Whangateau, Kaipara (Hoteo), Greater Tamaki, Henderson – Huruheru (Project Twin Streams), South Waitemata (Meola, Oakley and Motions) and Manukau (Papakura/Pahurehure). Two of these, the Mahurangi Action Plan and Project Twin Streams, are 10 years old this year.

Figure 17.7 provides some highlights of the programme.



Project Twin Streams, Auckland (Raewyn Peart)

Key area	Highlight
Regenerating blue-green areas	- More than 780,000 plants have been planted by Project Twin Streams since 2004. Over half a million of these have been planted by around 78,000 volunteers - Co-hosting planting days with Mahurangi College students and Whangateau Harbour Care to demonstrate riparian restoration best practice at sites in Whangateau and Mahurangi
Enhanced fisheries	- Annual shellfish monitoring workshops and field work in Whangateau with Mahurangi College students and Whangateau Harbour Care - Planting of riparian species providing stream shade and habitat for aquatic species along the Henderson Creek, Opanuku, Oratia, Swanson, Paremuka and Waikumete Streams, at La Rosa Reserve, downstream of Miranda Reserve and on the Meola and Oakley Creeks - Hoteo catchment work specifically targets areas which drain into high-value seagrass beds in the Kaipara Harbour located at the mouth of the Hoteo River
Manawhenua integration	- Trial site of the Ngāti Whatua holistic approach and management on the Anamata Stream - Mahurangi Catchment Strategic Plan written with and endorsed by Ngāti Whatua o Kaipara
Community ownership	- Community organisations manage work programme in Project Twin Streams catchments - Mahurangi community actively involved in planning implementation of the Mahurangi Action Plan strategy - Groups in Whangateau planning public land projects with council
Active land management	- Over 100 kilometres of stream and wetlands have been fenced and around 900 hectares of land have been protected since 2004 through grants across Mahurangi, Hoteo and Whangateau. The grants for fencing and riparian planting are provided to applicants in priority subcatchments to support land use management changes. - Ongoing active land management along the five Project Twin Streams catchments and in the South Waitemata catchment with multiple agencies and groups involved - Over 50 farm plans written to support rural land management
Knowledge generation	- Over 100 creative engagement art projects have been completed by the Project Twin Streams community - Erosion and Sediment Control Plans have been produced for the Hoteo, Mahurangi and Whangateau catchments to guide planning and implementation - Community open day to share science in the Hoteo catchment attended by over 100 people

Figure 17.7 Achievements of Auckland's Sustainable Catchments Programme

Issue 1: Coastal land and water management (including mangrove management)

Key concerns	Potential methods	Expected outcomes
• Sedimentation of harbour and decreased access even during high tide • Increased use of harbour and sometimes conflicting users • Access to harbour and coastal environment • Expansion of mangroves (last decade) into recreational area and other significant habitats such as seagrass • Rapid expansion of saltwater paspalum • Decline in coastal habitats • A reduction in number of fish caught in the harbour • Shellfish toxins leading to gathering and consumption restrictions	• Review of dredging consent and possible increased dredging • Land management practices implemented up stream to alleviate sediment input • Preservation of coastal habitats through restoration works including saltwater paspalum control, planting and retirement • Ongoing Beachcare works • Appropriate mangrove removal consents gained and implemented • Saltwater paspalum control consent sought and implemented • Enforcement of navigation safety bylaws • Re-establishment of harbour committee with multiple stakeholders	• Improved harbour access • Improved habitat • Historical recreation access point regained • Line held on mangrove spread • Intact dunes systems providing buffers for coastal erosion, landscape values and habitat • Better communication between agencies and harbour users

Figure 17.8 Example of an issue and associated key concerns, potential methods to manage and expected outcomes from the Tairua Harbour and Catchment Management Plan (Waikato Regional Council⁶⁰)

An essential part of the Plan is implementation and monitoring. A detailed implementation plan provides associated actions for achieving outcomes. Where needed, these should be able to feed into different annual and long-term planning processes in order to allocate the required funding. Monitoring is also important, and should identify changes over time and enable comparisons to be made.



Expected outcomes from the Tairua Harbour and Catchment Management Plan include improved harbour access and habitat (Raewyn Peart)

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Conclusion

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